

CSE 6220/CX 4220

Introduction to HPC

Lecture 5: Caching and Cache-Efficient Algorithms

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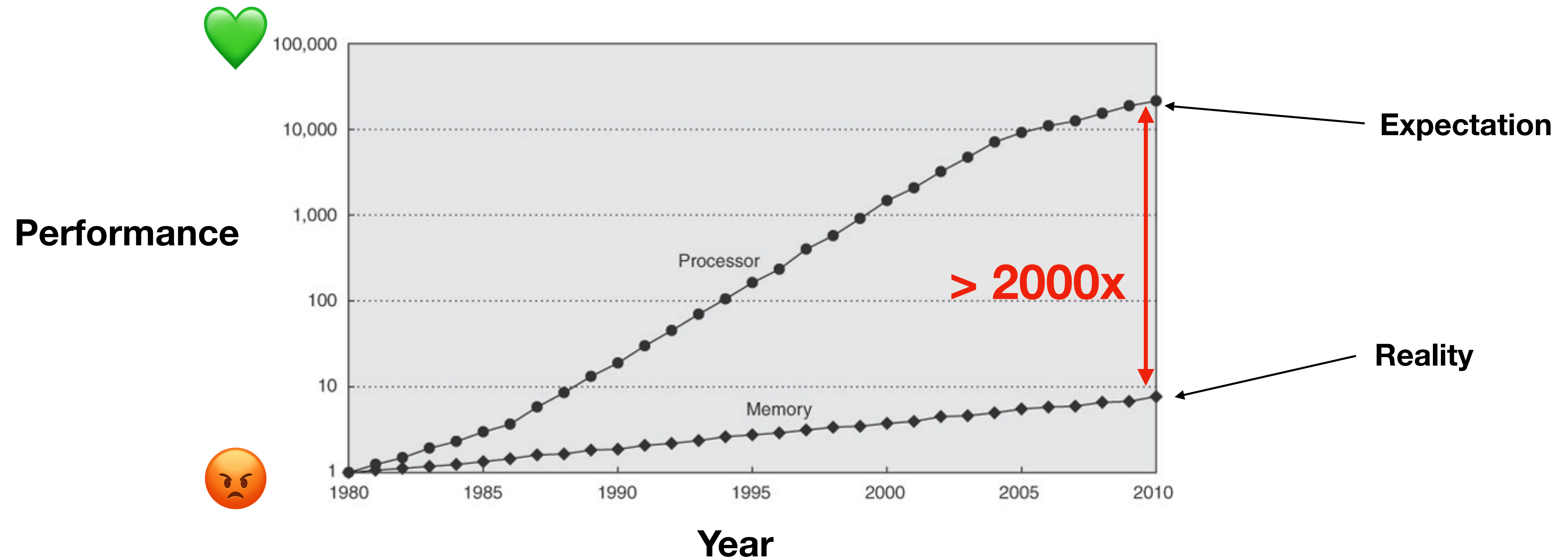
Georgia Tech College of Computing
School of Computational
Science and Engineering

**Let's discuss the memory hierarchy
on one processor - why?**

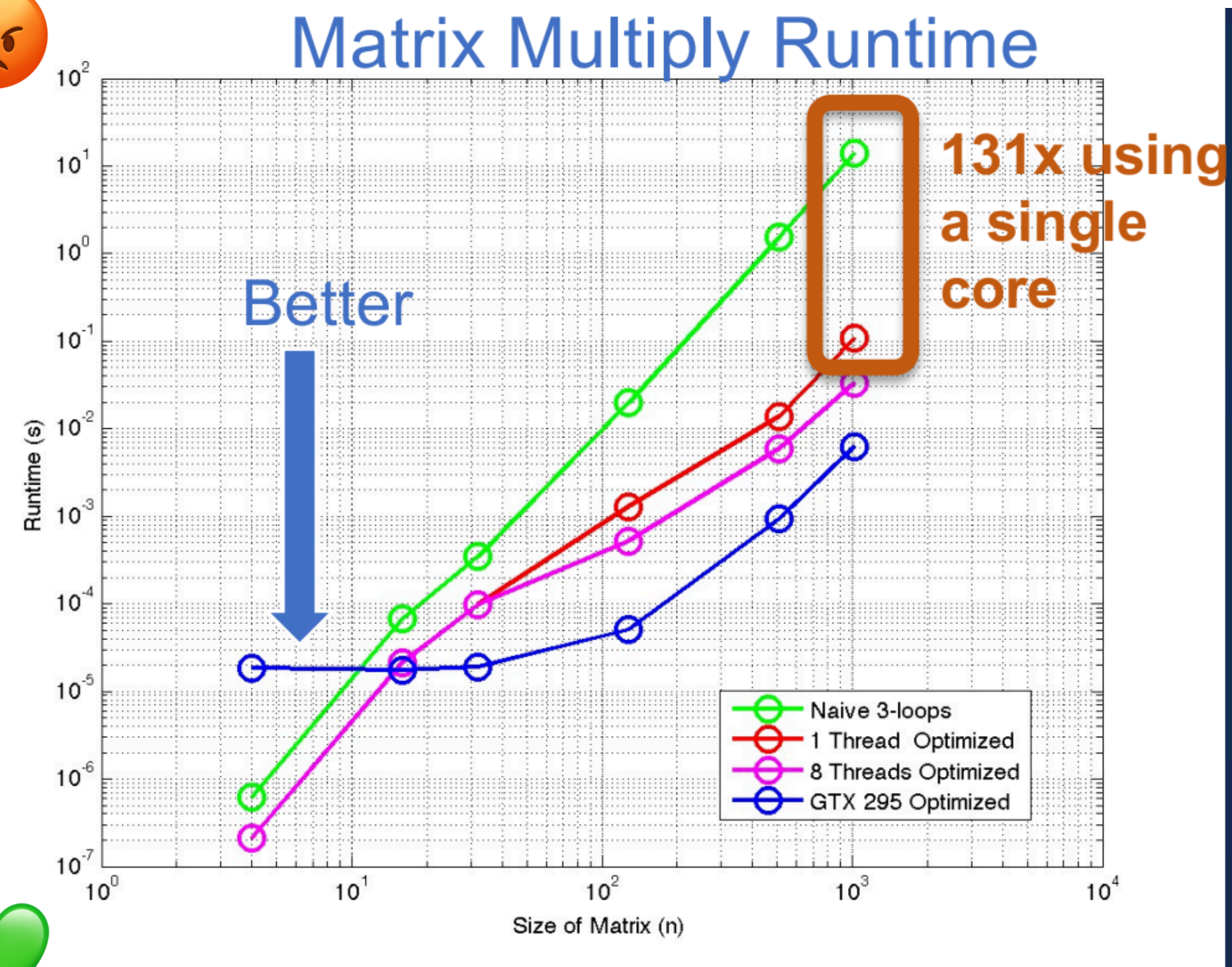
Memory limitations affect overall performance

Most applications run at **<10%** of “peak” performance (max possible rate of flops).

Much of the performance is lost on a single processor due to **data movement**.



Why should we care about serial performance in a course on high-performance computing?



Starting with a fast serial implementation is an **important first step** towards a fast parallel implementation.

Another way to say it: **If you parallelize slow serial code, you will get slow parallel code.**

Matrix multiply is extreme, but ~10x improvement from **sequential optimizations** is more common.

Memory Hierarchies

Memory access performance measurements

Memory accesses (load/store) have two costs:

Latency - the cost to load or store one word (τ)

Bandwidth - the average rate (bytes / sec) to load/store a large chunk of data (μ)

Bandwidth

$\approx$ data throughput (bytes/second)



Low Bandwidth



High Bandwidth

Latency

$\approx$ delay due data travel time (secs)

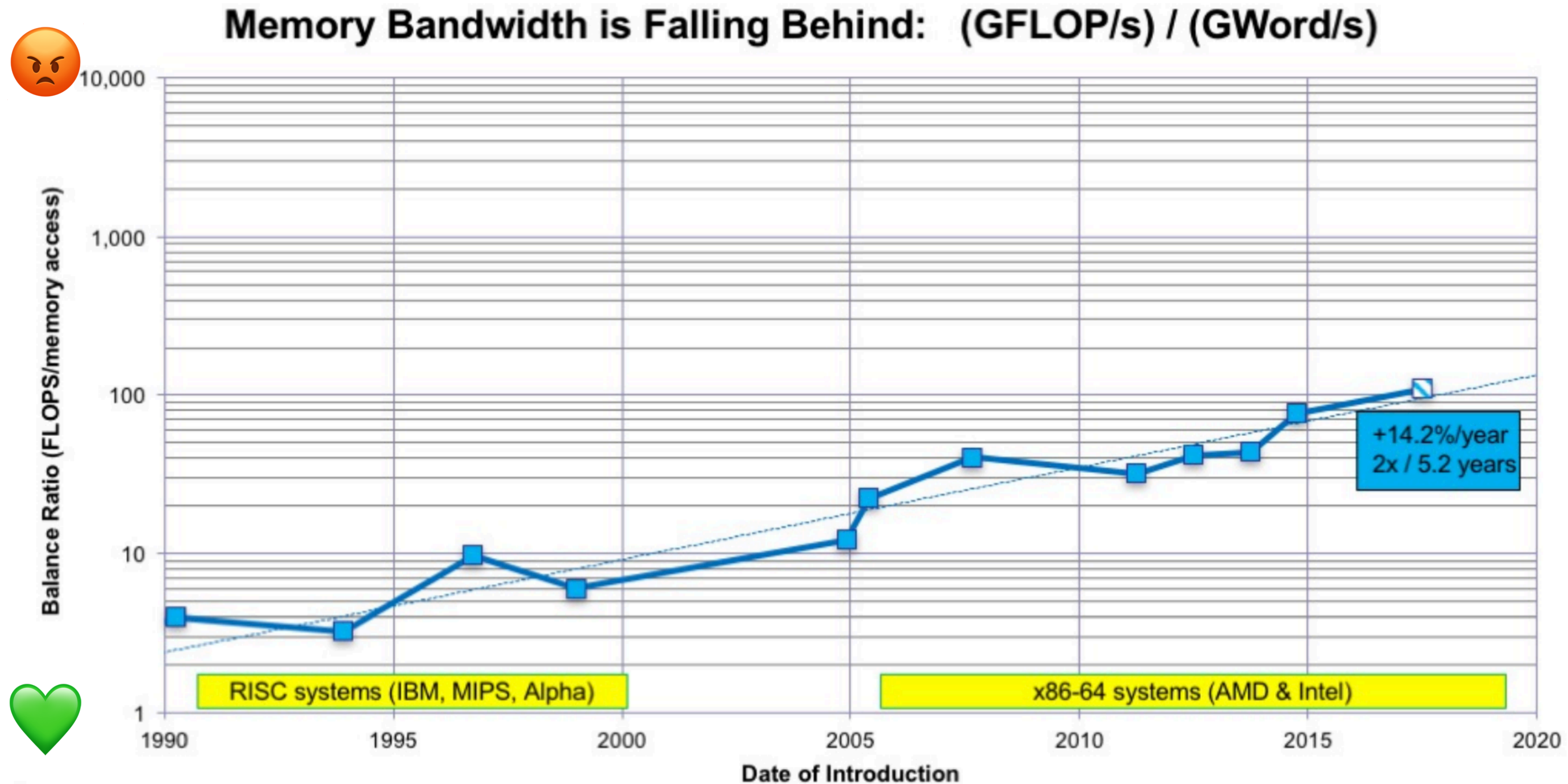


Low Latency



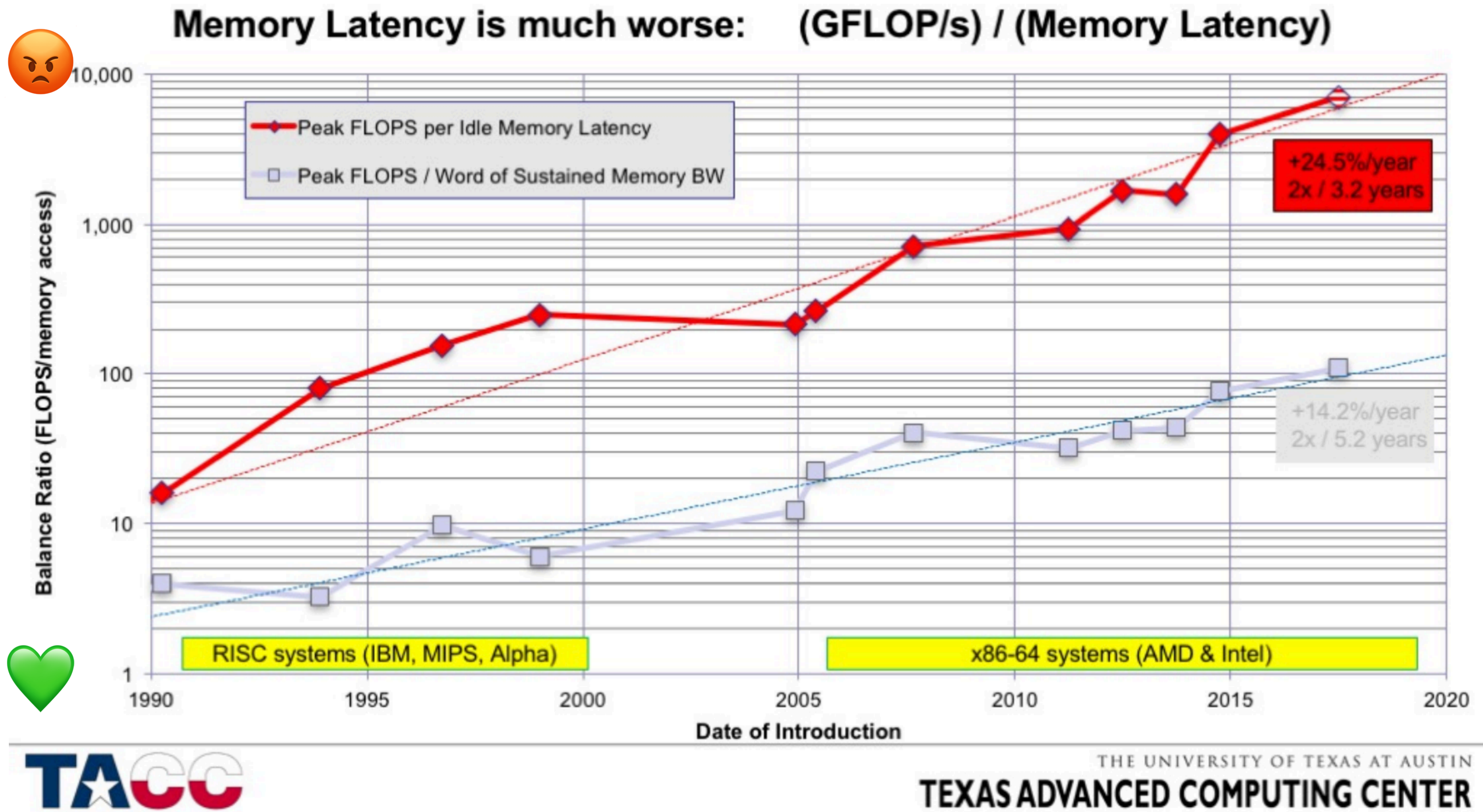
High Latency

Memory bandwidth gap



THE UNIVERSITY OF TEXAS AT AUSTIN
TEXAS ADVANCED COMPUTING CENTER

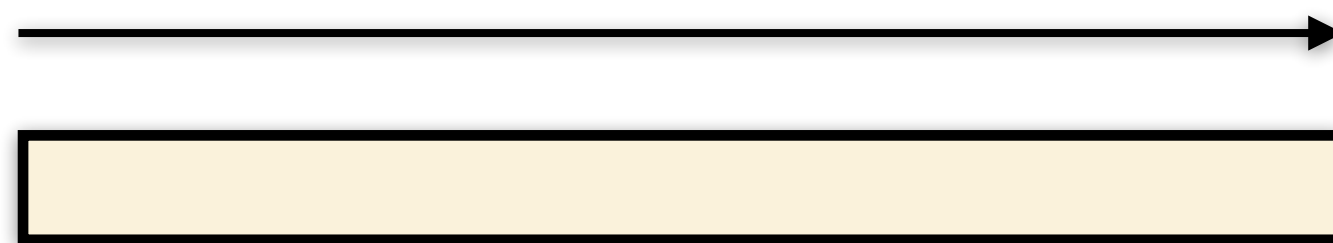
Memory latency gap is worse



Two main types of locality: Spatial and Temporal

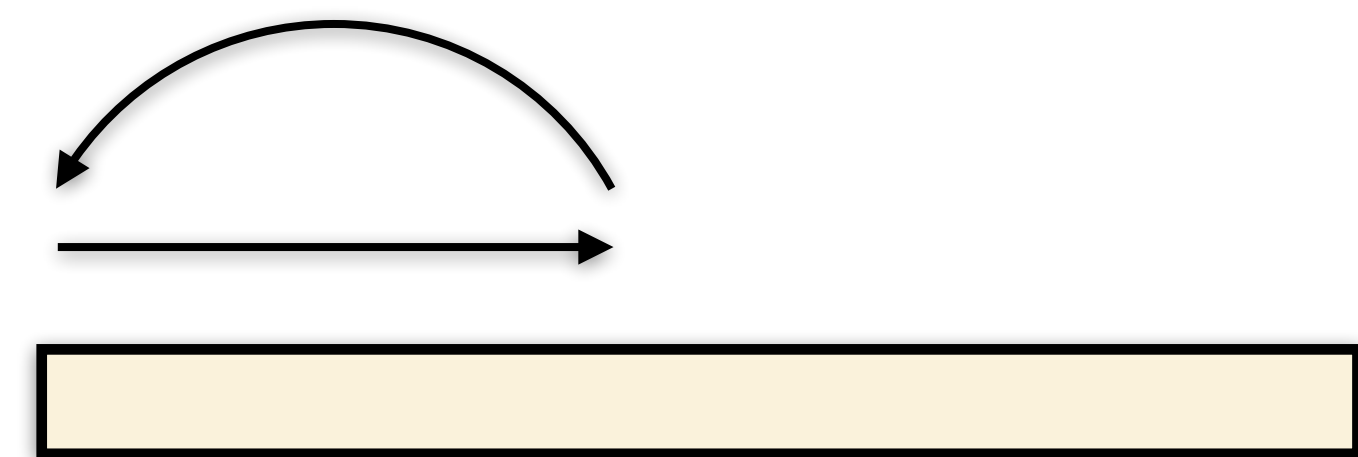
Most programs have a high degree of **locality**.

Spatial locality: how many accesses an algorithm makes to **nearby** data over a short period of time [Denning72, Denning05].



Makes use of **multiple elements transferred together**

Temporal locality: how many repeated accesses an algorithm makes to **the same** data over a short period of time [Denning72, Denning05].



Makes use of **efficient hierarchical accesses**

Memory hierarchy

The **memory hierarchy** takes advantage of locality to speed up the average case to handle memory latency.

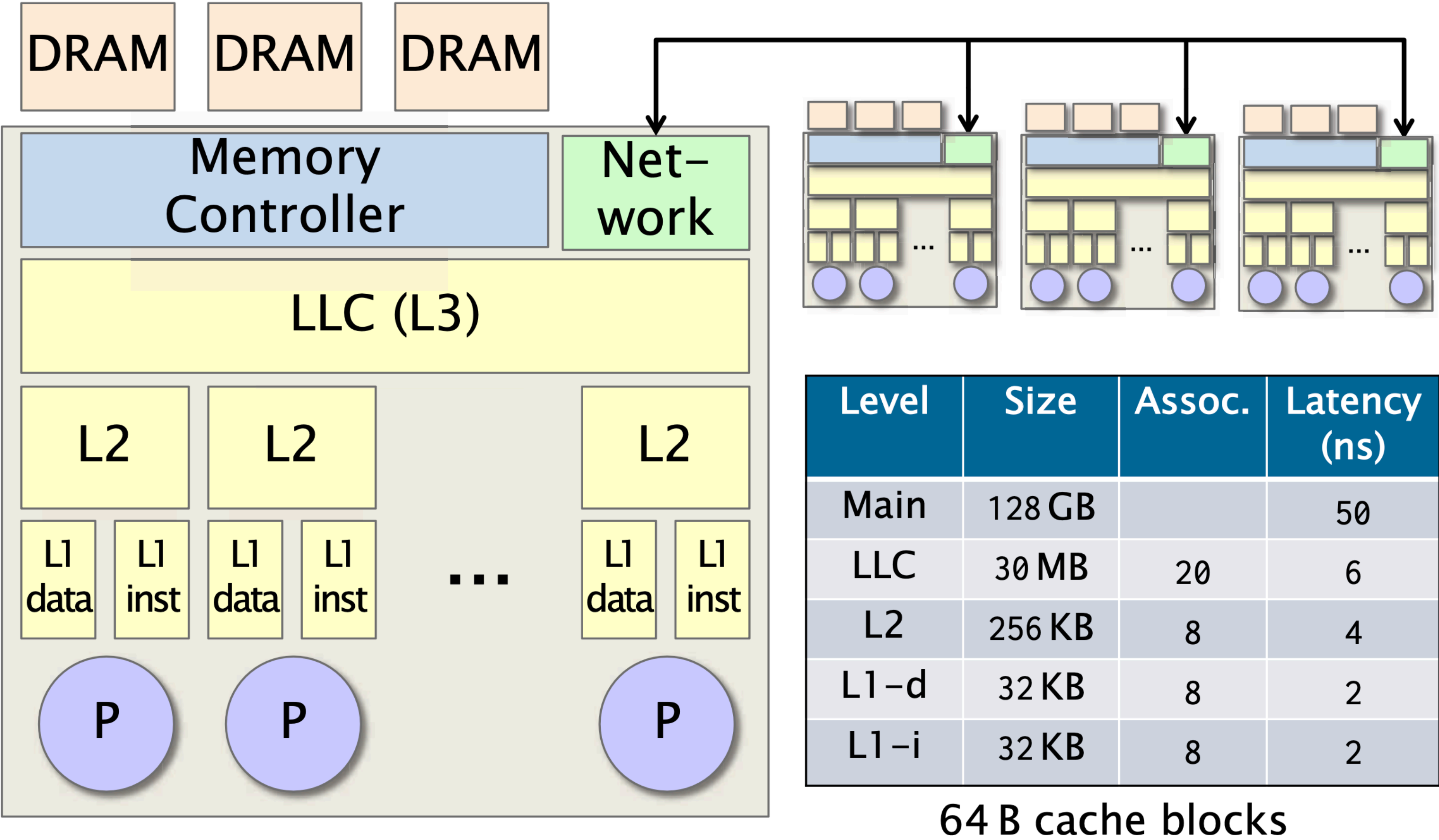


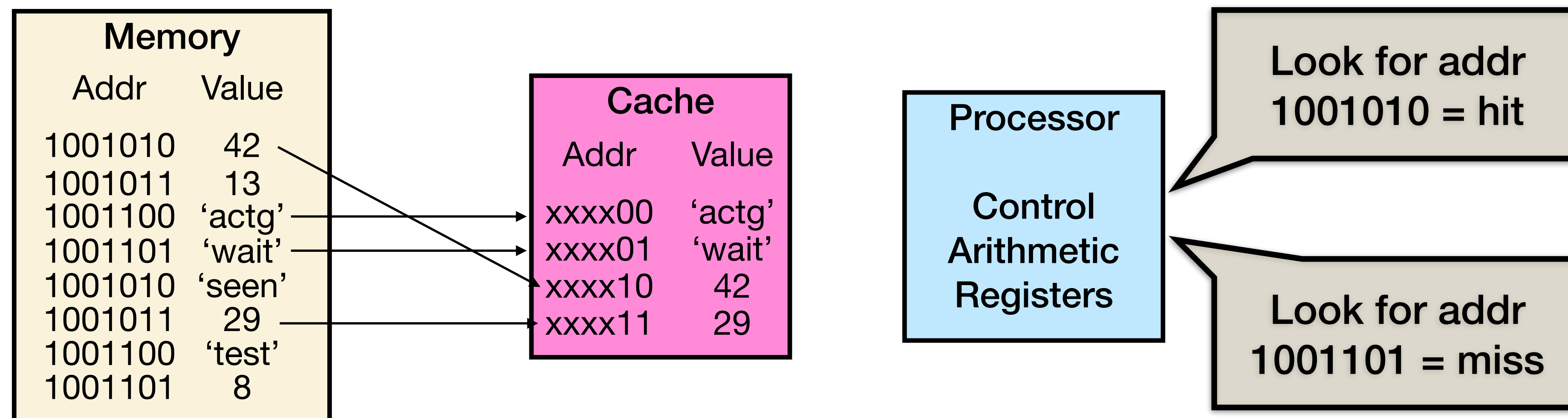
Figure from MIT 6.172

Cache basics

Cache is fast (expensive) memory which keeps a copy of the data; it is hidden from software.

Cache-line length: number of bytes loaded together in one entry (often 64 bytes).

Simple example: data at address xxxxx10 is stored at cache location 10.

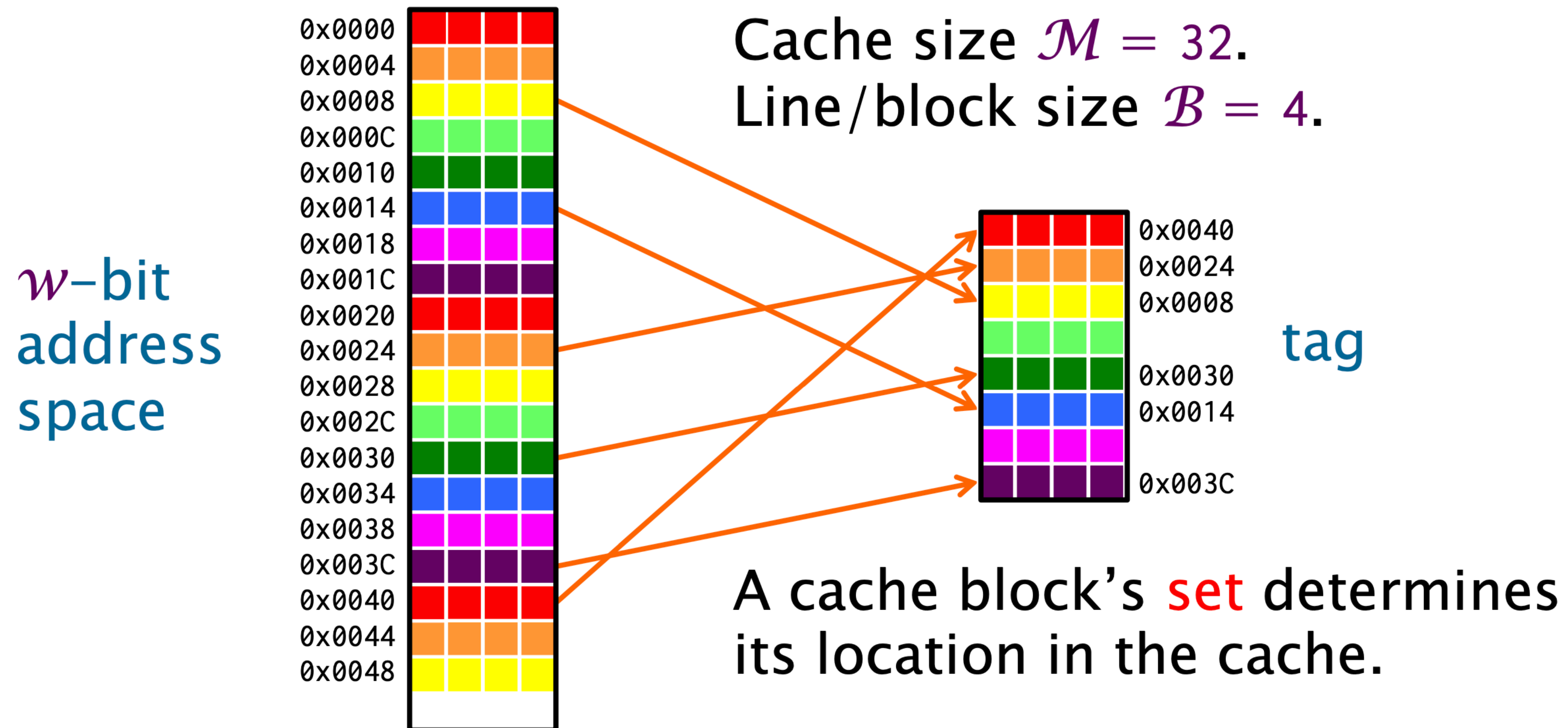


Cache **hit**: access to a memory address in cache - cheap

Cache **miss**: non-cached memory access - expensive

Need to look in next, slower level of memory.

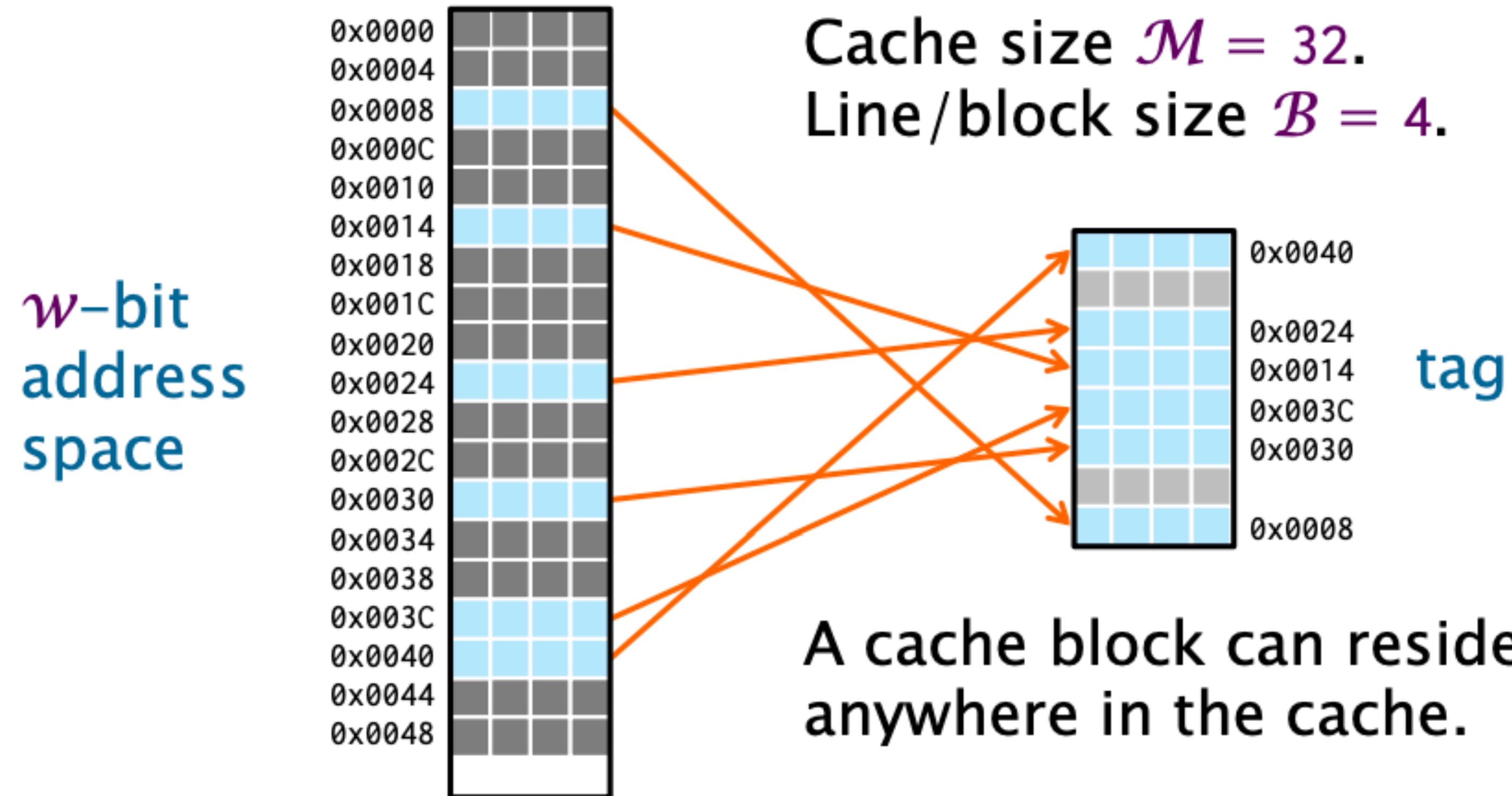
Direct-mapped cache



address		
	tag	set
bits	$w - \lg \mathcal{M}$	$\lg(\mathcal{M}/\mathcal{B})$
	offset	
bits	$\lg \mathcal{B}$	

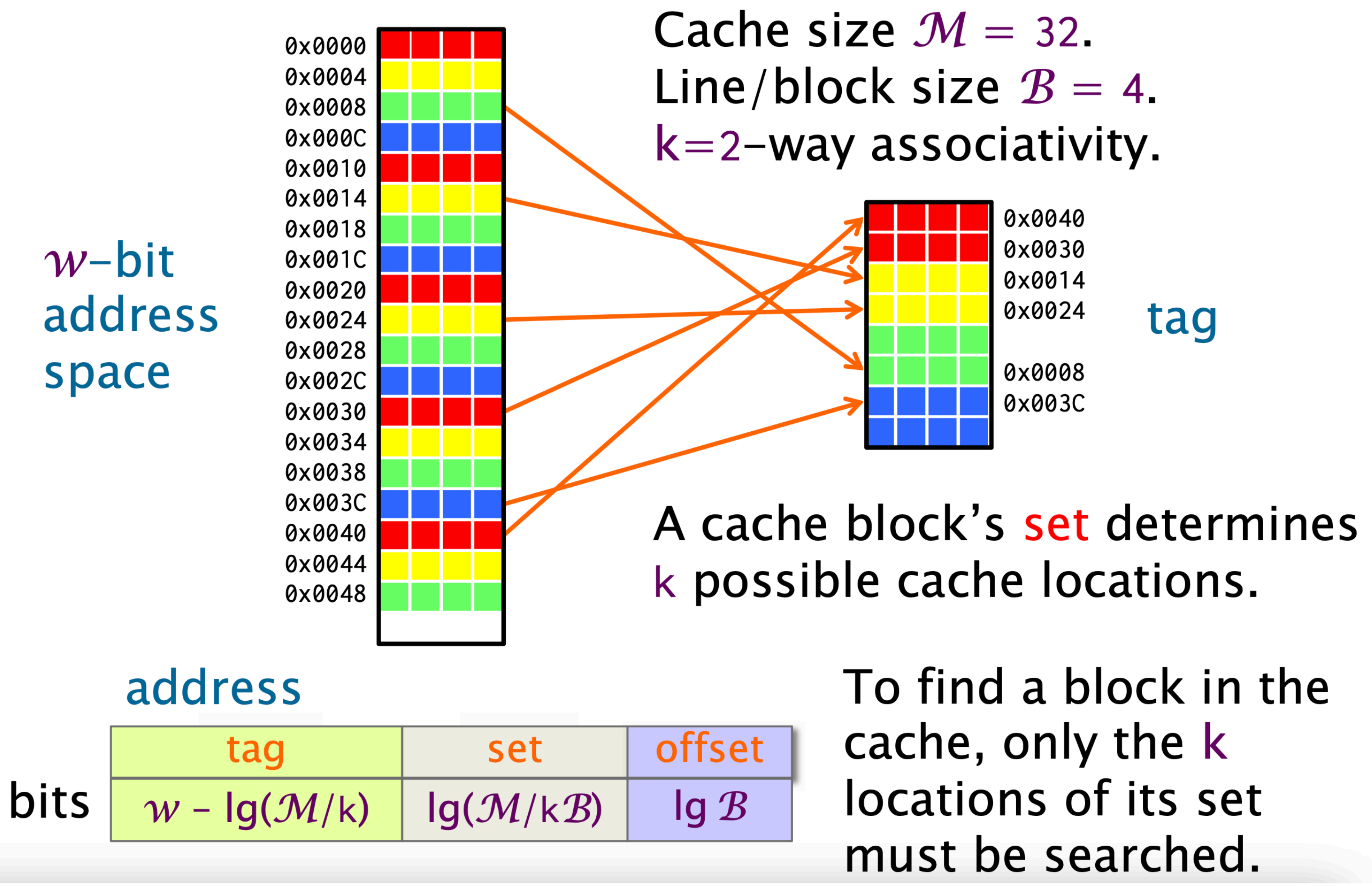
To find a block in the cache, only a single location in the cache need be searched.

Fully-associative cache



To find a block in the cache, the entire cache must be searched for the tag. When the cache becomes full, a block must be **evicted** to make room for a new block. The **replacement policy** determines which block to evict.

Set-associative cache



Type of cache misses - Three C's

Cold miss

The first time a cache block is accessed.

Capacity miss

The previous cached copy would have been evicted even with a fully-associative cache.

Conflict miss

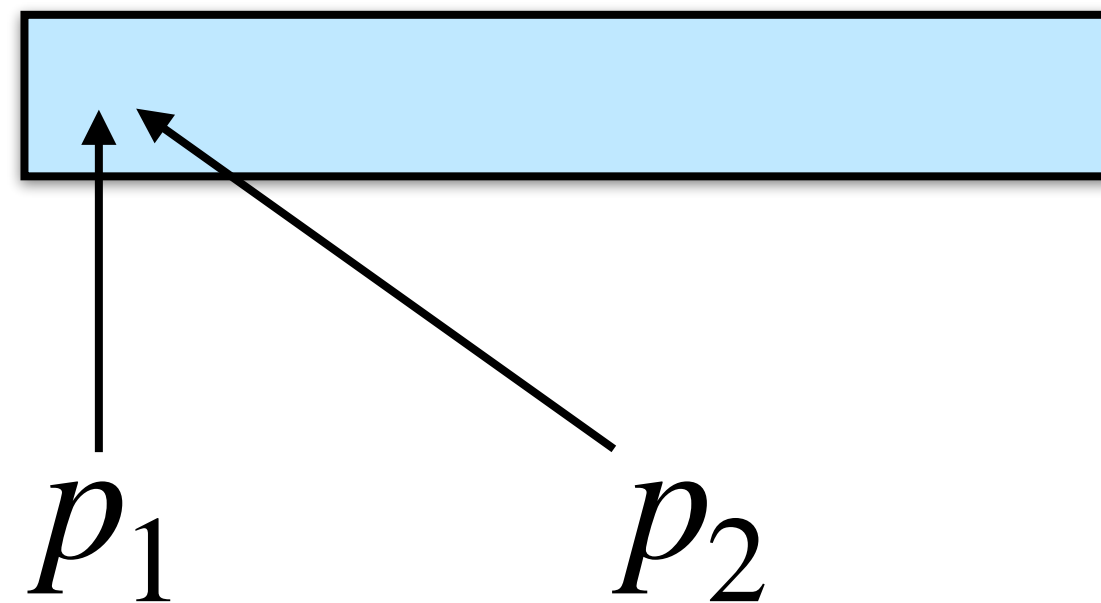
Too many blocks from the same set in cache. The block would not have been evicted with a fully-associative cache.

Sharing misses

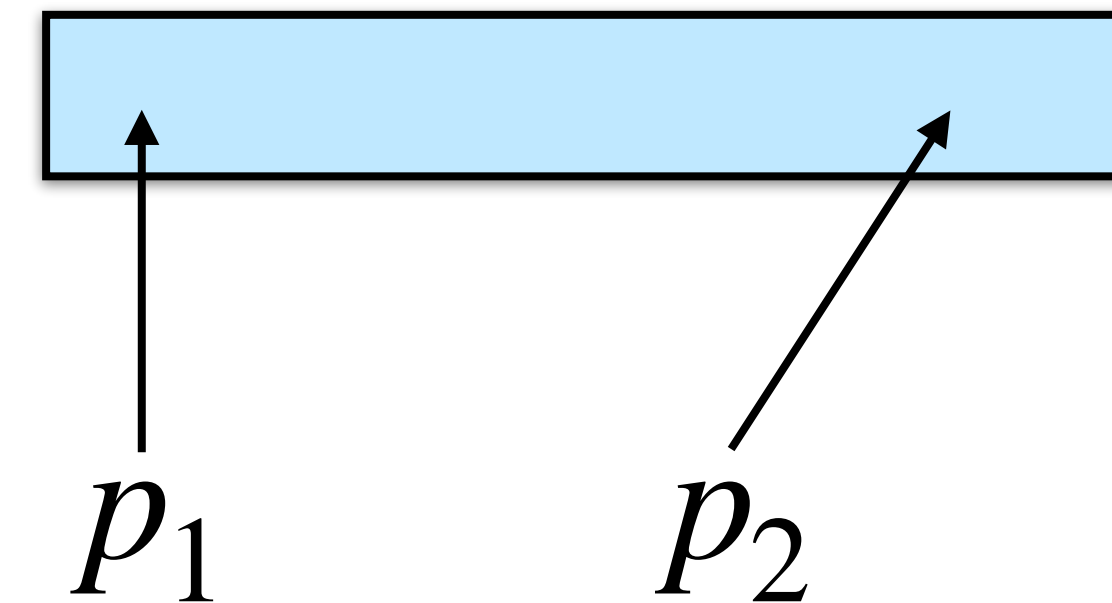
Sharing miss

- Another processor acquired exclusive access to the cache block.
- True-sharing miss: The two processors are accessing the same data on the cache line.
- False-sharing miss: The two processors are accessing different data that happen to reside on the same cache line.

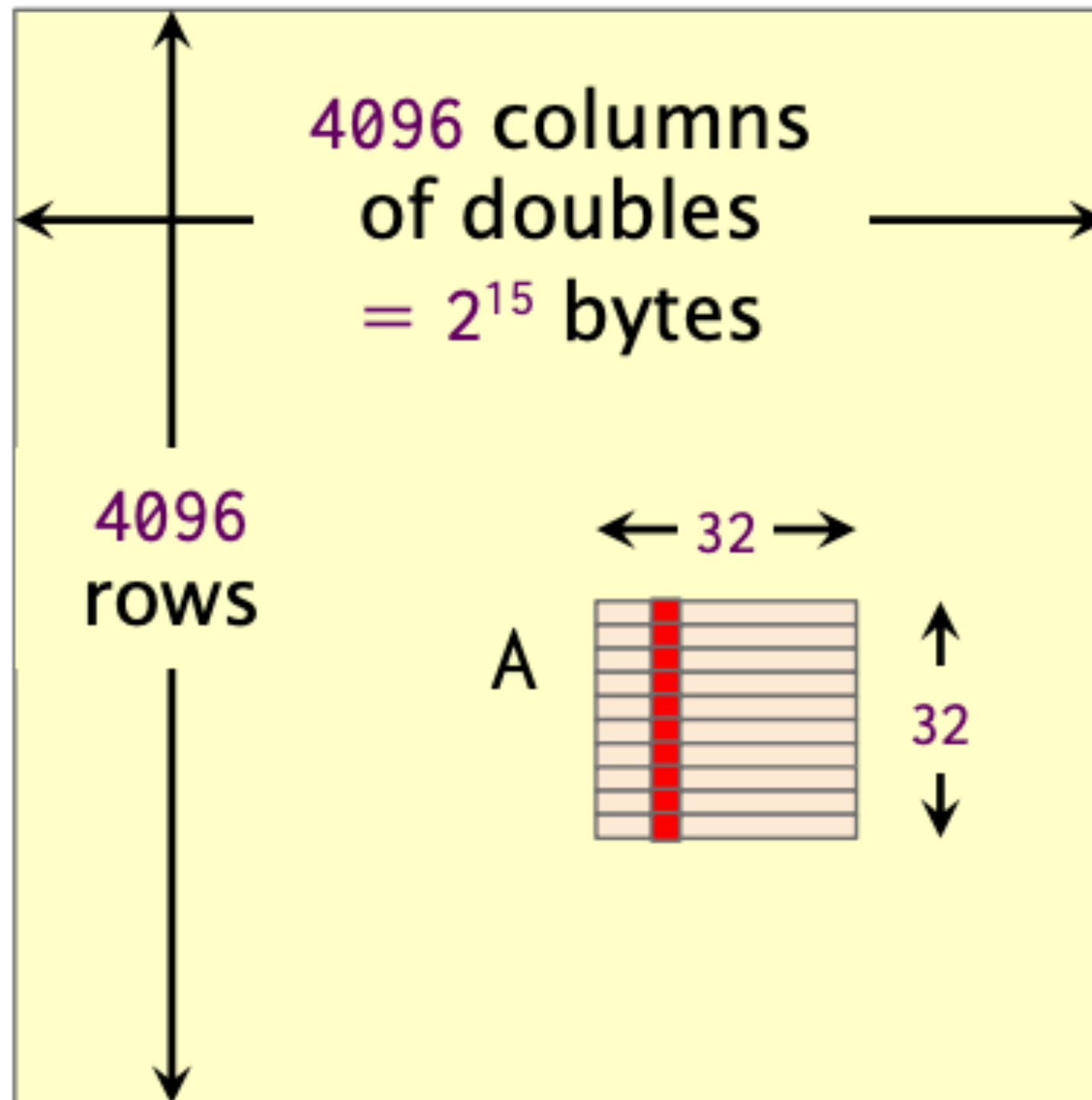
True Sharing



False Sharing



Conflict misses for submatrices



Assume:

- Word width $w = 64$.
- Cache size $\mathcal{M} = 32\text{K}$.
- Line (block) size $\mathcal{B} = 64$.
- $k=4$ -way associativity.

Conflict misses can be problematic for caches with limited associativity.

address		
tag	set	offset
$w - \lg(\mathcal{M}/k)$	$\lg(\mathcal{M}/k\mathcal{B})$	$\lg \mathcal{B}$
51	7	6

bits

Analysis

Look at a column of submatrix A. The addresses of the elements are $x, x+2^{15}, x+2 \cdot 2^{15}, \dots, x+31 \cdot 2^{15}$.

They all fall into the same set!

Solutions

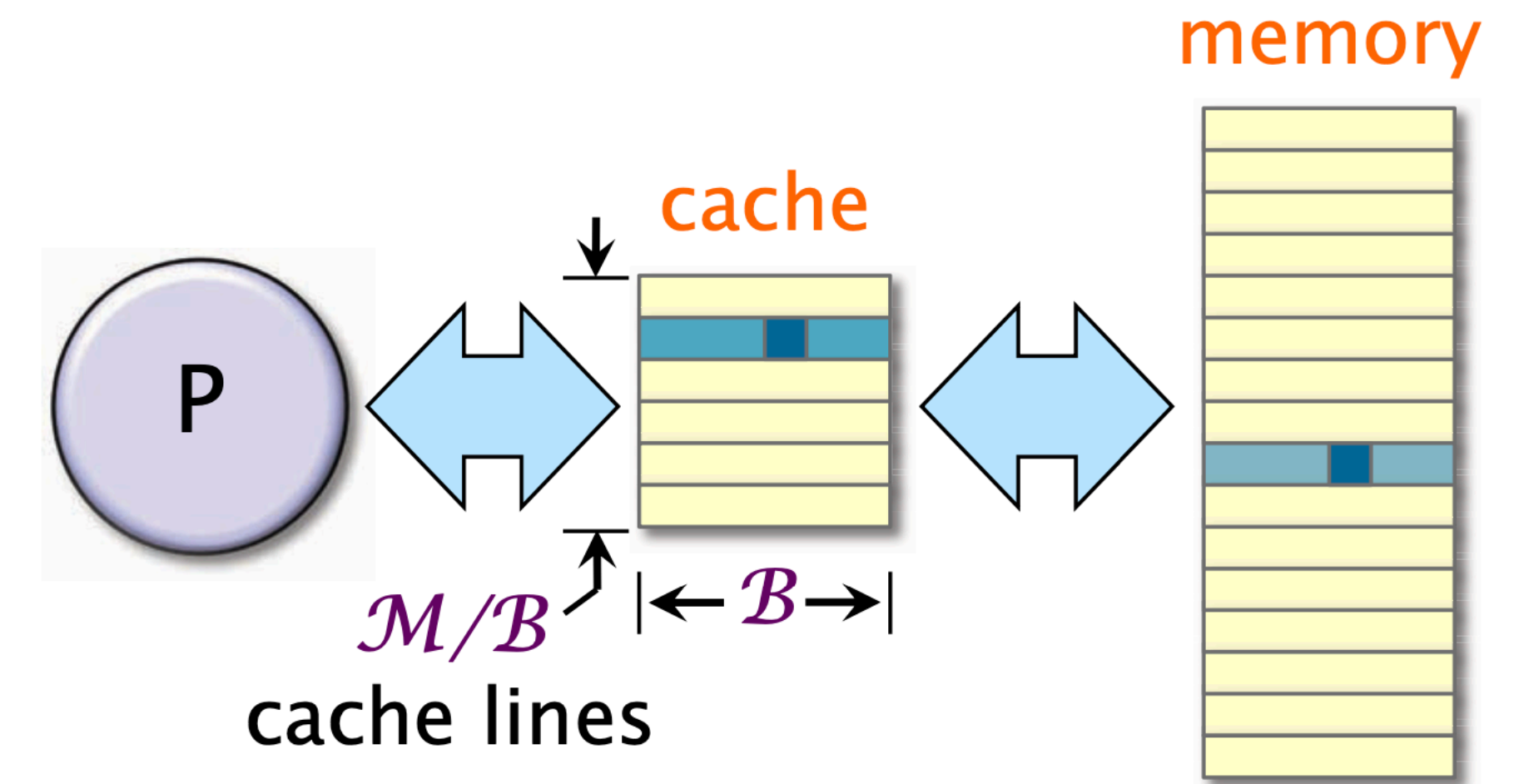
Copy A into a temporary 32×32 matrix, or pad rows.

Ideal-Cache Model

Ideal-Cache Model

Parameters

- Two-level hierarchy
- Cache size of $\mathcal{M}$ bytes
- Cache-line length of $\mathcal{B}$ bytes
- Fully associative
- Optimal, omniscient replacement.



Performance Measures

- **Work** W (ordinary running time)
- **Cache misses** Q (number of cache lines that need to be transferred between cache and memory)

How reasonable are ideal caches?

“LRU” Lemma [ST85]. Suppose that an algorithm incurs Q cache misses on an ideal cache of size $\mathcal{M}$. Then on a fully associative cache of size $2\mathcal{M}$ that uses the **least-recently used (LRU)** replacement policy, it incurs at most $2Q$ cache misses.

Implication

For asymptotic analyses, one can assume optimal or LRU replacement, as convenient.

Software engineering

- Design a **theoretically good** algorithm.
- **Engineer** for detailed performance.
- Real caches are not fully associative.
- Loads and stores have different costs with respect to bandwidth and latency.

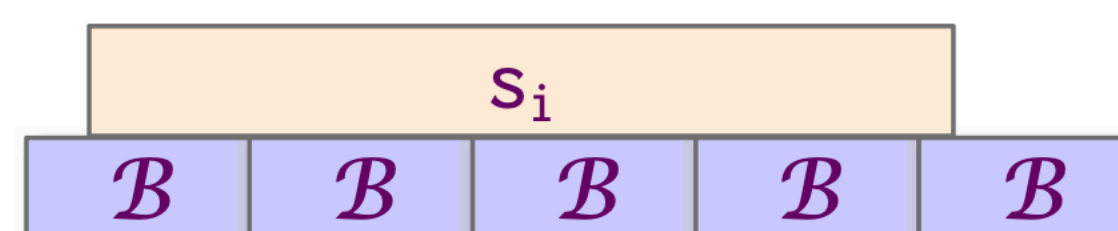
Cache-miss lemma

Lemma. Suppose that a program reads a set of r data segments, where the i th segment consists of s_i bytes, and suppose that

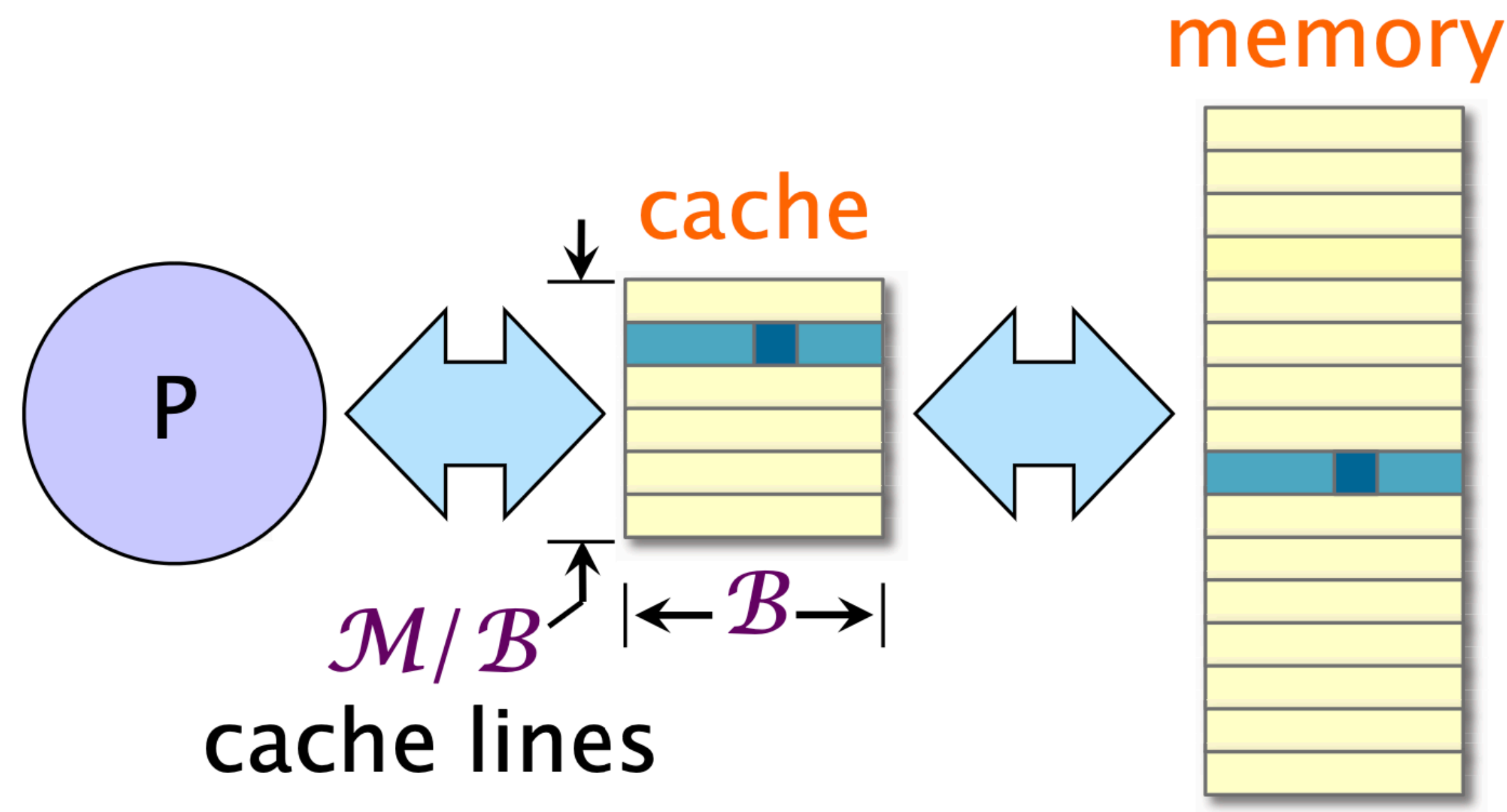
$$\sum_{i=1}^r s_i = N < \mathcal{M}/3 \text{ and } N/r \geq \mathcal{B}.$$

Then all of the segments fit into cache, and the number of misses to read them all is at most $3N/\mathcal{B}$.

Proof. Suppose that a program reads a set of r data segments, where the i th segment consists of A single segment s_i incurs at most $s_i/\mathcal{B}$ misses, and so

$$\begin{aligned} \sum_{i=1}^r (s_i/\mathcal{B} + 2) &= N/\mathcal{B} + 2r \\ &= (N/\mathcal{B} + 2r\mathcal{B})/\mathcal{B} \\ &\leq N/\mathcal{B} + 2N/\mathcal{B} \\ &= 3N/\mathcal{B}. \quad \blacksquare \end{aligned}$$


Tall caches



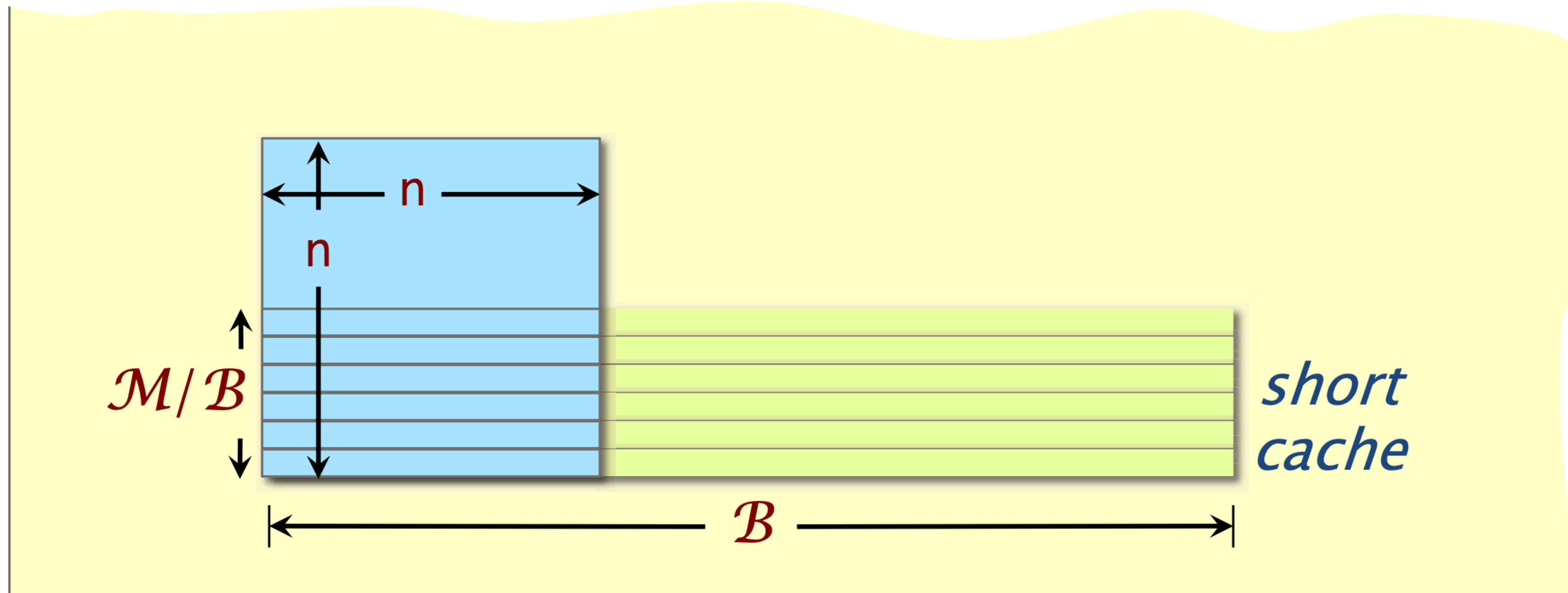
Tall-cache assumption

$\mathcal{B}^2 < c \mathcal{M}$ for some sufficiently small constant $c \leq 1$.

Example: Intel Xeon E5-2666 v3

- Cache-line length = 64 bytes.
- L1-cache size = 32 Kbytes.

What's wrong with short caches?

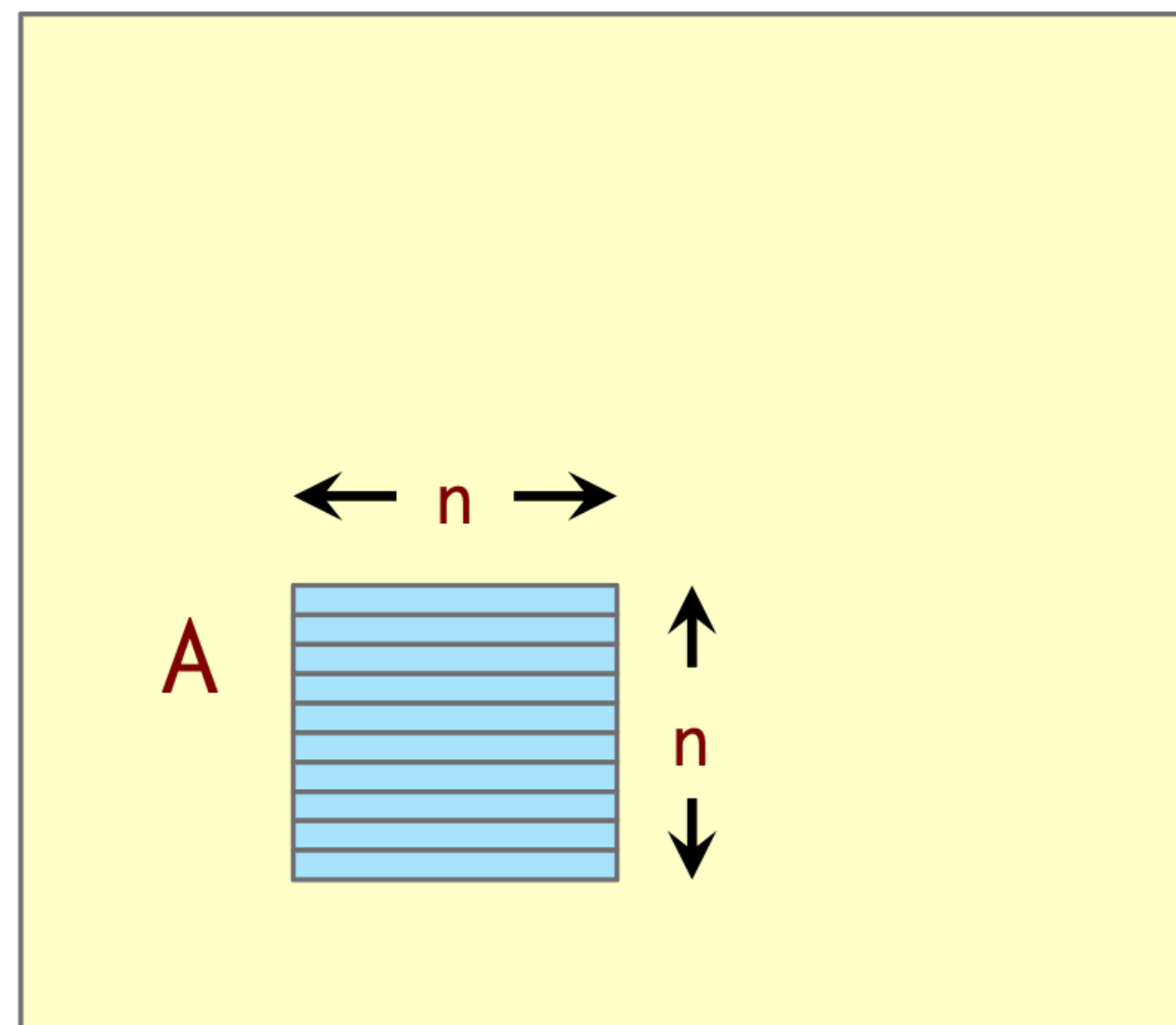


Tall-cache assumption

$\mathcal{B}^2 < c\mathcal{M}$ for some sufficiently small constant $c \leq 1$.

An $n \times n$ submatrix stored in row-major order may not fit in a short cache even if $n^2 < c\mathcal{M}$!

Submatrix caching lemma



Lemma. Suppose that an $n \times n$ submatrix A is read into a tall cache satisfying $\mathcal{B}^2 < c\mathcal{M}$, where $c \leq 1$ is constant, and suppose that $c\mathcal{M} \leq n^2 < \mathcal{M}/3$. Then A fits into cache, and the number of misses to read all A 's elements is at most $3n^2/\mathcal{B}$.

Proof. We have $N = n^2$, $n = r = s_i$, $\mathcal{B} \leq n = N/r$, and $N < \mathcal{M}/3$. Thus, the Cache-Miss Lemma applies. ■

PART 1 ENDED HERE

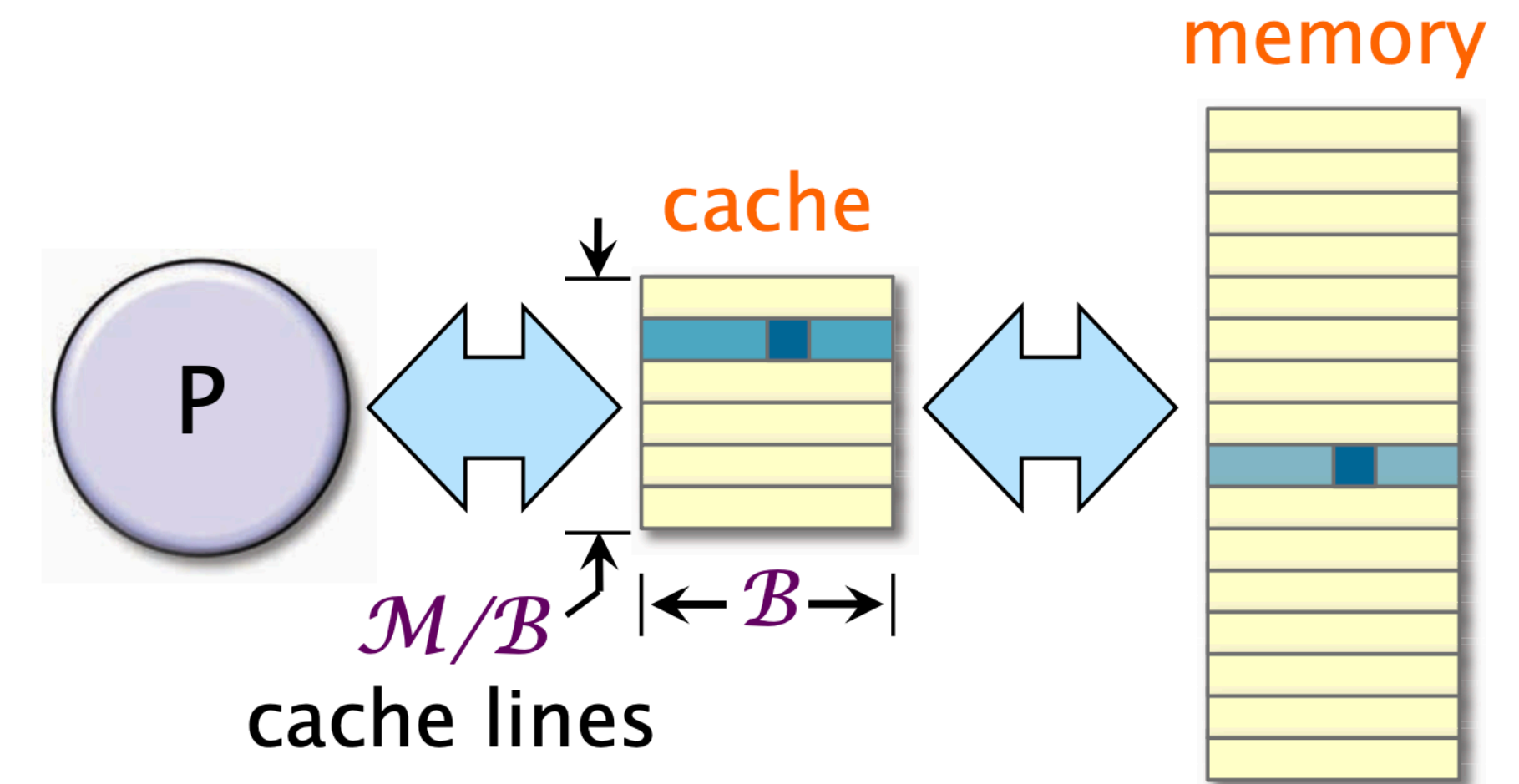
Announcements

- PA2 signup out later today, due this Friday 1/31 by end of day
- PA1 due this Fri, Jan 31 - late deadline +2 days to Sunday 2/2

Recall: Ideal-Cache Model

Parameters

- Two-level hierarchy
- Cache size of $\mathcal{M}$ bytes
- Cache-line length of $\mathcal{B}$ bytes
- Fully associative
- Optimal, omniscient replacement.



Performance Measures

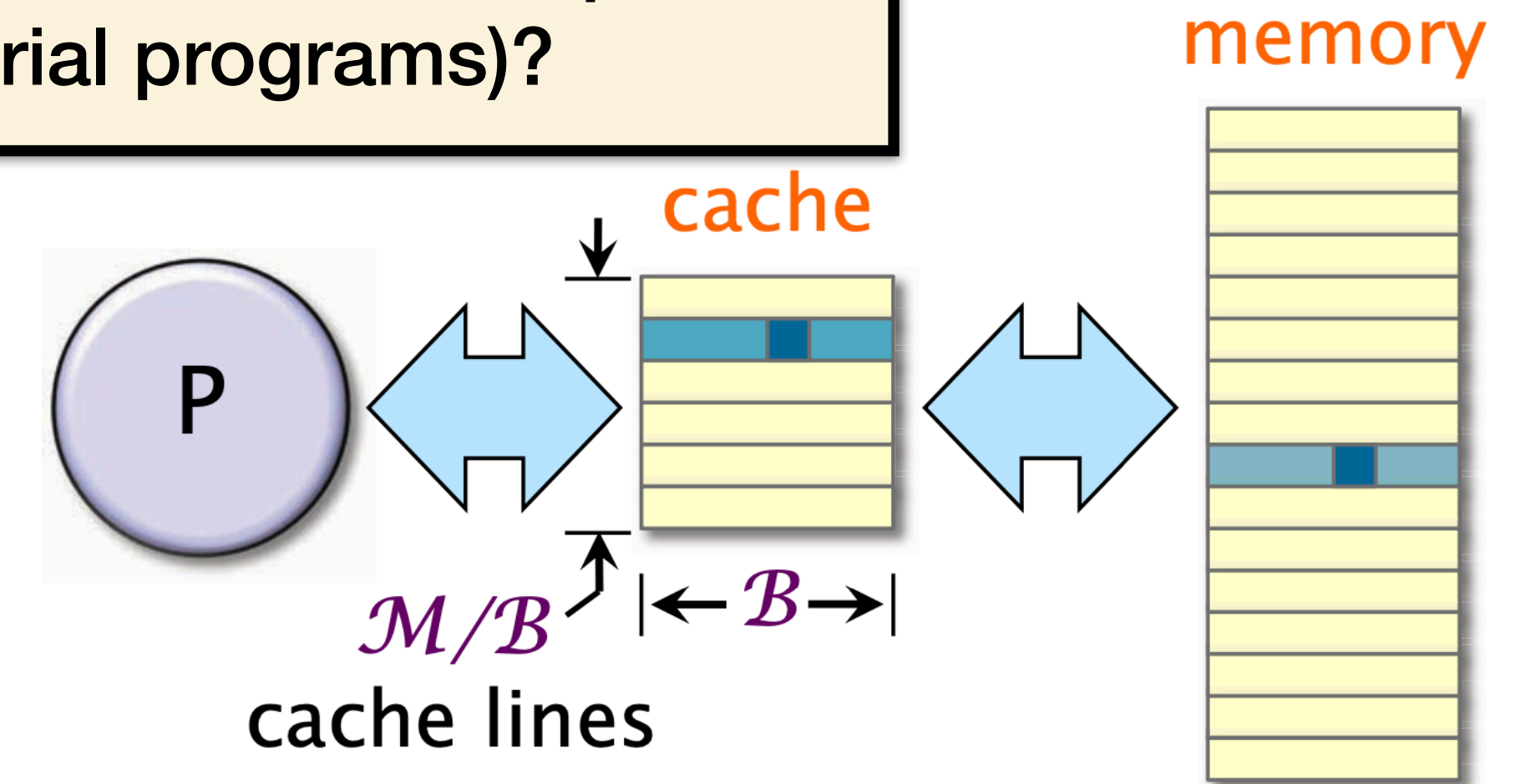
- **Work** W (ordinary running time)
- **Cache misses** Q (number of cache lines that need to be transferred between cache and memory)

Recall: Ideal-Cache Model

Quiz: What type of cache misses are not captured by this model (for serial programs)?

Parameters

- Two-level hierarchy
- Cache size of $\mathcal{M}$ bytes
- Cache-line length of $\mathcal{B}$ bytes
- Fully associative
- Optimal, omniscient replacement.



Performance Measures

- **Work** W (ordinary running time)
- **Cache misses** Q (number of cache lines that need to be transferred between cache and memory)

Cache Analysis of Matrix Multiplication

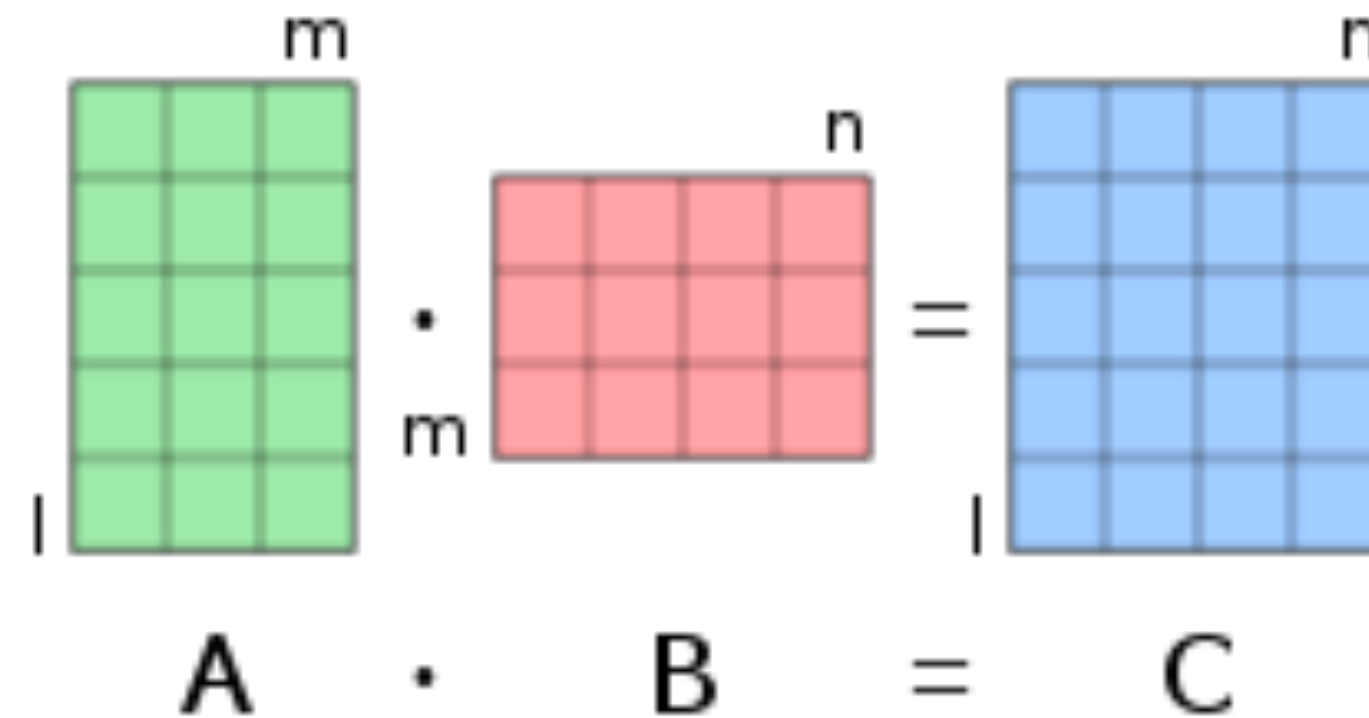
Why matrix multiplication?

Matrix multiplication is an **important kernel** in many problems:

- Dense linear algebra is a motif in every list,
- Closely related to other algorithms, e.g., transitive closure on a graph,
- And dominates training time in deep learning (CNNs)

Good model problem (well-studied, illustrates ideas).

Easy to find good libraries that are hard to beat! (e.g., Intel MKL, etc.)



Multiply square matrices

```
void Mult(double *C, double *A, double *B, int64_t n) {  
    for (int64_t i=0; i < n; i++)  
        for (int64_t j=0; j < n; j++)  
            for (int64_t k=0; k < n; k++)  
                C[i*n+j] += A[i*n+k] * B[k*n+j];  
}
```

Analysis of work:

$$W(n) = \Theta(n^3)$$

Memory layout of matrices

In this matrix-multiplication code, matrices are laid out in memory in **row-major order**.

Matrix

Row 1
Row 2
Row 3
Row 4
Row 5
Row 6
Row 7
Row 8

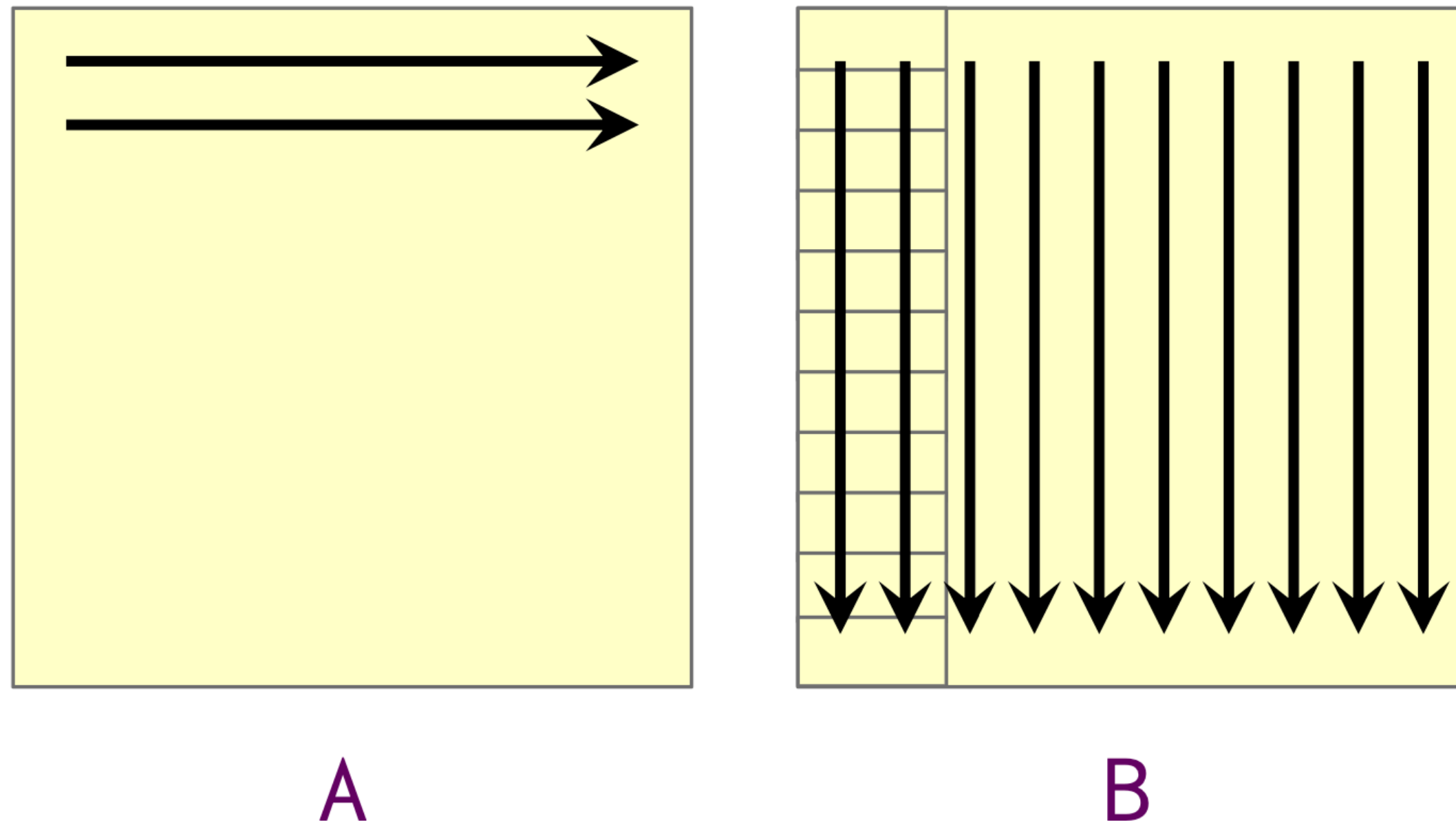
Memory

Row 1	Row 2	Row 3
-------	-------	-------

Analysis of cache misses

```
void Mult(double *C, double *A, double *B, int64_t n) {  
    for (int64_t i=0; i < n; i++)  
        for (int64_t j=0; j < n; j++)  
            for (int64_t k=0; k < n; k++)  
                C[i*n+j] += A[i*n+k] * B[k*n+j];  
}
```

Assume row major and tall cache



Case 1

$n > c\mathcal{M}/\mathcal{B}$. Analyze matrix **B**.

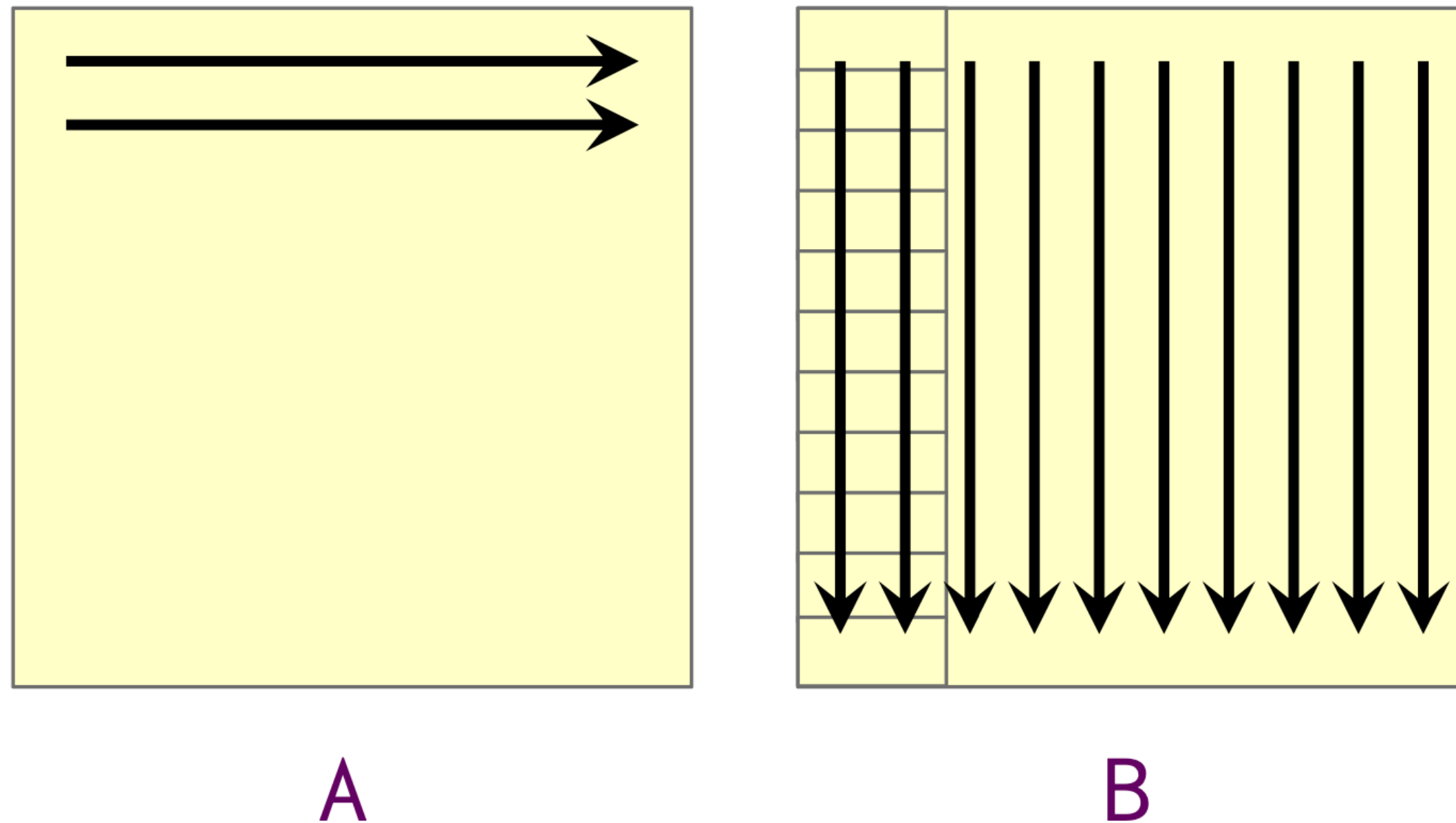
Assume LRU.

$Q(n) = \Theta(n^3)$, since matrix **B** misses on every access.

Analysis of cache misses

```
void Mult(double *C, double *A, double *B, int64_t n) {  
    for (int64_t i=0; i < n; i++)  
        for (int64_t j=0; j < n; j++)  
            for (int64_t k=0; k < n; k++)  
                C[i*n+j] += A[i*n+k] * B[k*n+j];  
}
```

Assume row major and tall cache



Case 2

$c' \mathcal{M}^{1/2} < n < c \mathcal{M} / \mathcal{B}$. Analyze matrix **B**.

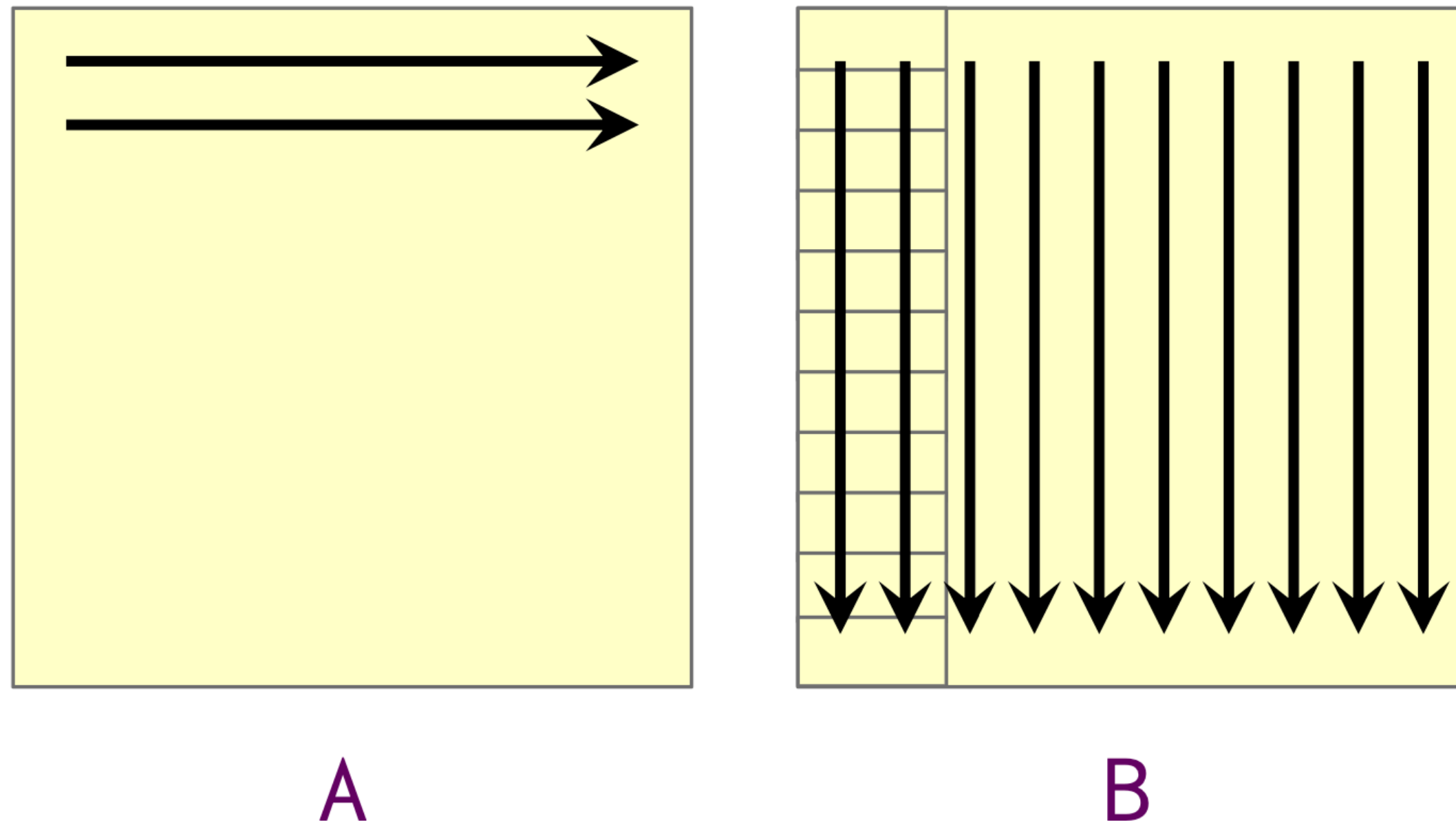
Assume LRU.

$Q(n) = n \cdot \Theta(n^2 / \mathcal{B}) = \Theta(n^3 / \mathcal{B})$,
since matrix **B** can exploit spatial locality.

Analysis of cache misses

```
void Mult(double *C, double *A, double *B, int64_t n) {  
    for (int64_t i=0; i < n; i++)  
        for (int64_t j=0; j < n; j++)  
            for (int64_t k=0; k < n; k++)  
                C[i*n+j] += A[i*n+k] * B[k*n+j];  
}
```

Assume row major and tall cache



Case 3

$n < c' \mathcal{M}^{1/2}$. Analyze matrix **B**.

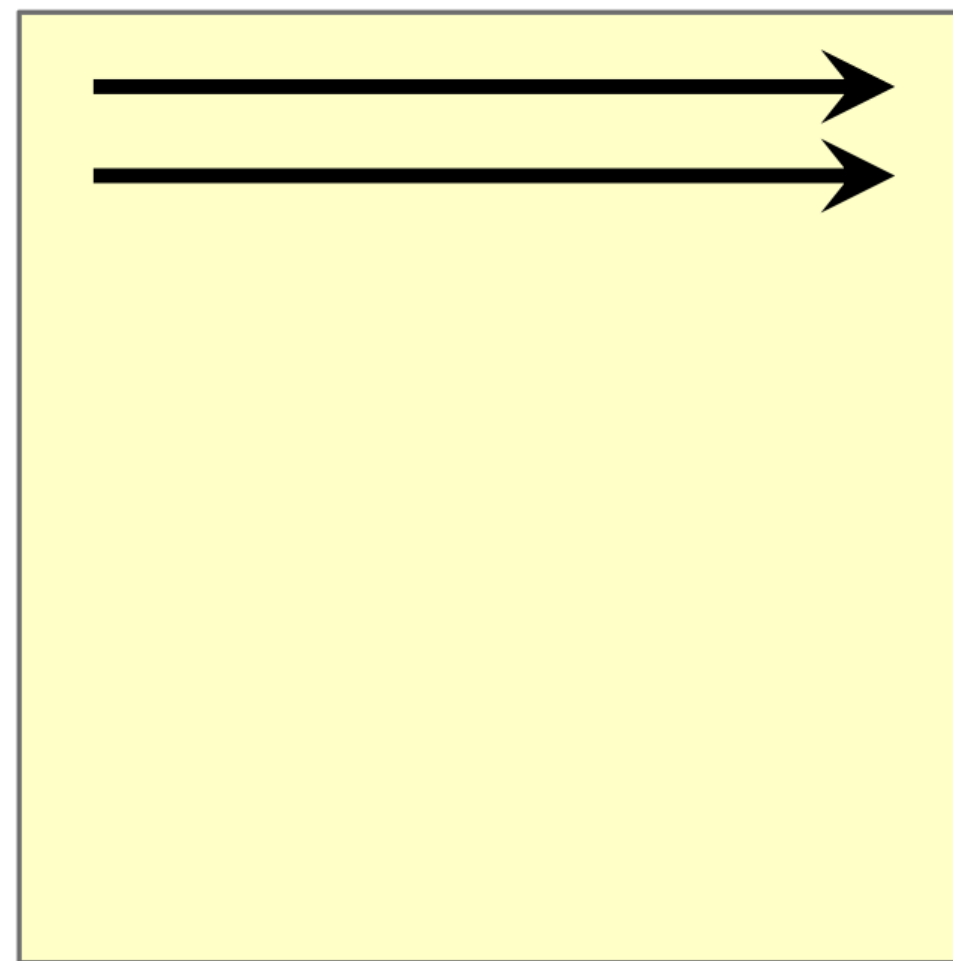
Assume LRU.

$Q(n) = \Theta(n^2 / \mathcal{B})$, since everything fits in cache!

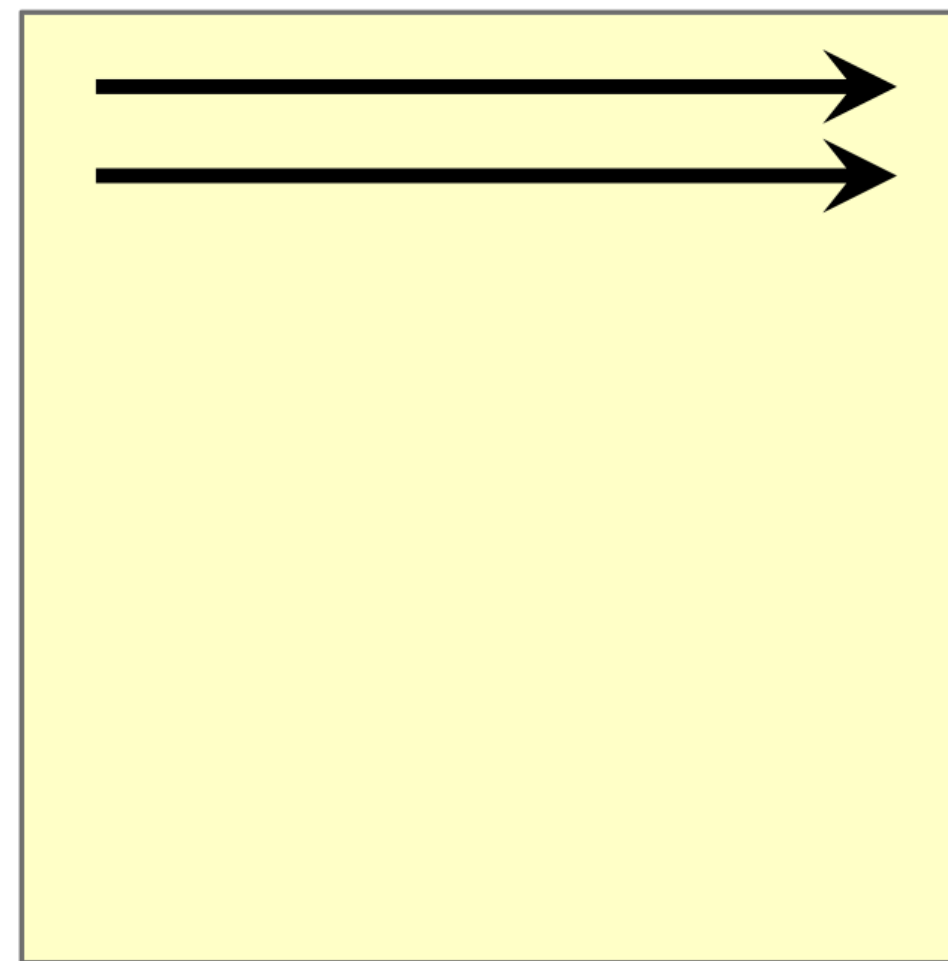
Swapping inner loop order

```
void Mult(double *C, double *A, double *B, int64_t n) {  
    for (int64_t i=0; i < n; I++)  
        for (int64_t k=0; k < n; k++)  
            for (int64_t j=0; j < n; j++)  
                C[i*n+j] += A[i*n+k] * B[k*n+j];  
}
```

Assume row major and tall cache



C



B

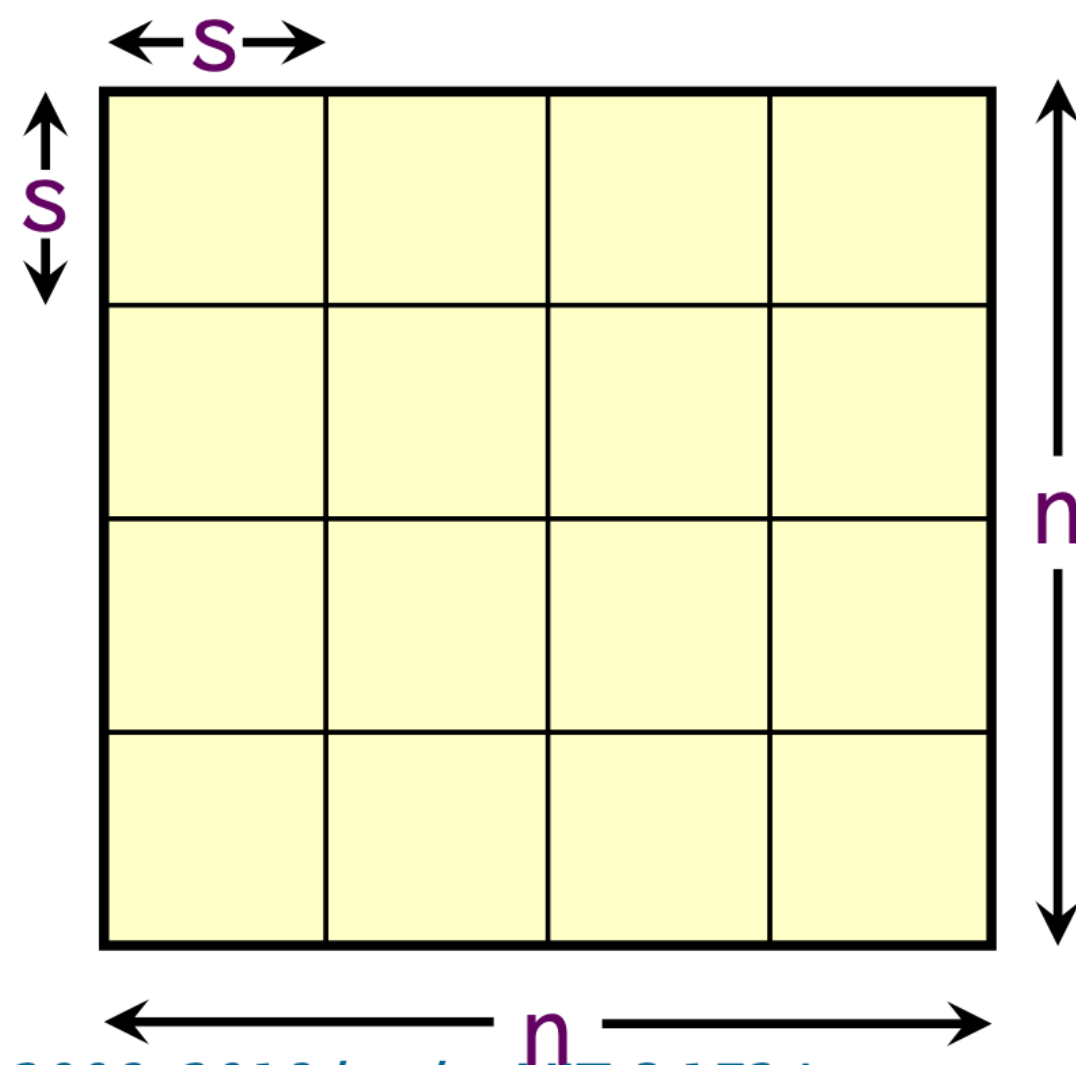
Assume matrix B. Assume LRU.

$Q(n) = n \cdot \Theta(n^2/\mathcal{B}) = \Theta(n^3/\mathcal{B})$,
since matrix **B** can exploit spatial
locality.

Tiling (aka Blocking)

Tiled (Blocked) matrix multiply

```
void Tiled_Mult(double *C, double *A, double *B, int64_t n) {  
    for (int64_t i1=0; i1<n; i1+=s)  
        for (int64_t j1=0; j1<n; j1+=s)  
            for (int64_t k1=0; k1<n; k1+=s)  
                for (int64_t i=i1; i<i1+s && i<n; i++)  
                    for (int64_t j=j1; j<j1+s && j<n; j++)  
                        for (int64_t k=k1; k<k1+s && k<n; k++)  
                            C[i*n+j] += A[i*n+k] * B[k*n+j];  
}
```



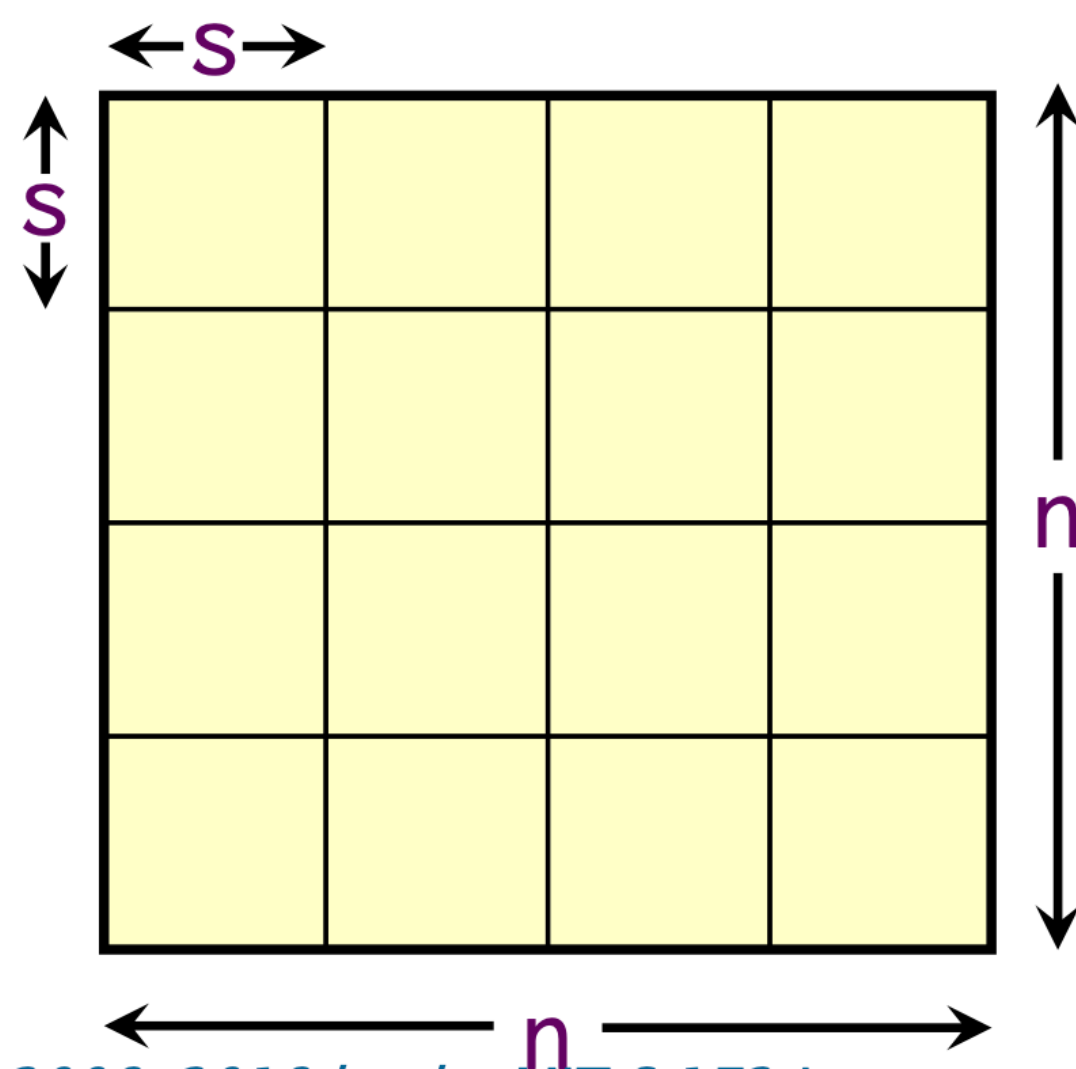
Analysis of work:

$$\begin{aligned} W(n) &= \Theta((n/s)^3 (s)^3) \\ &= \Theta(n^3) \end{aligned}$$

Tile size (or
block size)

Tiled (Blocked) matrix multiply

```
void Tiled_Mult(double *C, double *A, double *B, int64_t n) {  
    for (int64_t i1=0; i1<n; i1+=s)  
        for (int64_t j1=0; j1<n; j1+=s)  
            for (int64_t k1=0; k1<n; k1+=s)  
                for (int64_t i=i1; i<i1+s && i<n; i++)  
                    for (int64_t j=j1; j<j1+s && j<n; j++)  
                        for (int64_t k=k1; k<k1+s && k<n; k++)  
                            C[i*n+j] += A[i*n+k] * B[k*n+j];  
}
```



Analysis of cache misses

Tune s so that the submatrices just fit into cache:

$$s = \Theta(\mathcal{M}^{1/2}).$$

Submatrix Caching Lemma implies $\Theta(s^2/\mathcal{B})$ misses per submatrix.

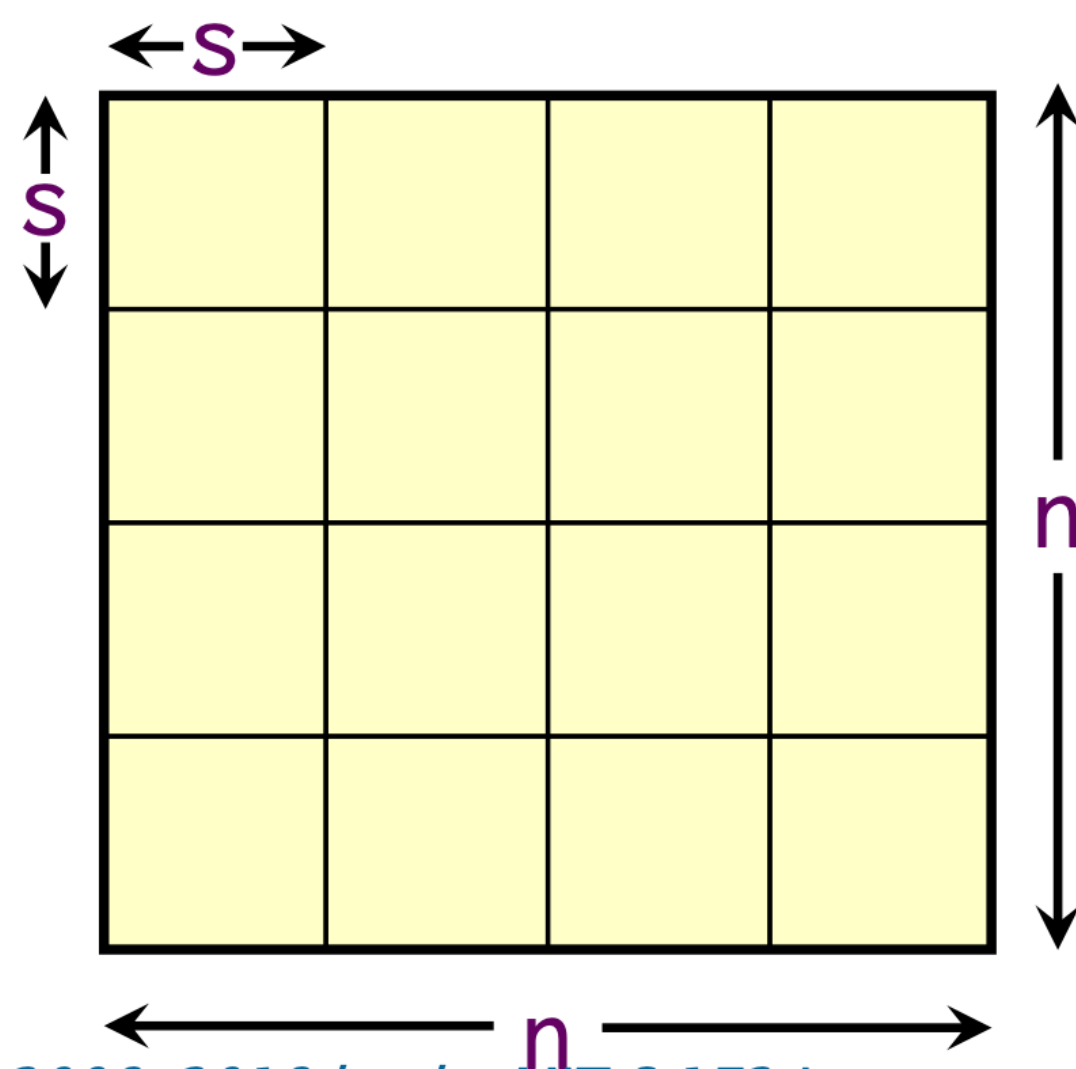
$$Q(n) = \Theta((n/s)^3(s^2/\mathcal{B})) = \Theta(n^3/(\mathcal{B}\mathcal{M}^{1/2})).$$

Optimal
[HK81]

Tiled (Blocked) matrix multiply

```
void Tiled_Mult(double *C, double *A, double *B, int64_t n) {  
    for (int64_t i1=0; i1<n; i1+=s)  
        for (int64_t j1=0; j1<n; j1+=s)  
            for (int64_t k1=0; k1<n; k1+=s)  
                for (int64_t i=i1; i<n; i++)  
                    for (int64_t j=j1; j<n; j++)  
                        for (int64_t k=k1; k<n; k++)  
                            C[i*n+j] += A[i*n+k] * B[k*n+j];  
}
```

How?



Analysis of cache misses

Tune s so that the submatrices just fit into cache:

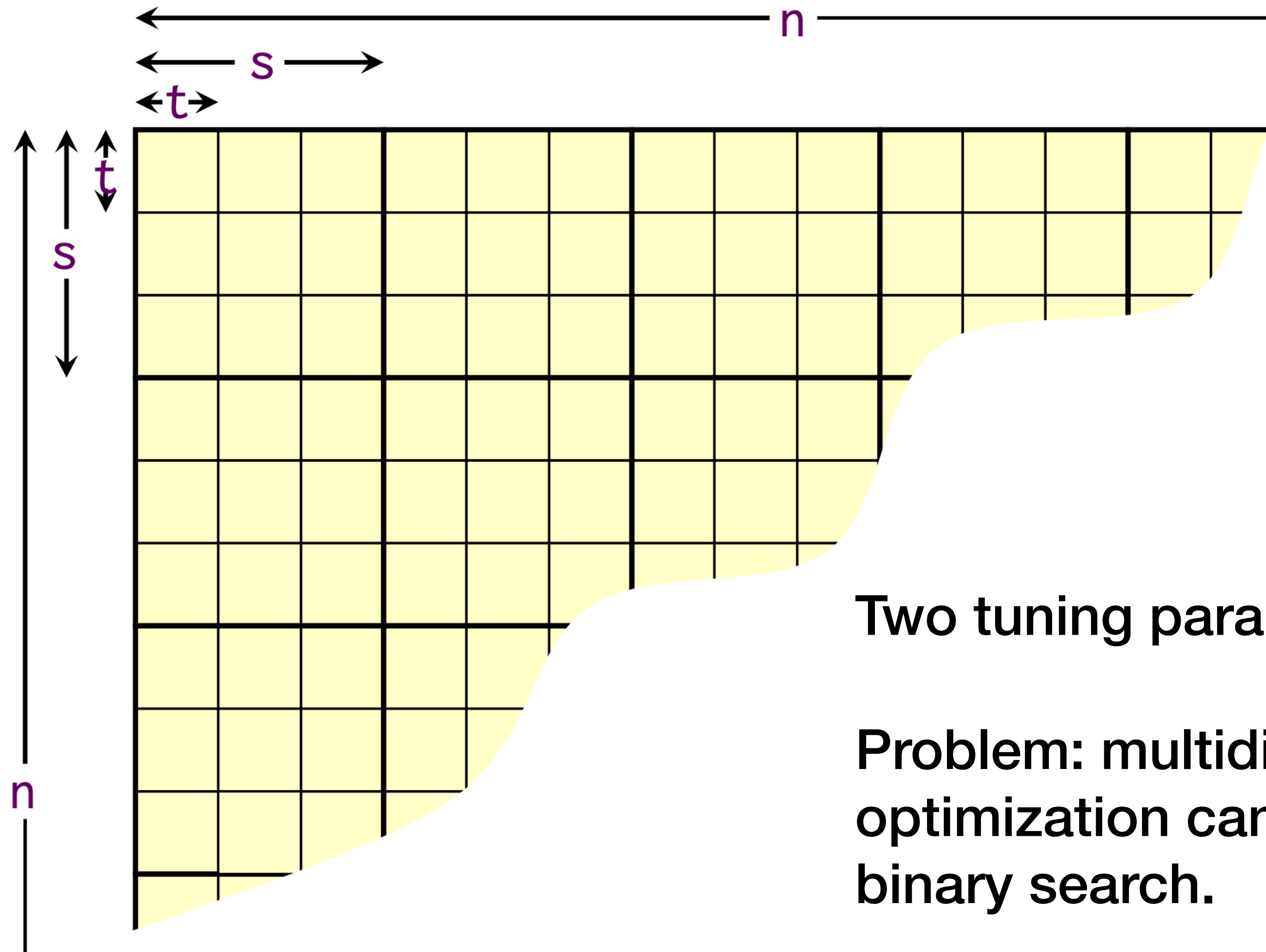
$$s = \Theta(\mathcal{M}^{1/2}).$$

Submatrix Caching Lemma implies $\Theta(s^2/\mathcal{B})$ misses per submatrix.

$$Q(n) = \Theta((n/s)^3(s^2/\mathcal{B})) = \Theta(n^3/(\mathcal{B}\mathcal{M}^{1/2})).$$

Optimal
[HK81]

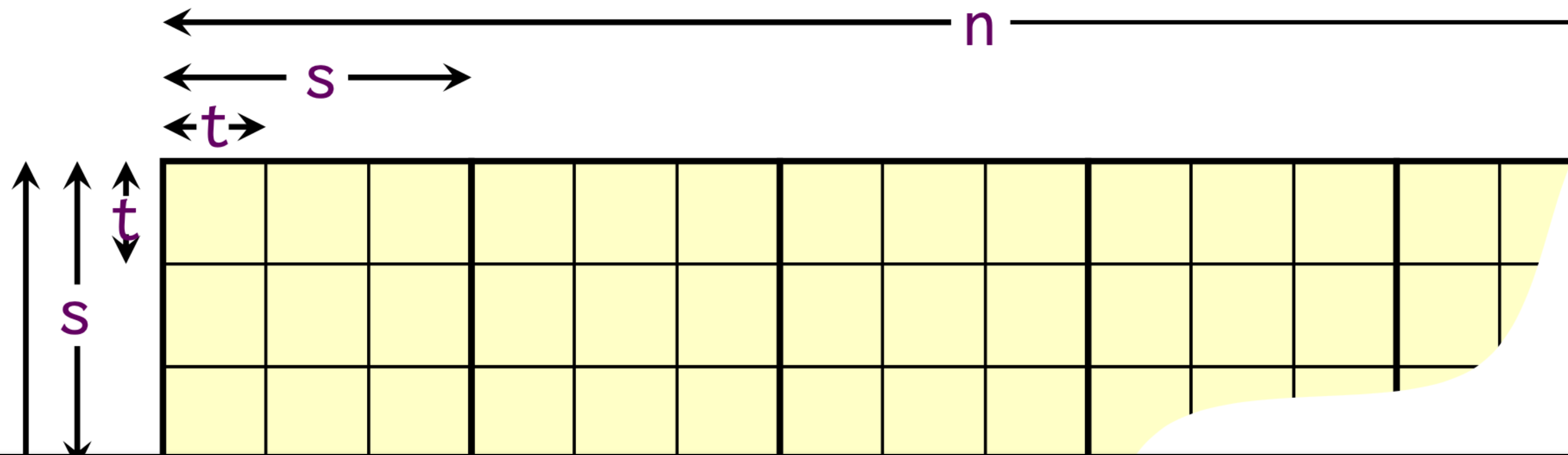
Two-level cache



Two tuning parameters: s and t .

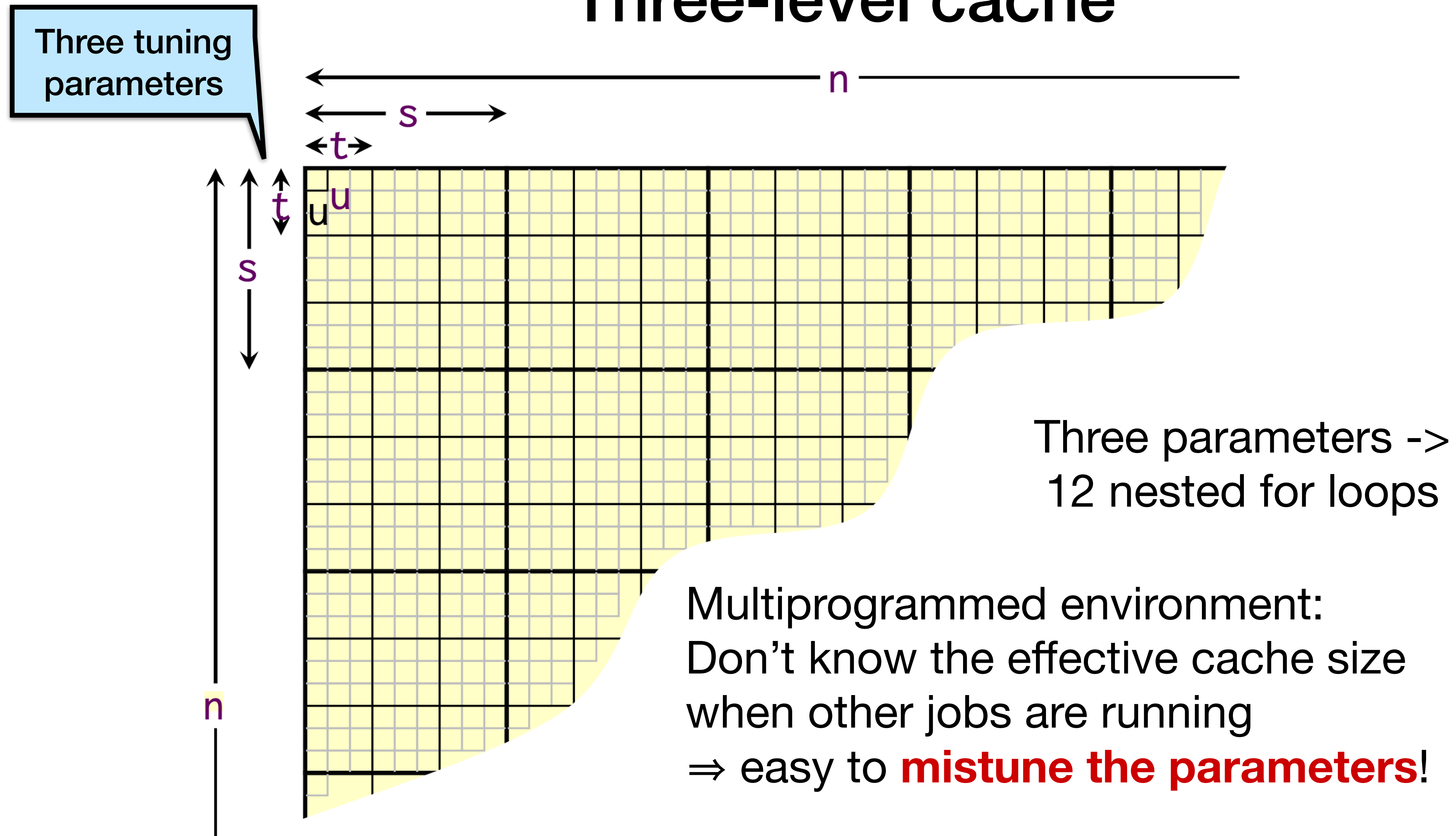
Problem: multidimensional tuning optimization cannot be done with binary search.

Two-level cache



```
void Tiled_Mult2(double *C, double *A, double *B, int64_t n) {  
    for (int64_t i2=0; i2<n; i2+=s)  
        for (int64_t j2=0; j2<n; j2+=s)  
            for (int64_t k2=0; k2<n; k2+=s)  
                for (int64_t i1=i2; i1<i2+s && i1<n; i1+=t)  
                    for (int64_t j1=j2; j1<j2+s && j1<n; j1+=t)  
                        for (int64_t k1=k2; k1<k2+s && k1<n; k1+=t)  
                            for (int64_t i=i1; i<i1+s && i<i2+t && i<n; i++)  
                                for (int64_t j=j1; j<j1+s && j<j2+t && j<n; j++)  
                                    for (int64_t k=k1; k<k1+s && k<k2+t && k<n; k++)  
                                        C[i*n+j] += A[i*n+k] * B[k*n+j];  
}
```


Three-level cache



Divide and Conquer

Recursive matrix multiplication

Divide-and-conquer on $n \times n$ matrices:

$$\begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \times \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$$
$$= \begin{bmatrix} A_{11}B_{11} & A_{11}B_{12} \\ A_{21}B_{11} & A_{21}B_{12} \end{bmatrix} + \begin{bmatrix} A_{12}B_{21} & A_{12}B_{22} \\ A_{22}B_{21} & A_{22}B_{22} \end{bmatrix}$$

8 multiply-adds of $(n/2) \times (n/2)$ matrices.

Recursive code

```
// Assume that n is an exact power of 2.
void Rec_Mult(double *C, double *A, double *B,
              int64_t n, int64_t rowsize) {
    if (n == 1)
        C[0] += A[0] * B[0];
    else {
        int64_t d11 = 0;
        int64_t d12 = n/2;
        int64_t d21 = (n/2) * rowsize;
        int64_t d22 = (n/2) * (rowsize+1);

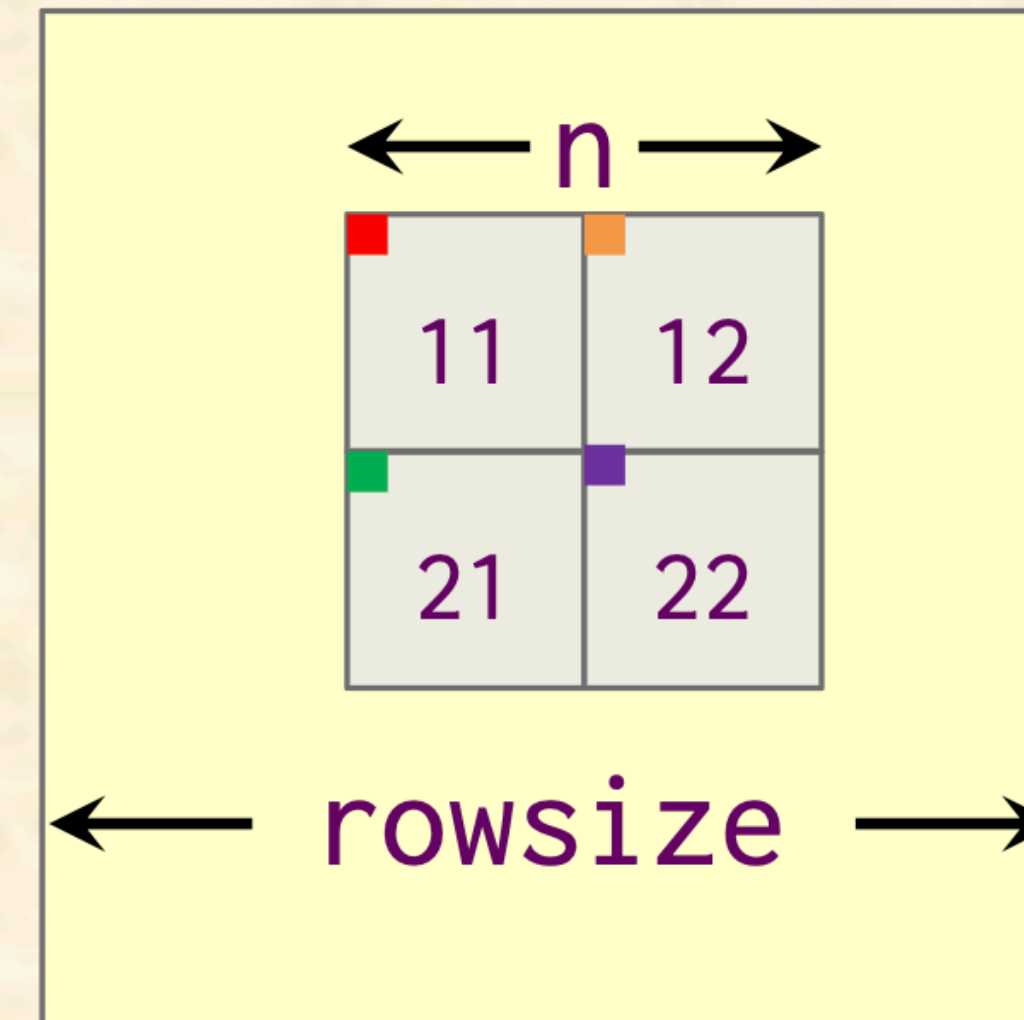
        Rec_Mult(C+d11, A+d11, B+d11, n/2, rowsize);
        Rec_Mult(C+d11, A+d12, B+d21, n/2, rowsize);
        Rec_Mult(C+d12, A+d11, B+d12, n/2, rowsize);
        Rec_Mult(C+d12, A+d12, B+d22, n/2, rowsize);
        Rec_Mult(C+d21, A+d21, B+d11, n/2, rowsize);
        Rec_Mult(C+d21, A+d22, B+d21, n/2, rowsize);
        Rec_Mult(C+d22, A+d21, B+d12, n/2, rowsize);
        Rec_Mult(C+d22, A+d22, B+d22, n/2, rowsize);
    }
}
```

Coarsen base case to overcome function-call overheads.

Recursive code

```
// Assume that n is an exact power of 2.
void Rec_Mult(double *C, double *A, double *B,
              int64_t n, int64_t rowsize) {
    if (n == 1)
        C[0] += A[0] * B[0];
    else {
        int64_t d11 = 0;
        int64_t d12 = n/2;
        int64_t d21 = (n/2) * rowsize;
        int64_t d22 = (n/2) * (rowsize+1);

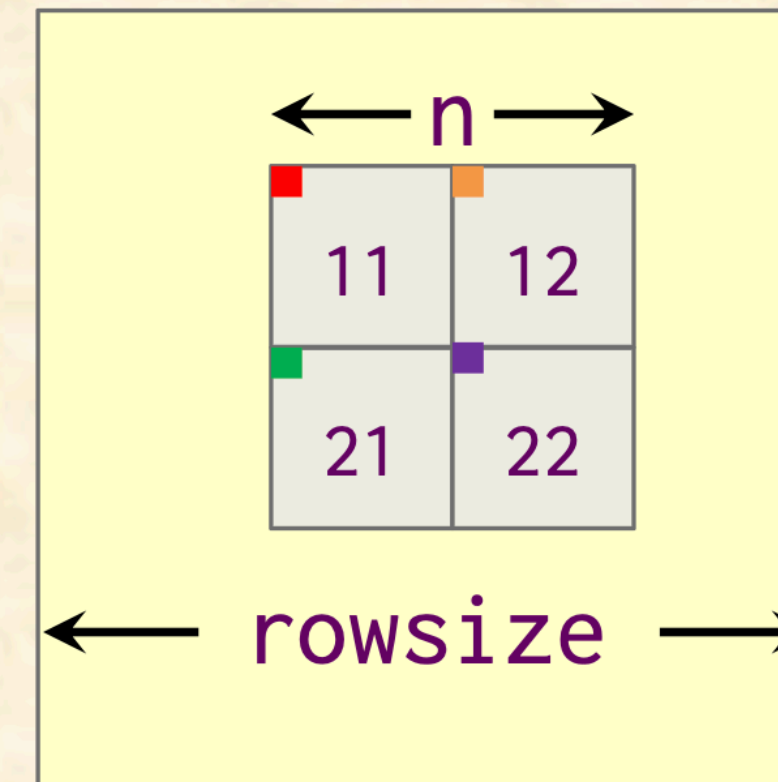
        Rec_Mult(C+d11, A+d11, B+d11, n/2, rowsize);
        Rec_Mult(C+d11, A+d12, B+d21, n/2, rowsize);
        Rec_Mult(C+d12, A+d11, B+d12, n/2, rowsize);
        Rec_Mult(C+d12, A+d12, B+d22, n/2, rowsize);
        Rec_Mult(C+d21, A+d21, B+d11, n/2, rowsize);
        Rec_Mult(C+d21, A+d22, B+d21, n/2, rowsize);
        Rec_Mult(C+d22, A+d21, B+d12, n/2, rowsize);
        Rec_Mult(C+d22, A+d22, B+d22, n/2, rowsize);
    }
}
```



Analysis of work

```
// Assume that n is an exact power of 2.
void Rec_Mult(double *C, double *A, double *B,
              int64_t n, int64_t rowsize) {
    if (n == 1)
        C[0] += A[0] * B[0];
    else {
        int64_t d11 = 0;
        int64_t d12 = n/2;
        int64_t d21 = (n/2) * rowsize;
        int64_t d22 = (n/2) * (rowsize+1);

        Rec_Mult(C+d11, A+d11, B+d11, n/2, rowsize);
        Rec_Mult(C+d11, A+d12, B+d21, n/2, rowsize);
        Rec_Mult(C+d12, A+d11, B+d12, n/2, rowsize);
        Rec_Mult(C+d12, A+d12, B+d22, n/2, rowsize);
        Rec_Mult(C+d21, A+d21, B+d11, n/2, rowsize);
        Rec_Mult(C+d21, A+d22, B+d21, n/2, rowsize);
        Rec_Mult(C+d22, A+d21, B+d12, n/2, rowsize);
        Rec_Mult(C+d22, A+d22, B+d22, n/2, rowsize);
    }
}
```



$$\begin{aligned} W(n) &= 8W(n/2) + \Theta(1) \\ &= \Theta(n^3) \end{aligned}$$

Analysis of work

$$W(n) = 8W(n/2) + \Theta(1)$$

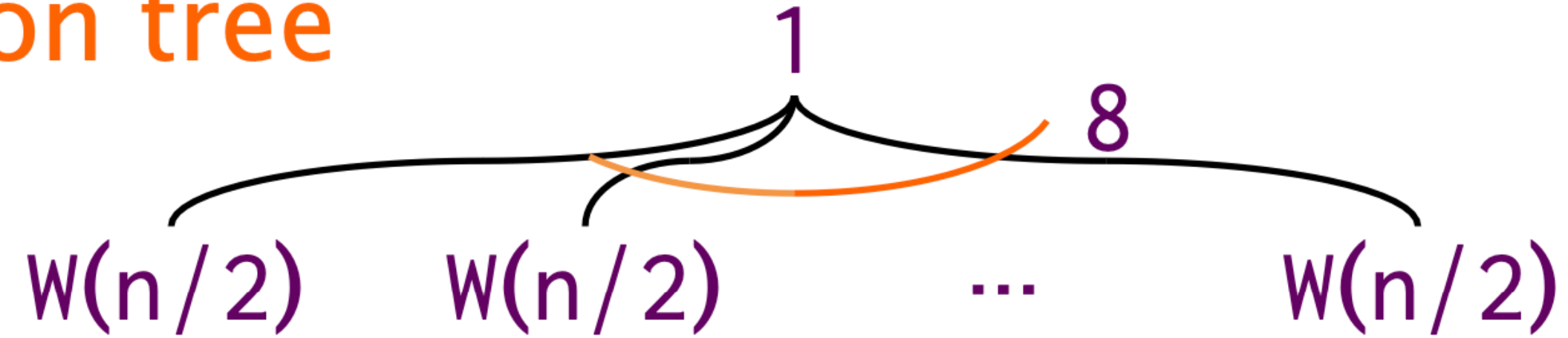
recursion tree

$W(n)$

Analysis of work

$$W(n) = 8W(n/2) + \Theta(1)$$

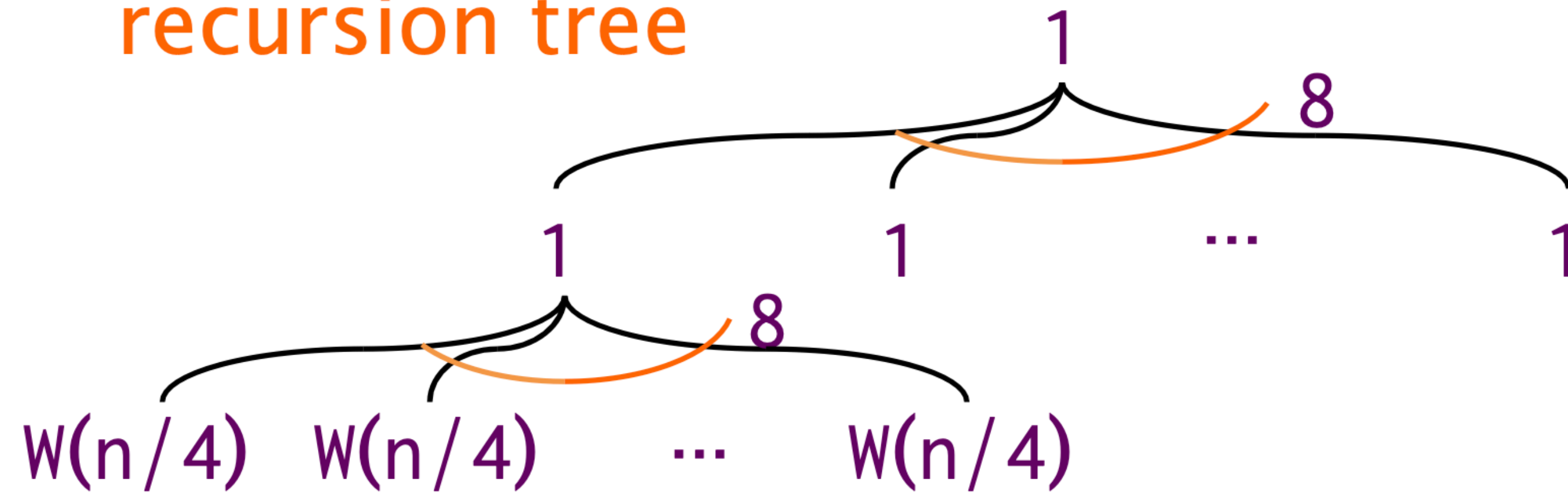
recursion tree



Analysis of work

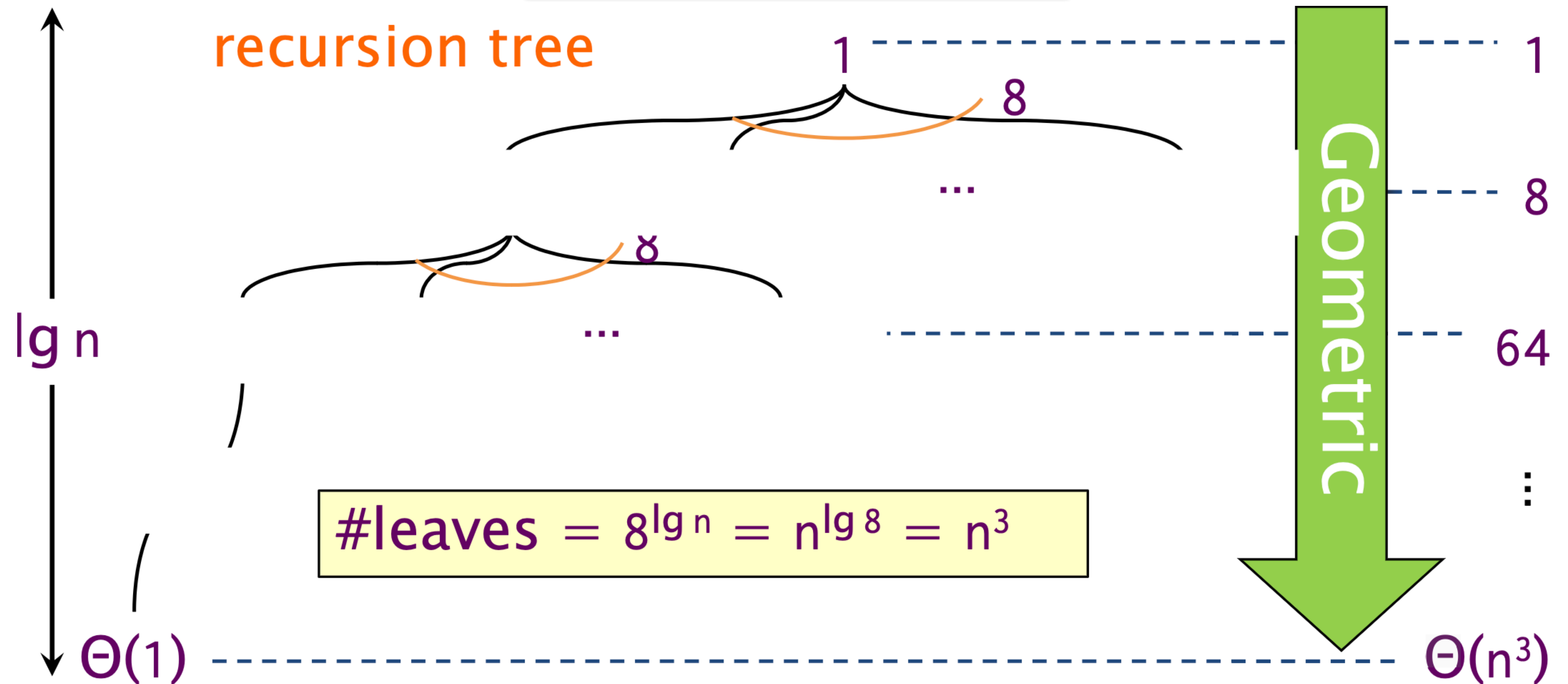
$$W(n) = 8W(n/2) + \Theta(1)$$

recursion tree



Analysis of work

$$W(n) = 8W(n/2) + \Theta(1)$$



Note: Same work as looping versions.

$$W(n) = \Theta(n^3)$$

Analysis of cache misses

```
// Assume that n is an exact power of 2.
void Rec_Mult(double *C, double *A, double *B,
              int64_t n, int64_t rowsize) {
    if (n == 1)
        C[0] += A[0] * B[0];
    else {
        int64_t d11 = 0;
        int64_t d12 = n/2;
        int64_t d21 = (n/2) * rowsize;
        int64_t d22 = (n/2) * (rowsize+1);

        Rec_Mult(C+d11, A+d11, B+d11, n/2, rowsize);
        Rec_Mult(C+d11, A+d12, B+d21, n/2, rowsize);
        Rec_Mult(C+d12, A+d11, B+d12, n/2, rowsize);
        Rec_Mult(C+d12, A+d12, B+d22, n/2, rowsize);
        Rec_Mult(C+d21, A+d21, B+d11, n/2, rowsize);
        Rec_Mult(C+d21, A+d22, B+d21, n/2, rowsize);
        Rec_Mult(C+d22, A+d21, B+d12, n/2, rowsize);
        Rec_Mult(C+d22, A+d22, B+d22, n/2, rowsize);
    }
}
```

Submatrix
Caching
Lemma

$$Q(n) = \begin{cases} \Theta(n^2/\mathcal{B}) & \text{if } n^2 < c\mathcal{M} \text{ for suff. small const } c \leq 1, \\ 8Q(n/2) + \Theta(1) & \text{otherwise.} \end{cases}$$

Analysis of cache misses

$$Q(n) = \begin{cases} \Theta(n^2/\mathcal{B}) & \text{if } n^2 < c\mathcal{M} \text{ for suff. small const } c \leq 1, \\ 8Q(n/2) + \Theta(1) & \text{otherwise.} \end{cases}$$

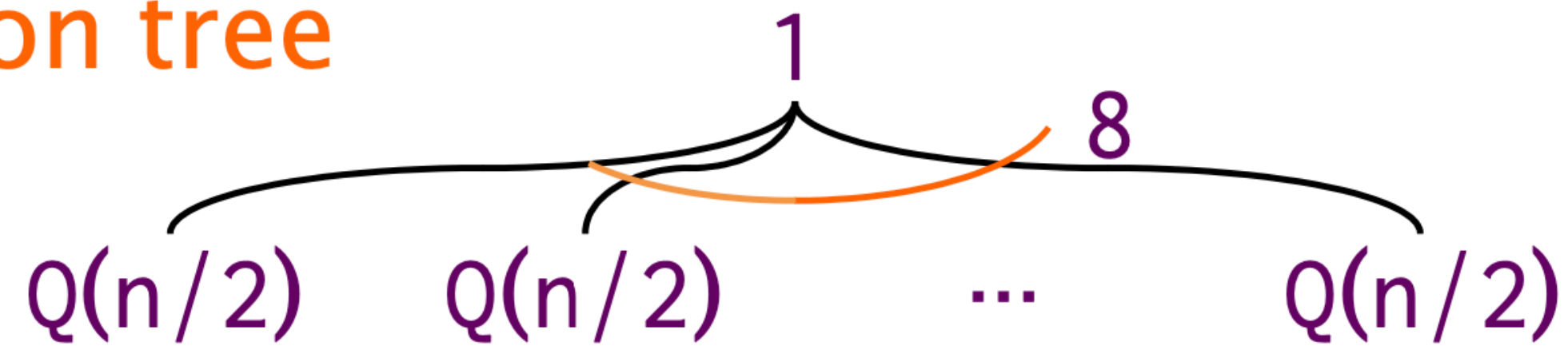
recursion tree

$Q(n)$

Analysis of cache misses

$$Q(n) = \begin{cases} \Theta(n^2/\mathcal{B}) & \text{if } n^2 < c\mathcal{M} \text{ for suff. small const } c \leq 1, \\ 8Q(n/2) + \Theta(1) & \text{otherwise.} \end{cases}$$

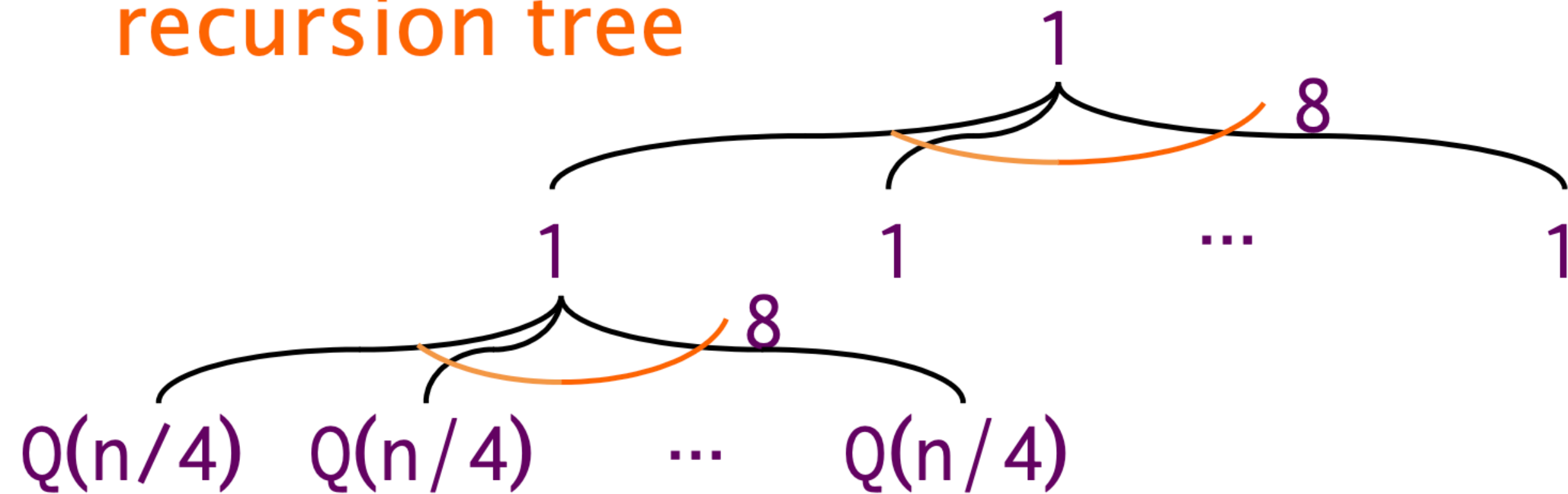
recursion tree



Analysis of cache misses

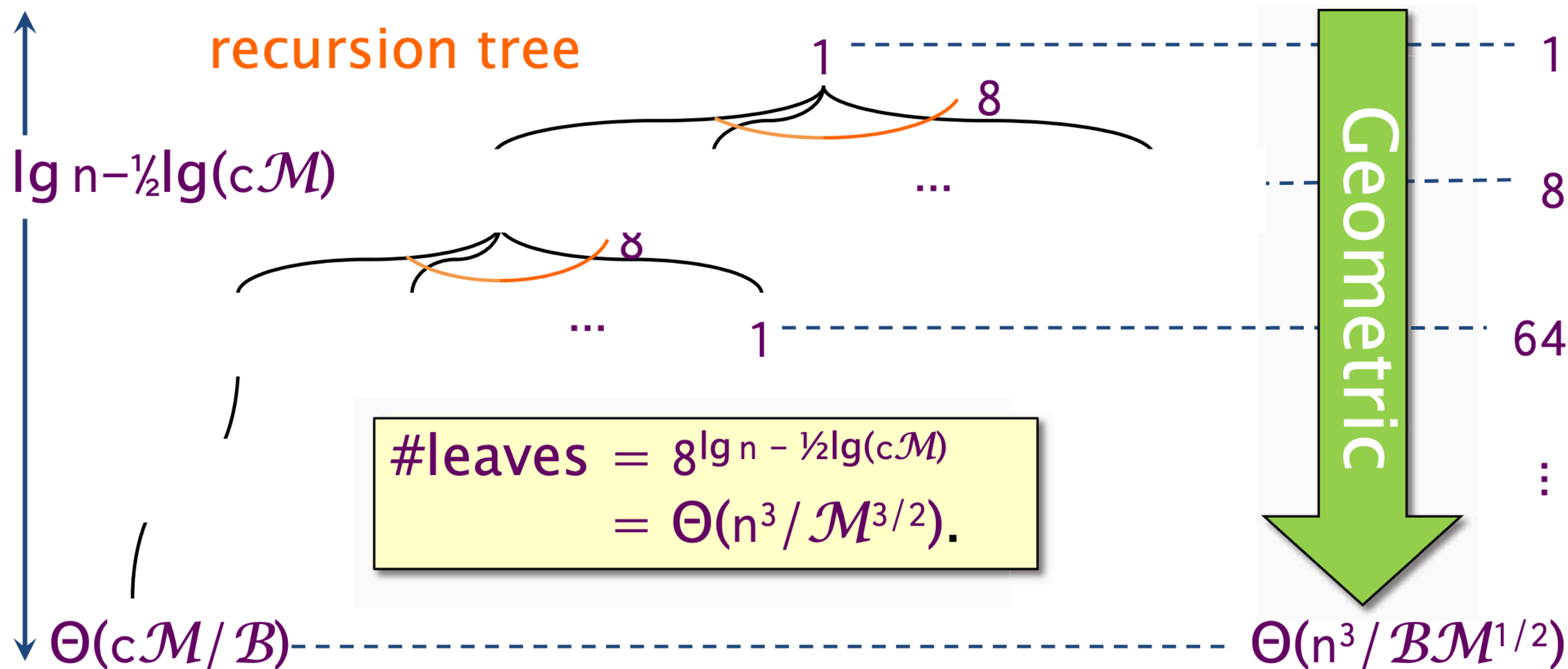
$$Q(n) = \begin{cases} \Theta(n^2/\mathcal{B}) & \text{if } n^2 < c\mathcal{M} \text{ for suff. small const } c \leq 1, \\ 8Q(n/2) + \Theta(1) & \text{otherwise.} \end{cases}$$

recursion tree



Analysis of cache misses

$$Q(n) = \begin{cases} \Theta(n^2/\mathcal{B}) & \text{if } n^2 < c\mathcal{M} \text{ for suff. small const } c \leq 1, \\ 8Q(n/2) + \Theta(1) & \text{otherwise.} \end{cases}$$

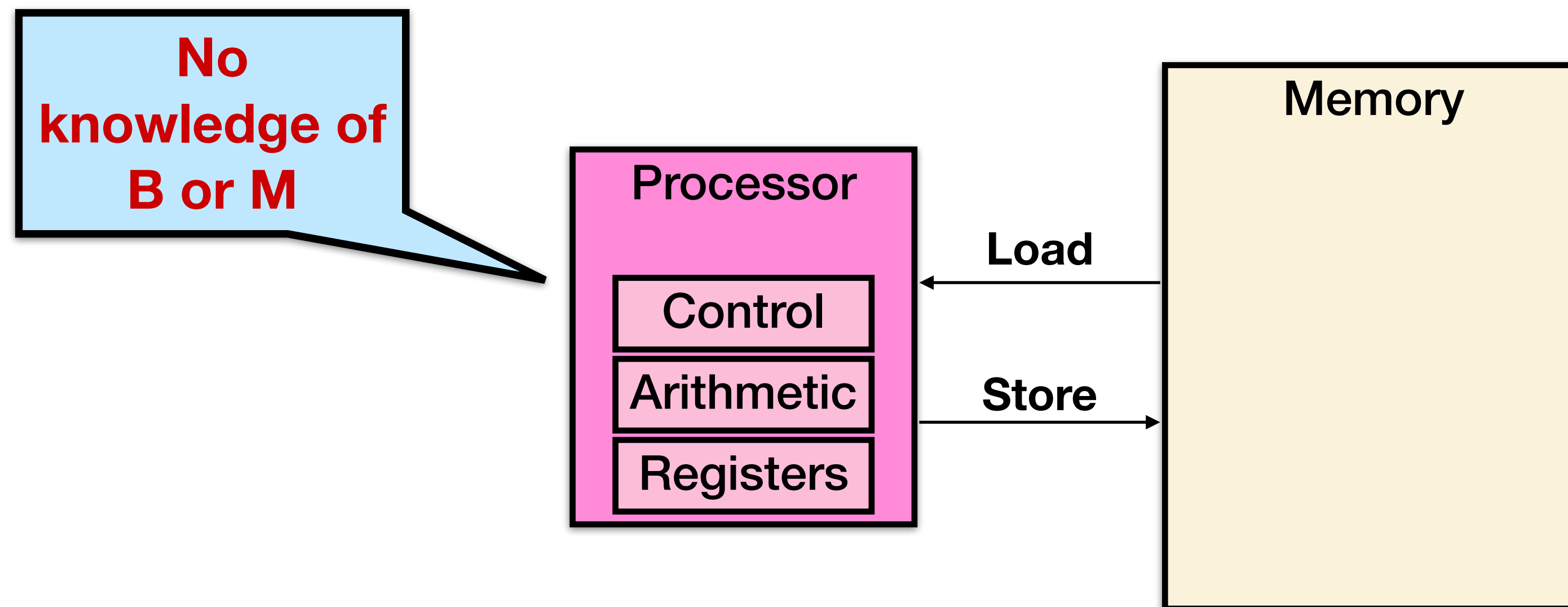


Same cache misses as with tiling!

$$Q(n) = \Theta(n^3/\mathcal{B}\mathcal{M}^{1/2})$$

Efficient cache-oblivious algorithms

- No tuning parameters.
- No explicit knowledge of cache sizes.
- Handle multilevel caches automatically (asymptotically optimally).
- Good in multiprogrammed environments (see: cache-adaptive algorithms).



CSE 6220/CX 4220

Introduction to HPC

Lecture 6: Cache-Oblivious Algorithms

Helen Xu

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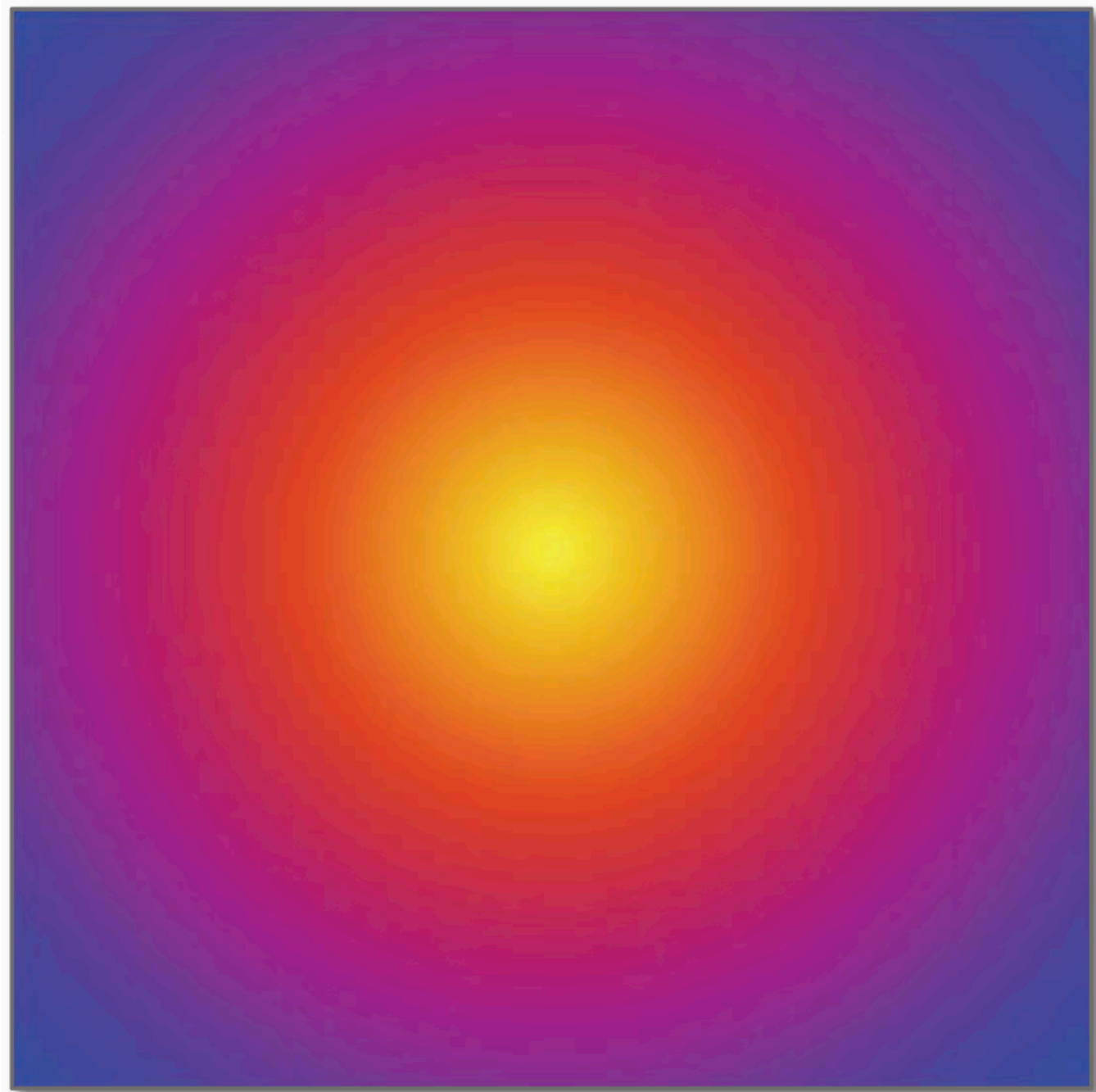
Georgia Tech College of Computing
School of Computational
Science and Engineering

Simulation of Heat Diffusion

Heat Diffusion

2D heat equation

Let $u(t, x, y)$ = temperature at time t of point (x, y) .

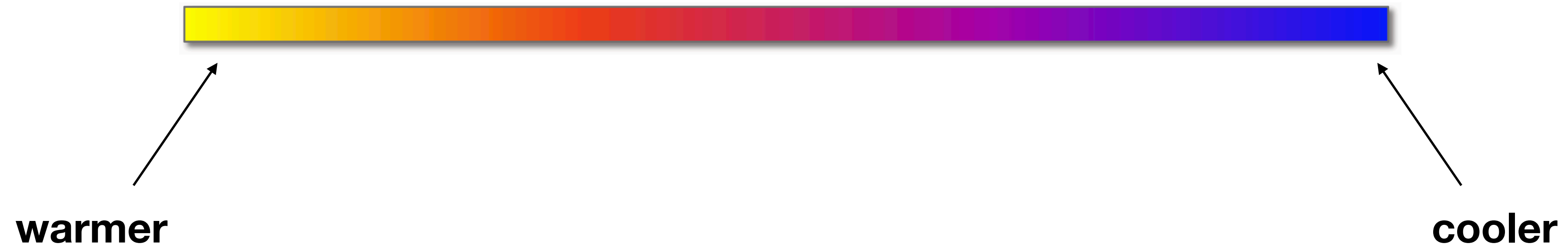


$$\frac{\partial u}{\partial t} = \alpha \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

α is the thermal diffusivity (constant).

1D Heat Equation

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}$$



Finite-Difference Approximation

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}$$

$$\frac{\partial}{\partial t} u(t, x) \approx \frac{u(t + \Delta t, x) - u(t, x)}{\Delta t},$$

$$\frac{\partial}{\partial x} u(t, x) \approx \frac{u(t, x + \Delta x/2) - u(t, x - \Delta x/2)}{\Delta x},$$

$$\frac{\partial^2}{\partial x^2} u(t, x) \approx \frac{\frac{\partial}{\partial x} u(t, x + \Delta x/2) - \frac{\partial}{\partial x} u(t, x - \Delta x/2)}{\Delta x},$$

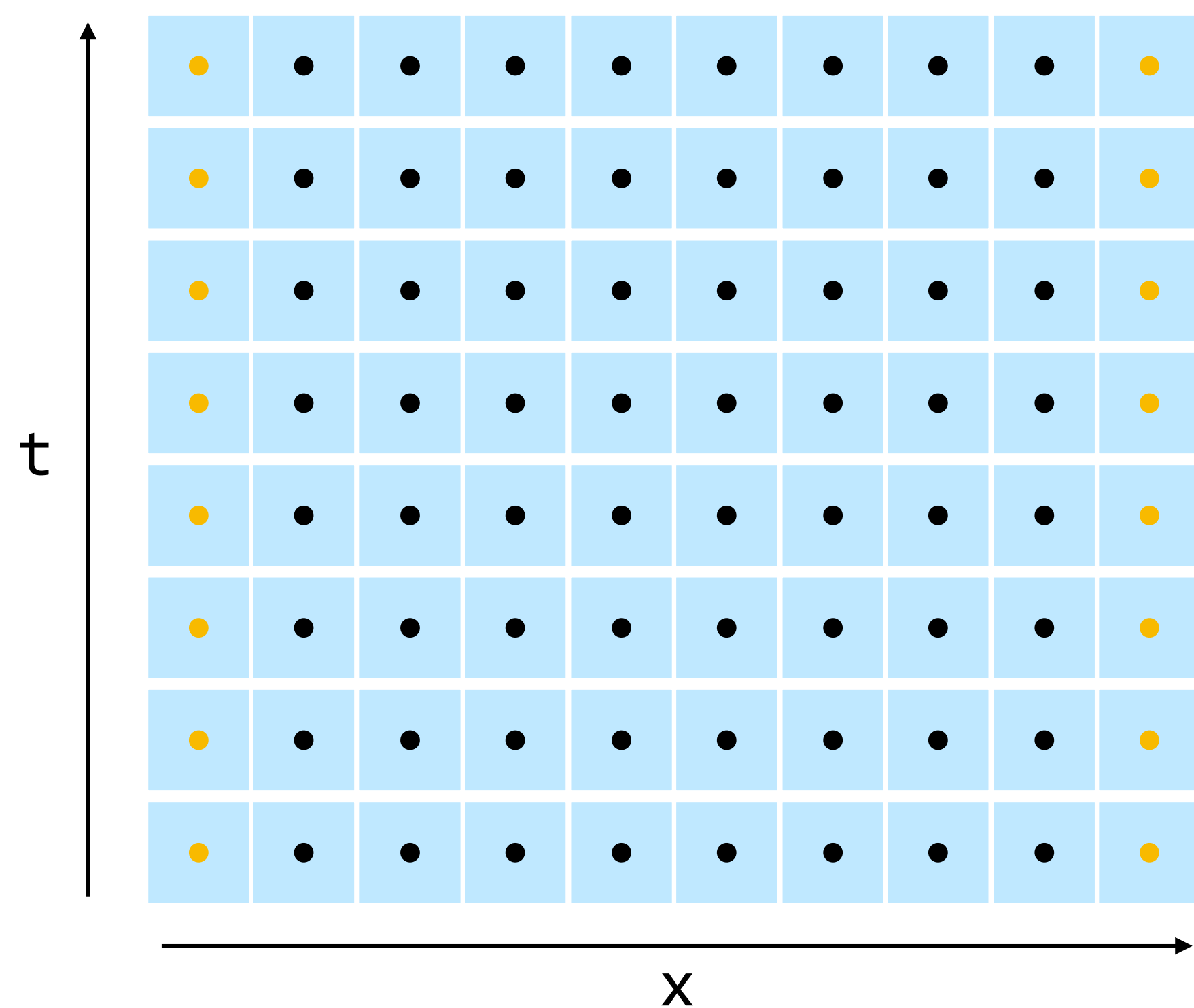
$$\frac{\partial^2}{\partial x^2} u(t, x) \approx \frac{u(t, x + \Delta x) - 2u(t, x) + u(t, x - \Delta x)}{(\Delta x)^2}.$$

So the 1D heat equation reduces to:

$$\frac{u(t + \Delta t, x) - u(t, x)}{\Delta t} \approx \alpha \left(\frac{u(t, x + \Delta x) - 2u(t, x) + u(t, x - \Delta x)}{(\Delta x)^2} \right).$$

3-Point Stencil

$$\frac{u(t + \Delta t, x) - u(t, x)}{\Delta t} \approx \alpha \left(\frac{u(t, x + \Delta x) - 2u(t, x) + u(t, x - \Delta x)}{(\Delta x)^2} \right)$$



A **stencil computation** updates each point in an array by a fixed pattern, called a stencil.

iteration space
boundary

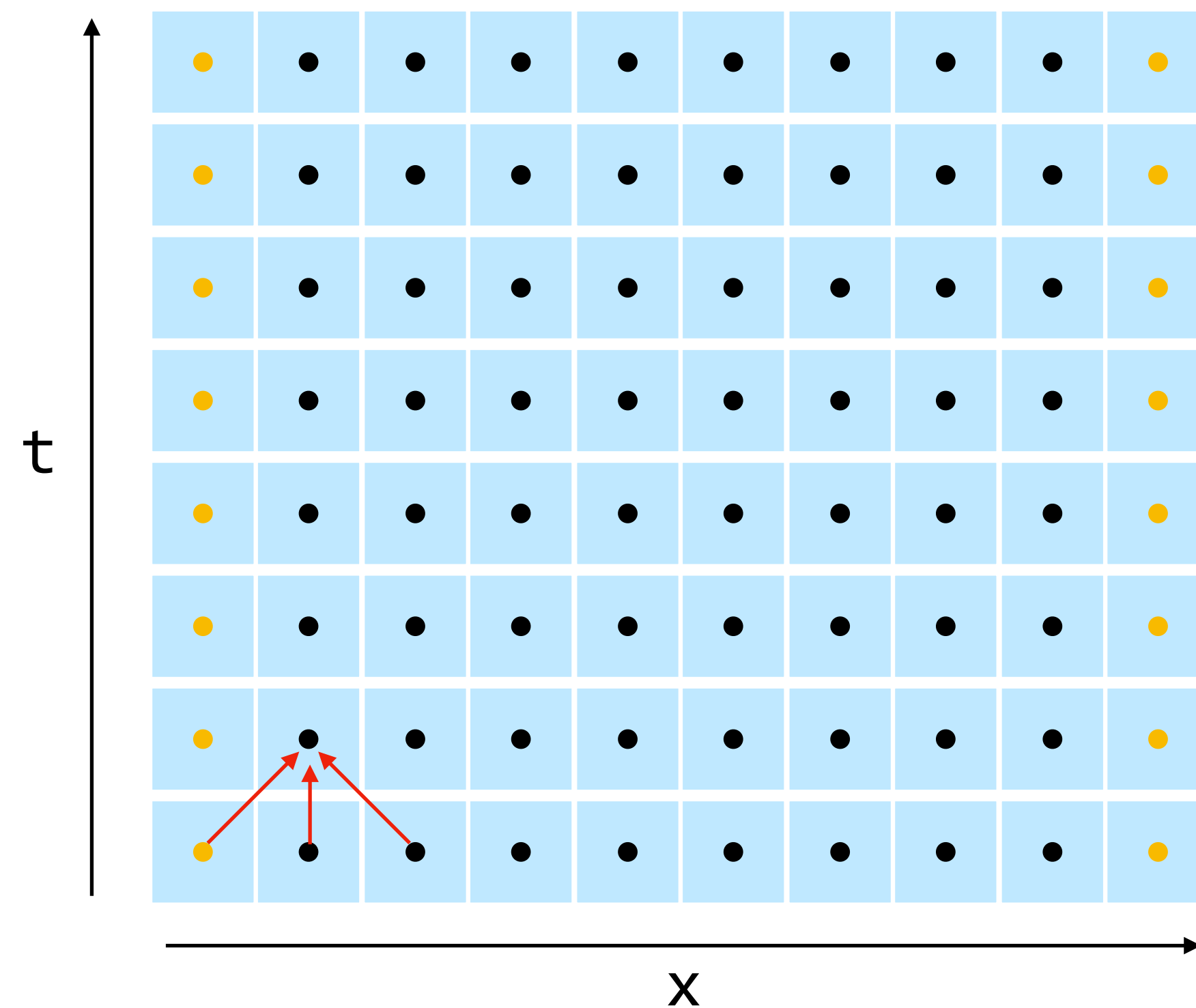
Update rule

$$\Delta t = 1, \Delta x = 1$$

$$u[t+1][x] = u[t][x] + \text{ALPHA} * (u[t][x+1] - 2*u[t][x] + u[t][x-1])$$

3-Point Stencil

$$\frac{u(t + \Delta t, x) - u(t, x)}{\Delta t} \approx \alpha \left(\frac{u(t, x + \Delta x) - 2u(t, x) + u(t, x - \Delta x)}{(\Delta x)^2} \right)$$



A **stencil computation** updates each point in an array by a fixed pattern, called a stencil.

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boundary

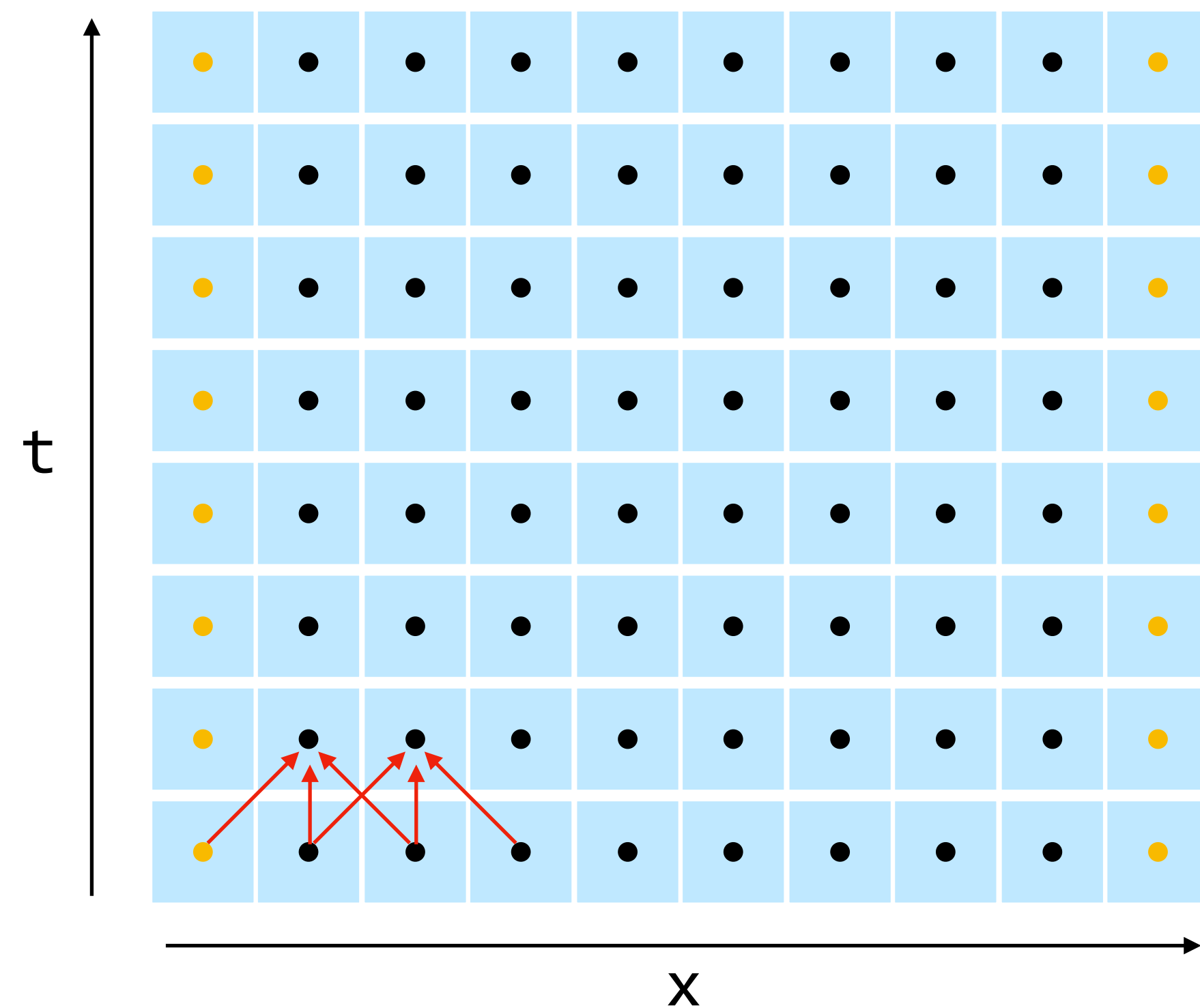
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$$u[t+1][x] = u[t][x] + \text{ALPHA} * (u[t][x+1] - 2*u[t][x] + u[t][x-1])$$

3-Point Stencil

$$\frac{u(t + \Delta t, x) - u(t, x)}{\Delta t} \approx \alpha \left(\frac{u(t, x + \Delta x) - 2u(t, x) + u(t, x - \Delta x)}{(\Delta x)^2} \right)$$



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iteration space
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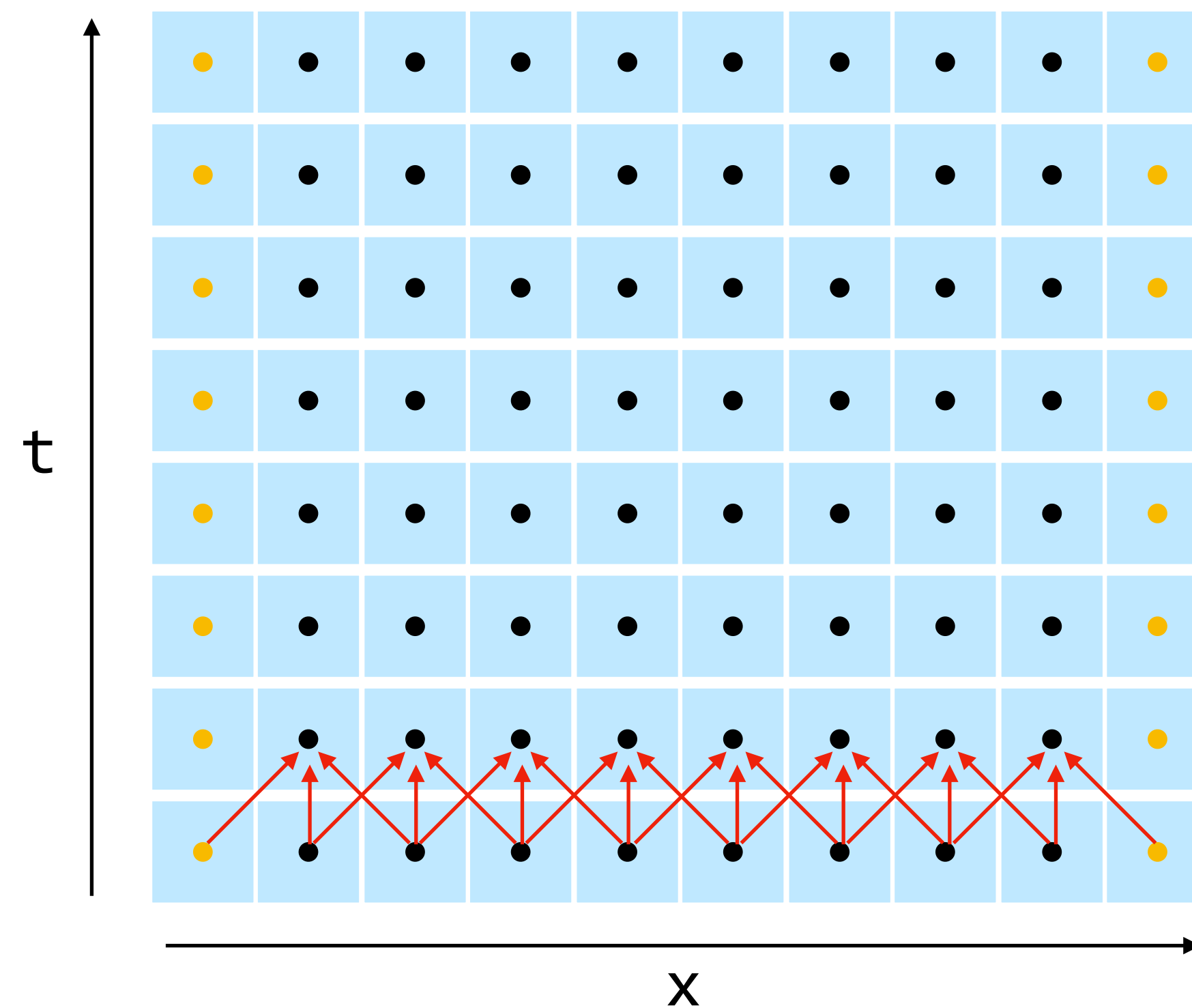
Update rule

$$\Delta t = 1, \Delta x = 1$$

$$u[t+1][x] = u[t][x] + \text{ALPHA} * (u[t][x+1] - 2*u[t][x] + u[t][x-1])$$

3-Point Stencil

$$\frac{u(t + \Delta t, x) - u(t, x)}{\Delta t} \approx \alpha \left(\frac{u(t, x + \Delta x) - 2u(t, x) + u(t, x - \Delta x)}{(\Delta x)^2} \right)$$



A **stencil computation** updates each point in an array by a fixed pattern, called a stencil.

iteration space

boundary

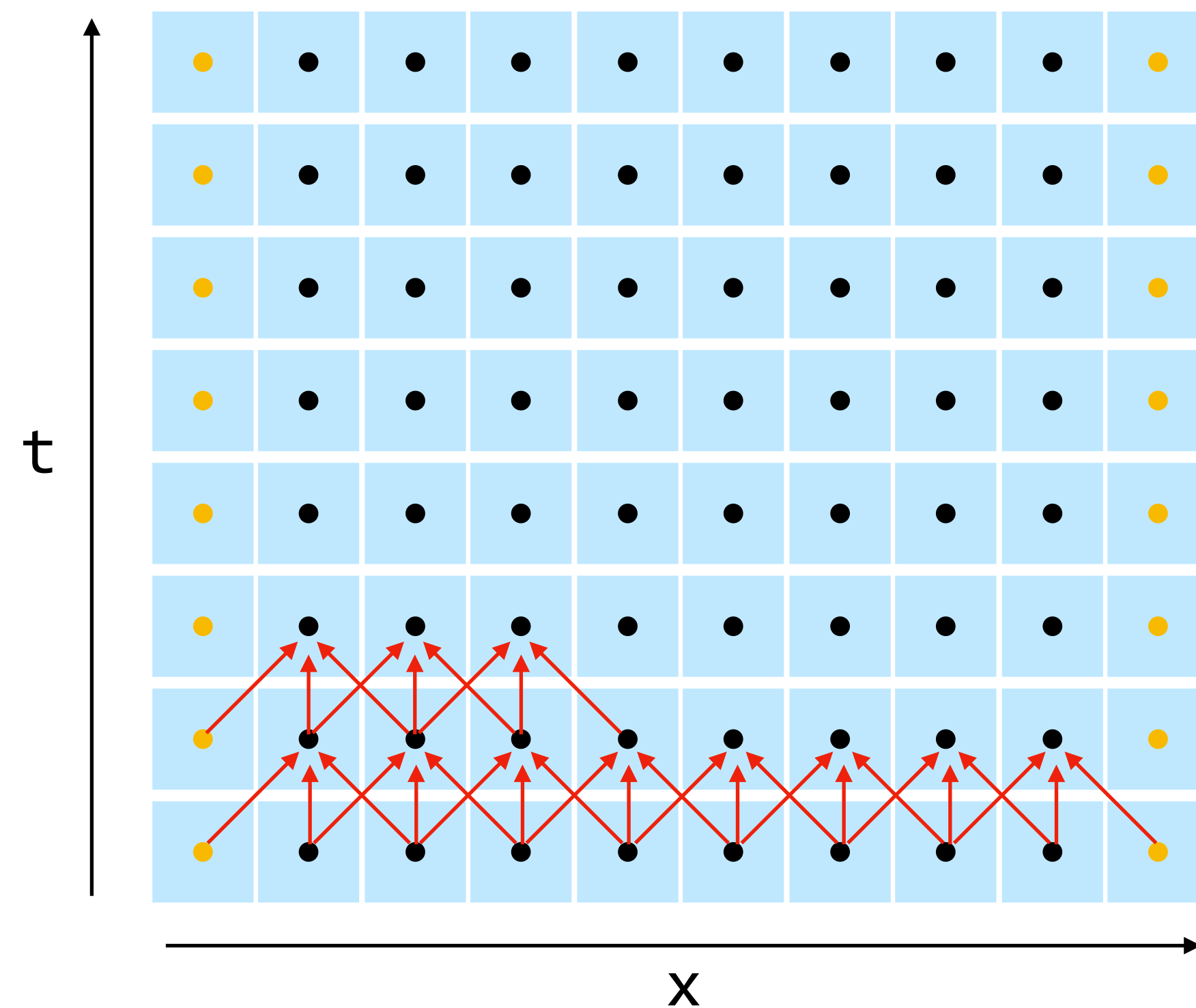
Update rule

$$\Delta t = 1, \Delta x = 1$$

$$u[t+1][x] = u[t][x] + \text{ALPHA} * (u[t][x+1] - 2*u[t][x] + u[t][x-1])$$

3-Point Stencil

$$\frac{u(t + \Delta t, x) - u(t, x)}{\Delta t} \approx \alpha \left(\frac{u(t, x + \Delta x) - 2u(t, x) + u(t, x - \Delta x)}{(\Delta x)^2} \right)$$



A **stencil computation** updates each point in an array by a fixed pattern, called a stencil.

iteration space
boundary

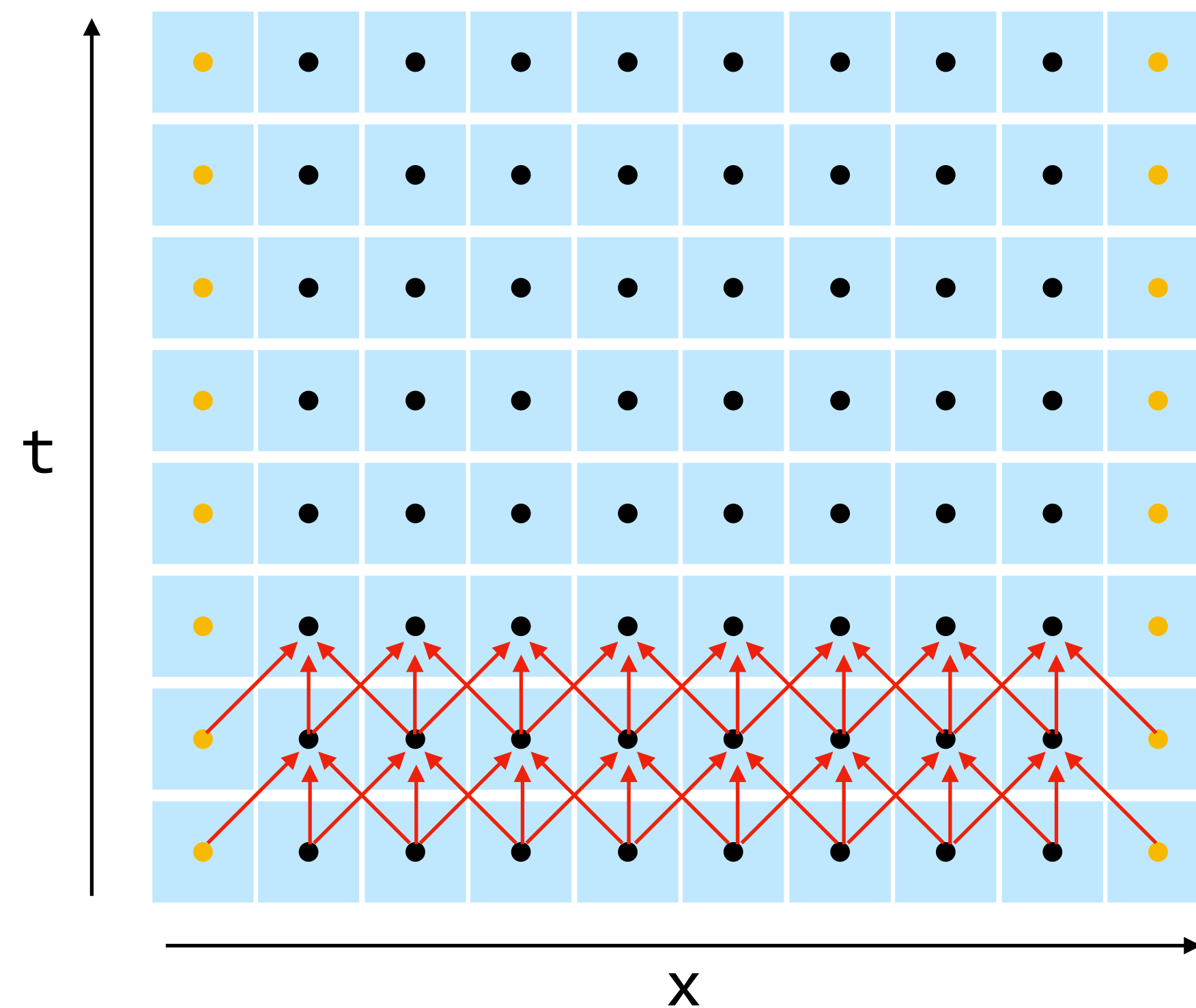
Update rule

$$\Delta t = 1, \Delta x = 1$$

$$u[t+1][x] = u[t][x] + \text{ALPHA} * (u[t][x+1] - 2*u[t][x] + u[t][x-1])$$

3-Point Stencil

$$\frac{u(t + \Delta t, x) - u(t, x)}{\Delta t} \approx \alpha \left(\frac{u(t, x + \Delta x) - 2u(t, x) + u(t, x - \Delta x)}{(\Delta x)^2} \right)$$



A **stencil computation** updates each point in an array by a fixed pattern, called a stencil.

iteration space
boundary

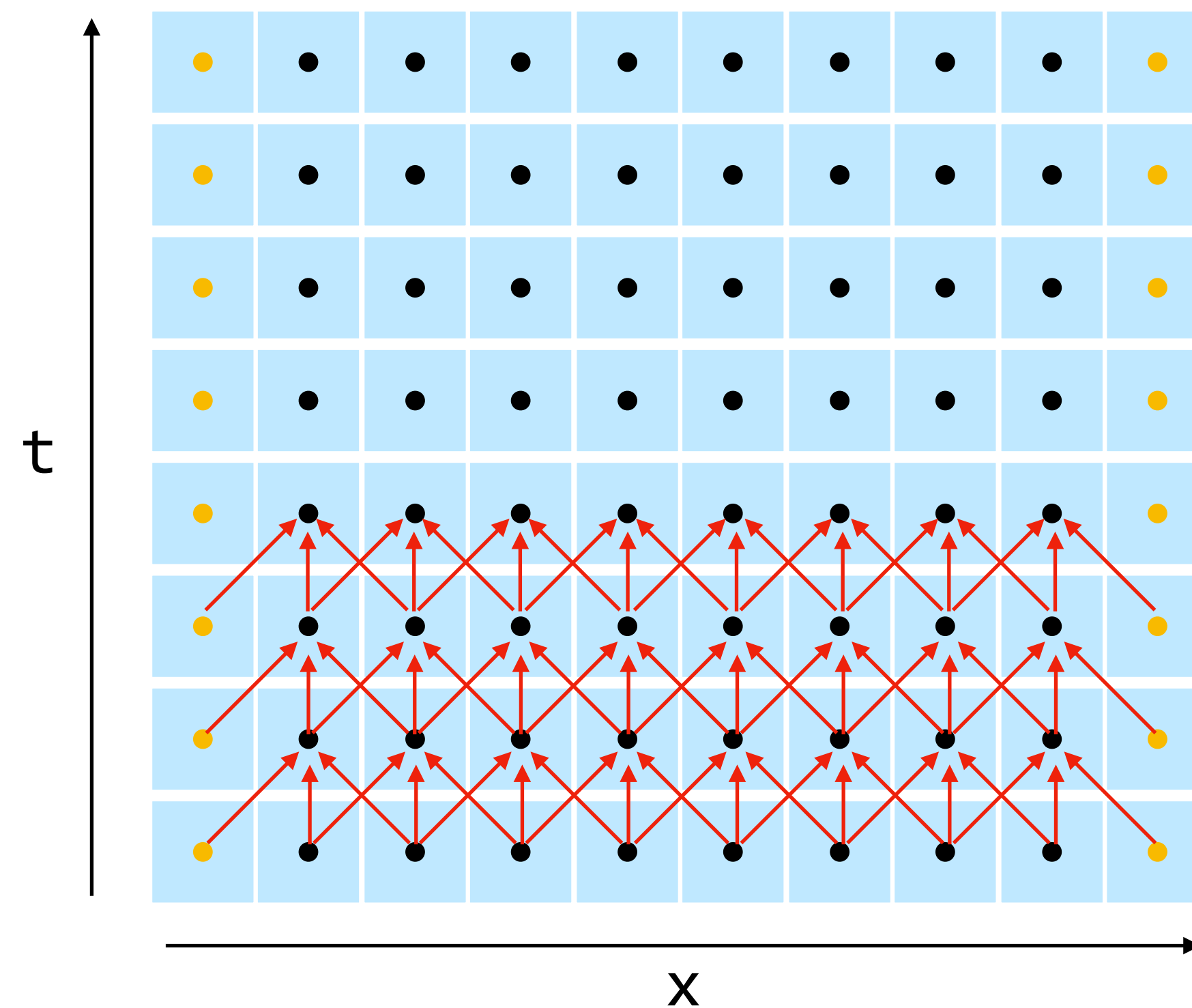
Update rule

$$\Delta t = 1, \Delta x = 1$$

$$u[t+1][x] = u[t][x] + \text{ALPHA} * (u[t][x+1] - 2*u[t][x] + u[t][x-1])$$

3-Point Stencil

$$\frac{u(t + \Delta t, x) - u(t, x)}{\Delta t} \approx \alpha \left(\frac{u(t, x + \Delta x) - 2u(t, x) + u(t, x - \Delta x)}{(\Delta x)^2} \right)$$



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iteration space
boundary

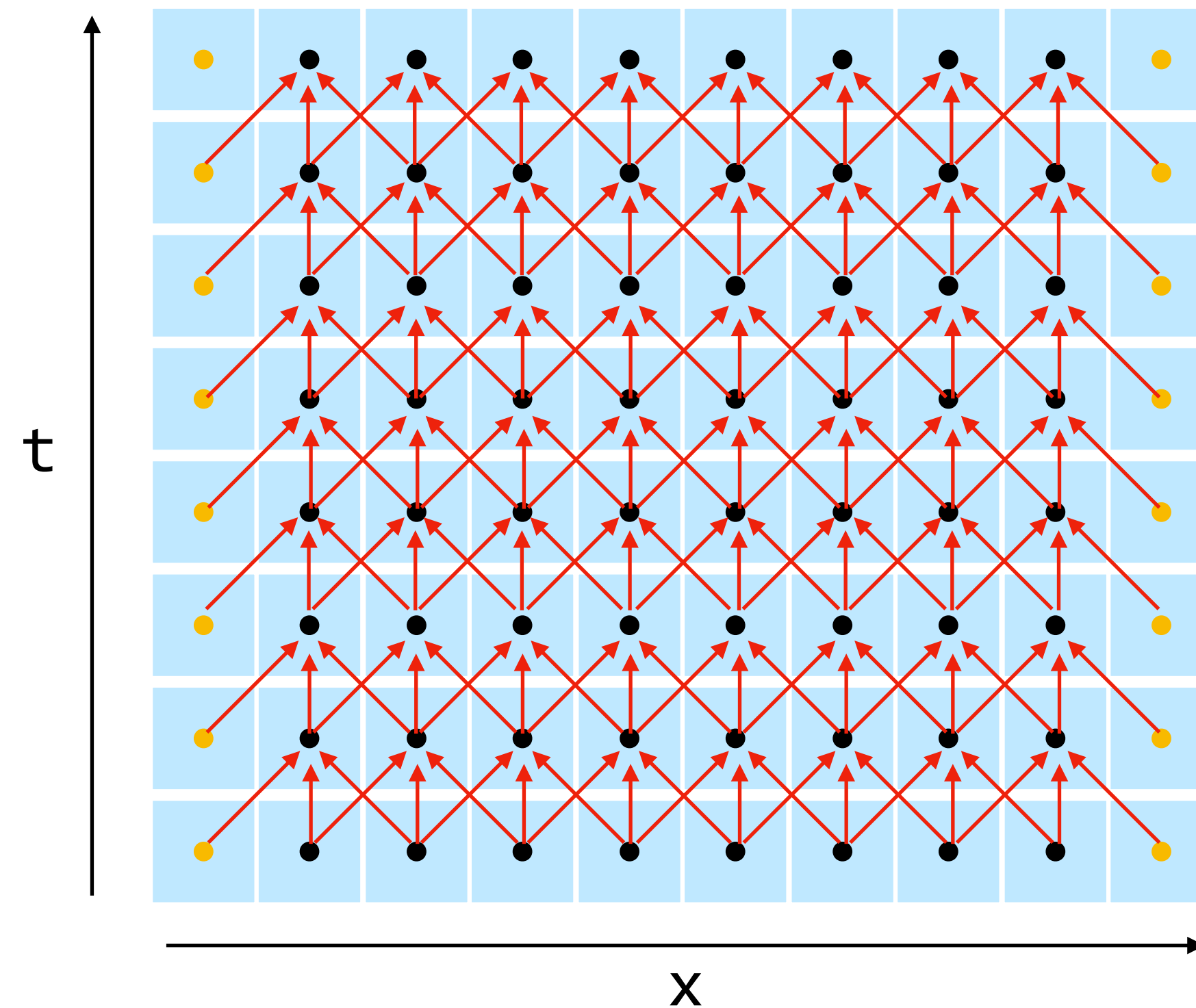
Update rule

$$\Delta t = 1, \Delta x = 1$$

$$u[t+1][x] = u[t][x] + \text{ALPHA} * (u[t][x+1] - 2*u[t][x] + u[t][x-1])$$

3-Point Stencil

$$\frac{u(t + \Delta t, x) - u(t, x)}{\Delta t} \approx \alpha \left(\frac{u(t, x + \Delta x) - 2u(t, x) + u(t, x - \Delta x)}{(\Delta x)^2} \right)$$



A **stencil computation** updates each point in an array by a fixed pattern, called a stencil.

iteration space
boundary

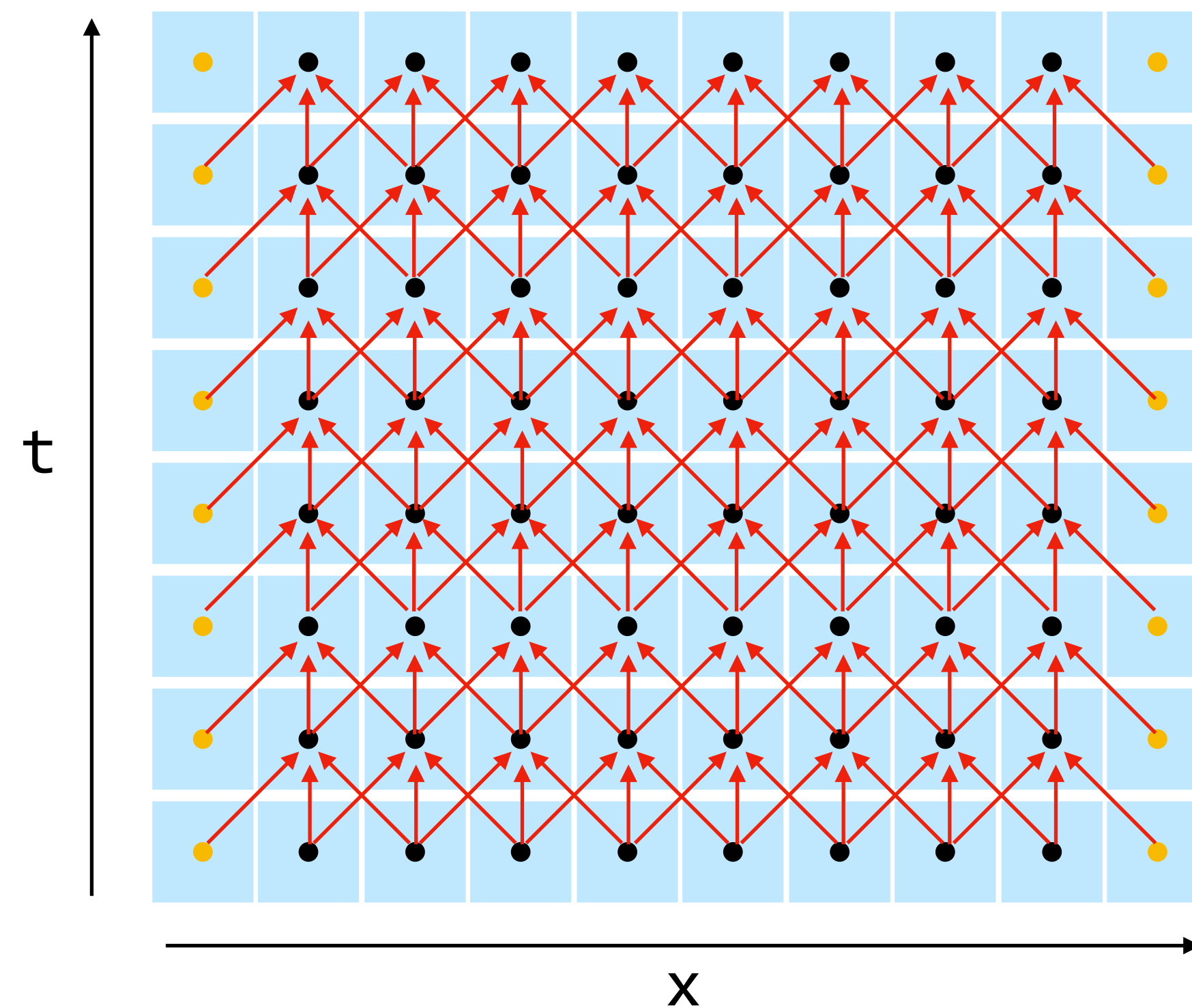
Update rule

$$\Delta t = 1, \Delta x = 1$$

$$u[t+1][x] = u[t][x] + \text{ALPHA} * (u[t][x+1] - 2*u[t][x] + u[t][x-1])$$

3-Point Stencil

$$\frac{u(t + \Delta t, x) - u(t, x)}{\Delta t} \approx \alpha \left(\frac{u(t, x + \Delta x) - 2u(t, x) + u(t, x - \Delta x)}{(\Delta x)^2} \right)$$



A **stencil computation** updates each point in an array by a fixed pattern, called a stencil.

iteration space
boundary

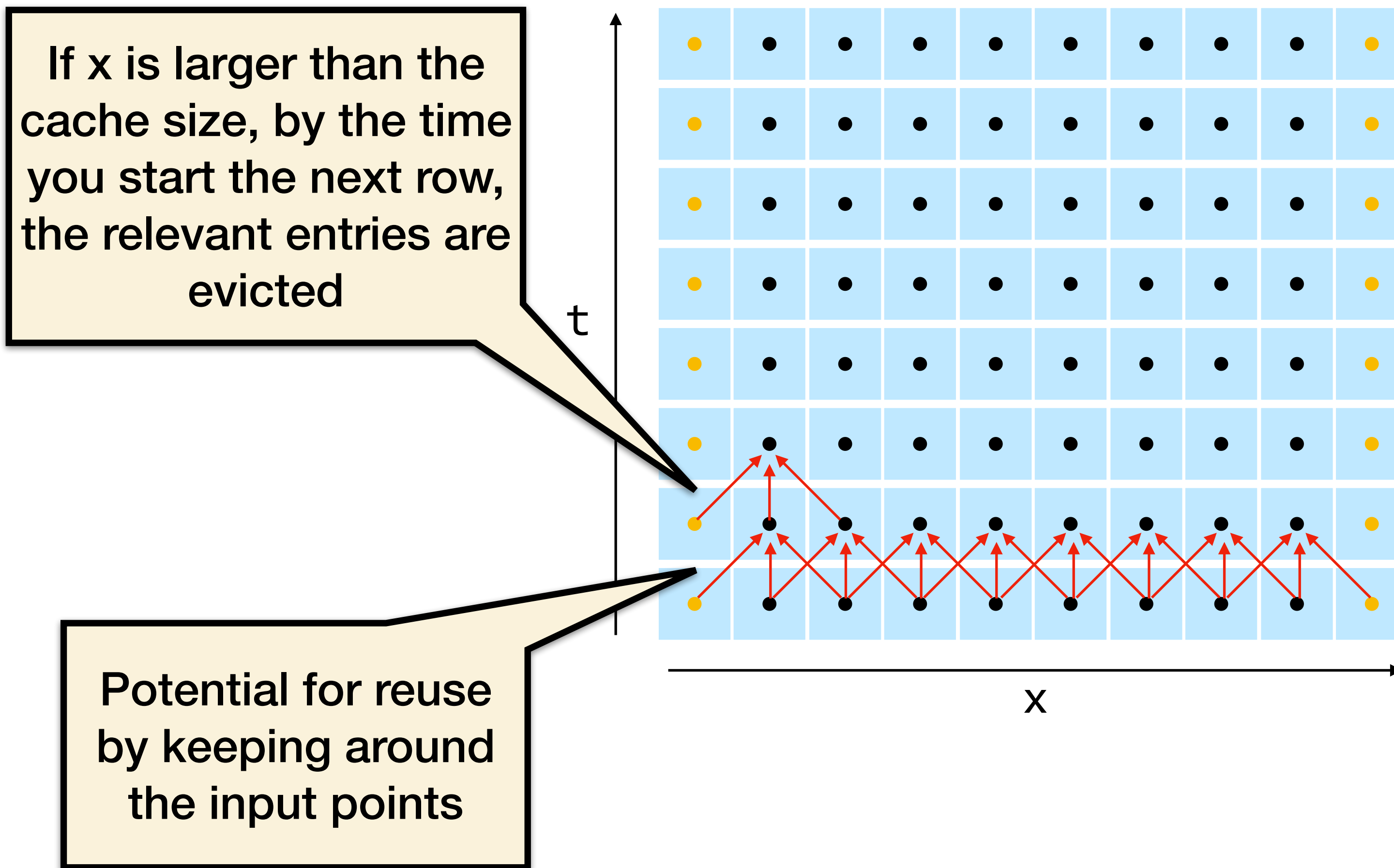
How would this code perform with respect to caching?

Update rule

$$\Delta t = 1, \Delta x = 1$$

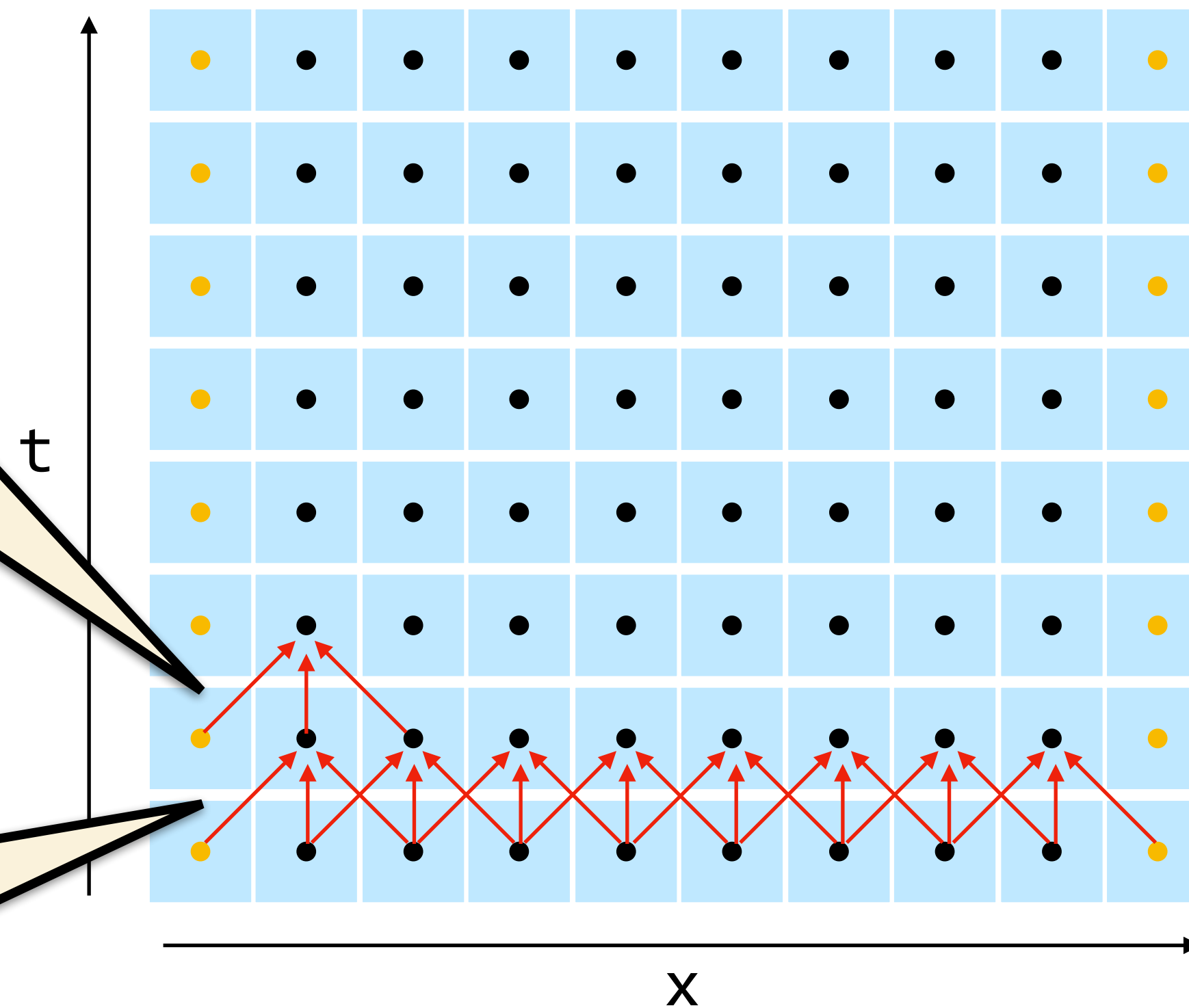
$$u[t+1][x] = u[t][x] + \text{ALPHA} * (u[t][x+1] - 2*u[t][x] + u[t][x-1])$$

Cache behavior of looping stencil



Cache behavior of looping stencil

If x is larger than the cache size, by the time you start the next row, the relevant entries are evicted

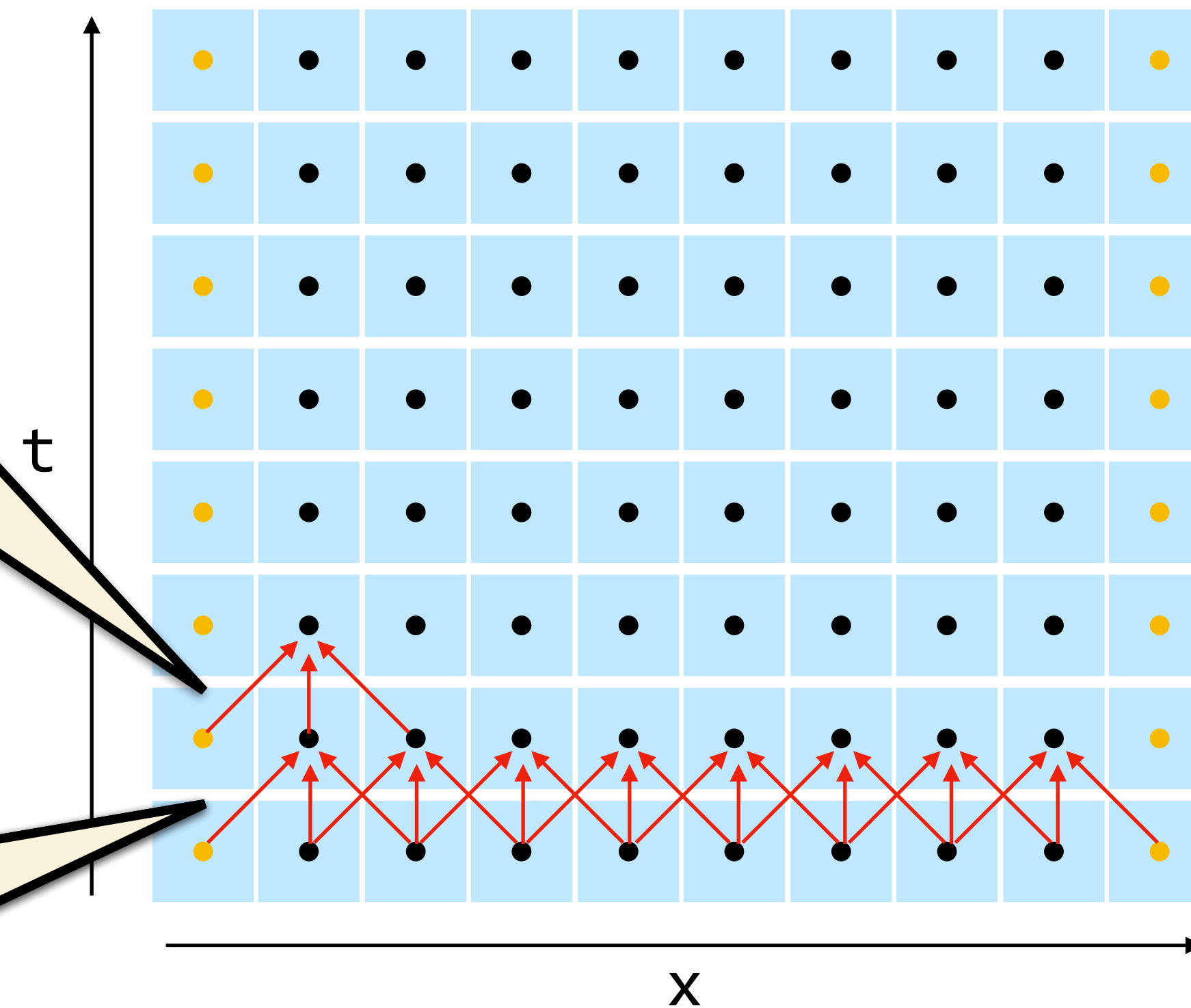


Potential for reuse by keeping around the input points

If we only care about the values of x at the most recent time step, do we need to keep around the whole 2D matrix? How many rows do we need?

Cache behavior of looping stencil

If x is larger than the cache size, by the time you start the next row, the relevant entries are evicted



Potential for reuse by keeping around the input points

If we only care about the values of x at the most recent time step, do we need to keep around the whole 2D matrix? How many rows do we need?

Can keep 2 rows - one for previous and one for current time

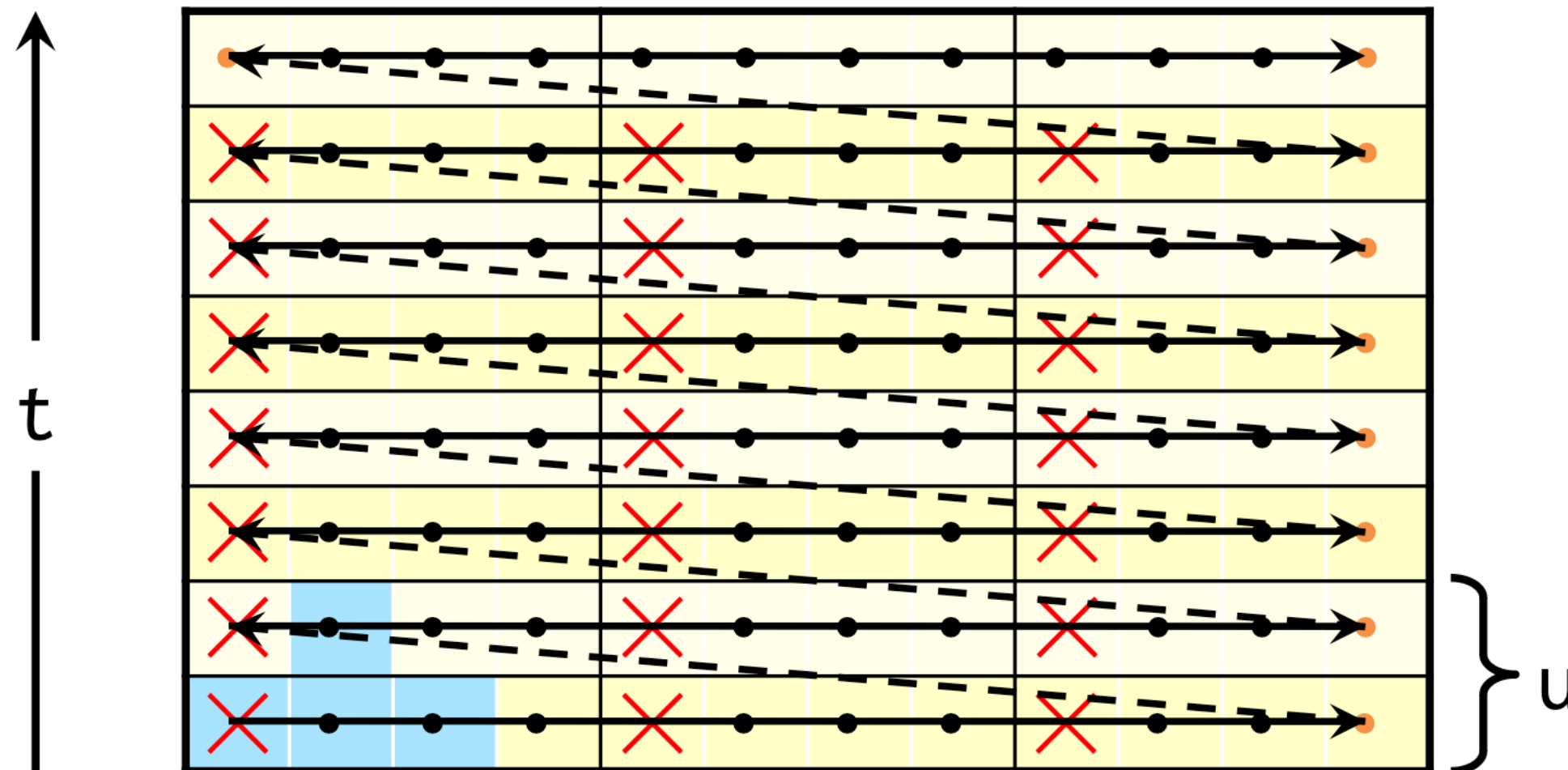
Cache-Oblivious Stencil Computations

Cache behavior of looping

```
double u[2][N]; // even-odd trick

static inline double kernel(double * w) {
    return w[0] + ALPHA * (w[-1] - 2*w[0] + w[1]);
}

for (size_t t = 1; t < T-1; ++t) { // time loop
    for(size_t x = 1; x < N-1; ++x) // space loop
        u[(t+1)%2][x] = kernel( &u[t%2][x] );
```



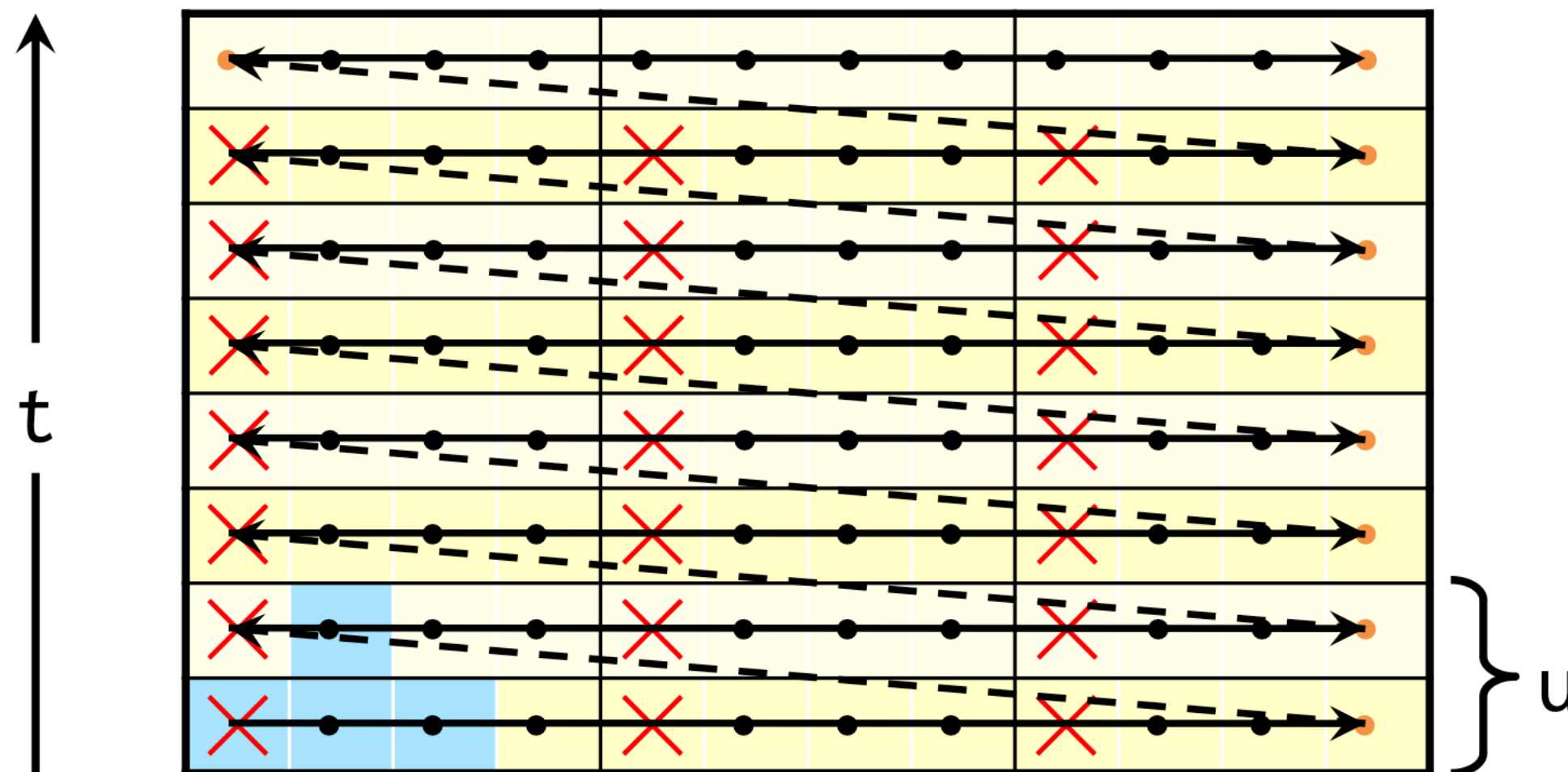
Assuming LRU, how many cache misses does the looping code incur (as a function of N and T)?

Cache behavior of looping

```
double u[2][N]; // even-odd trick

static inline double kernel(double * w) {
    return w[0] + ALPHA * (w[-1] - 2*w[0] + w[1]);
}

for (size_t t = 1; t < T-1; ++t) { // time loop
    for(size_t x = 1; x < N-1; ++x) // space loop
        u[(t+1)%2][x] = kernel( &u[t%2][x] );
```



Assuming LRU, if $N > \mathcal{M}$, then
 $Q = \Theta(NT/\mathcal{B})$

Cache-Oblivious 3-Point Stencil

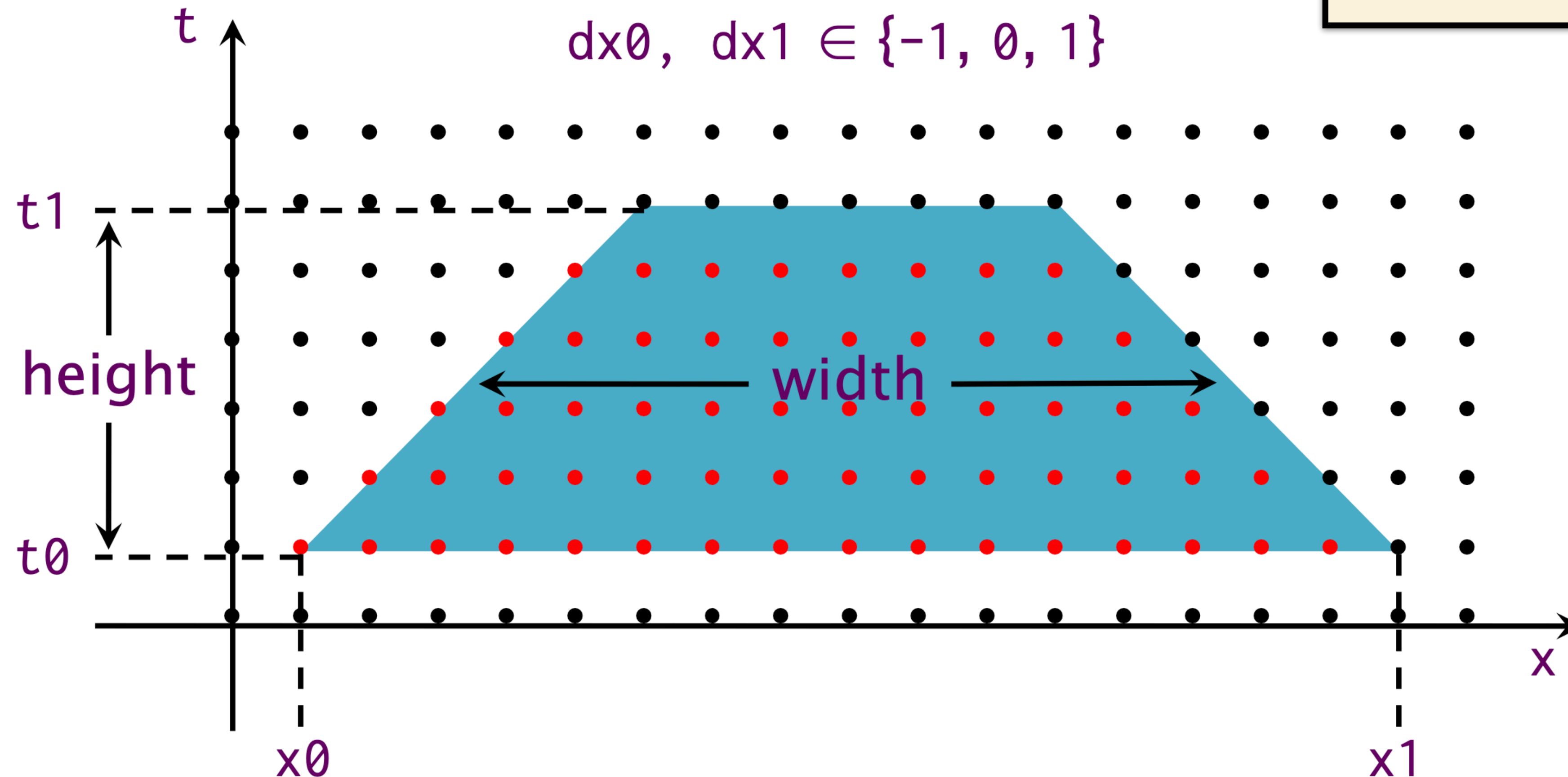
Recursively traverse trapezoidal regions of space-time points (t, x) such that

$$t_0 \leq t < t_1$$

$$x_0 + dx_0(t - t_0) \leq x < x_1 + dx_1(t - t_0)$$

$$dx_0, dx_1 \in \{-1, 0, 1\}$$

Why not quadrants like we did before in tableau construction?

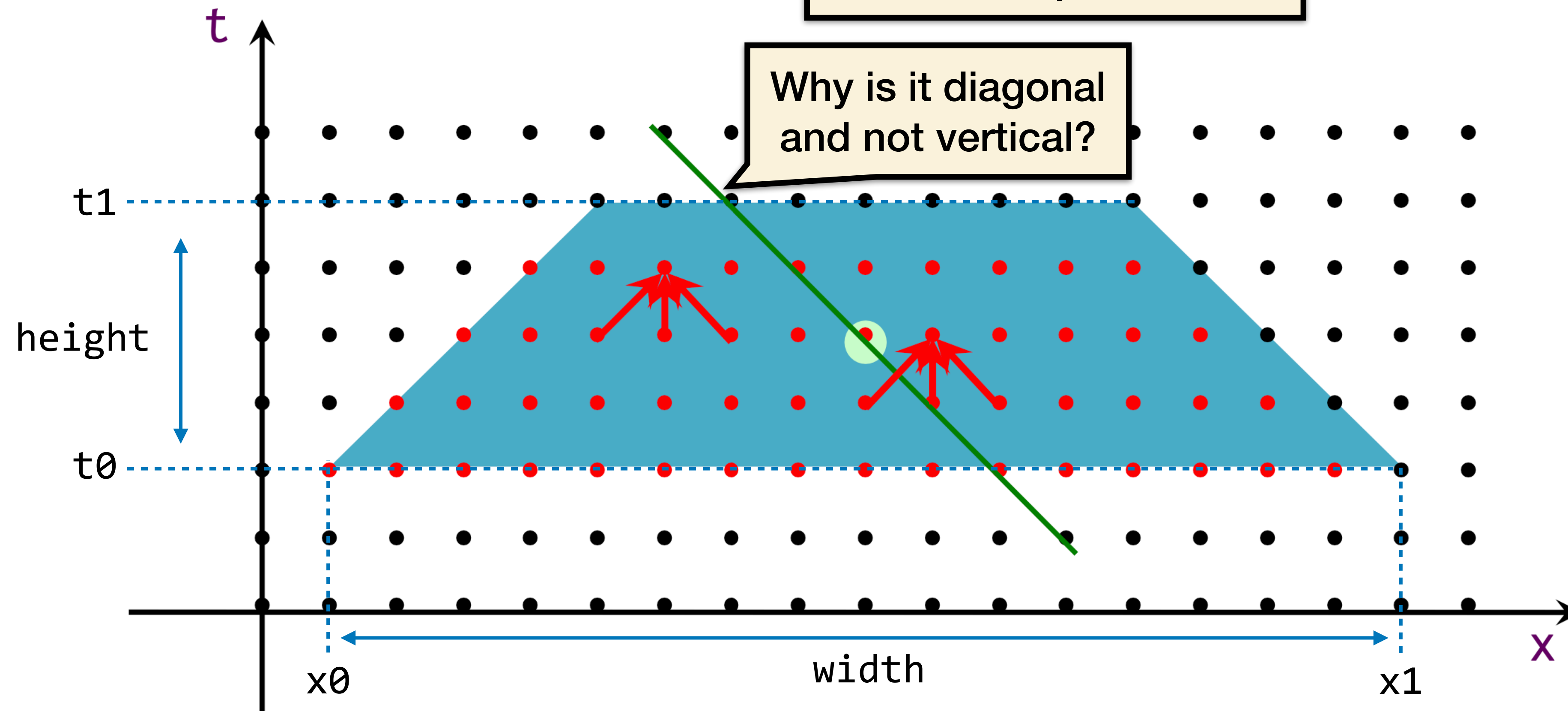


Squat Trapezoid: Space Cut

If $\text{width} \geq 2 \cdot \text{height}$, cut the trapezoid with a line of slope -1 through the center. Traverse the trapezoid on the left first, and then the one on the right.

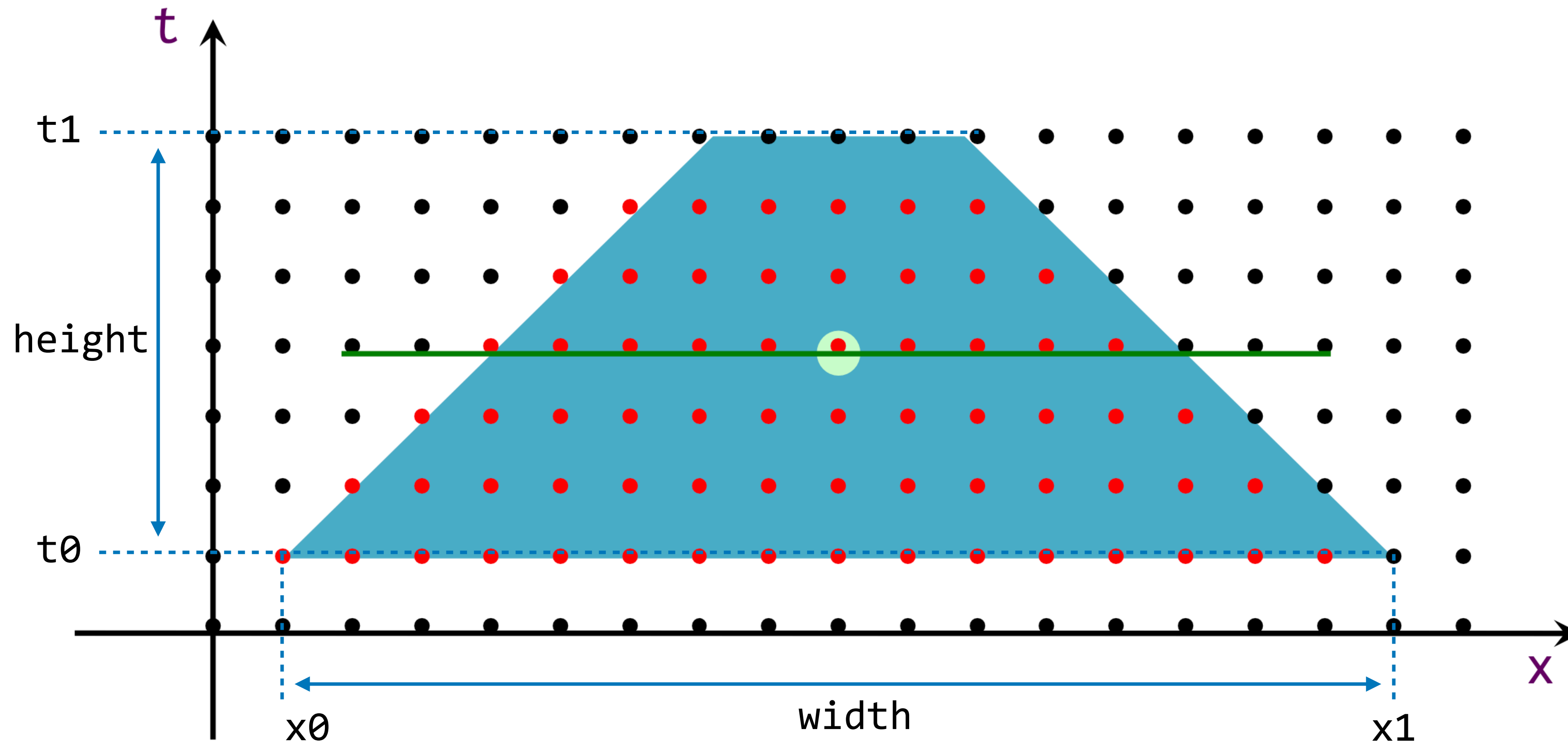
Can we swap the order?

Why is it diagonal and not vertical?



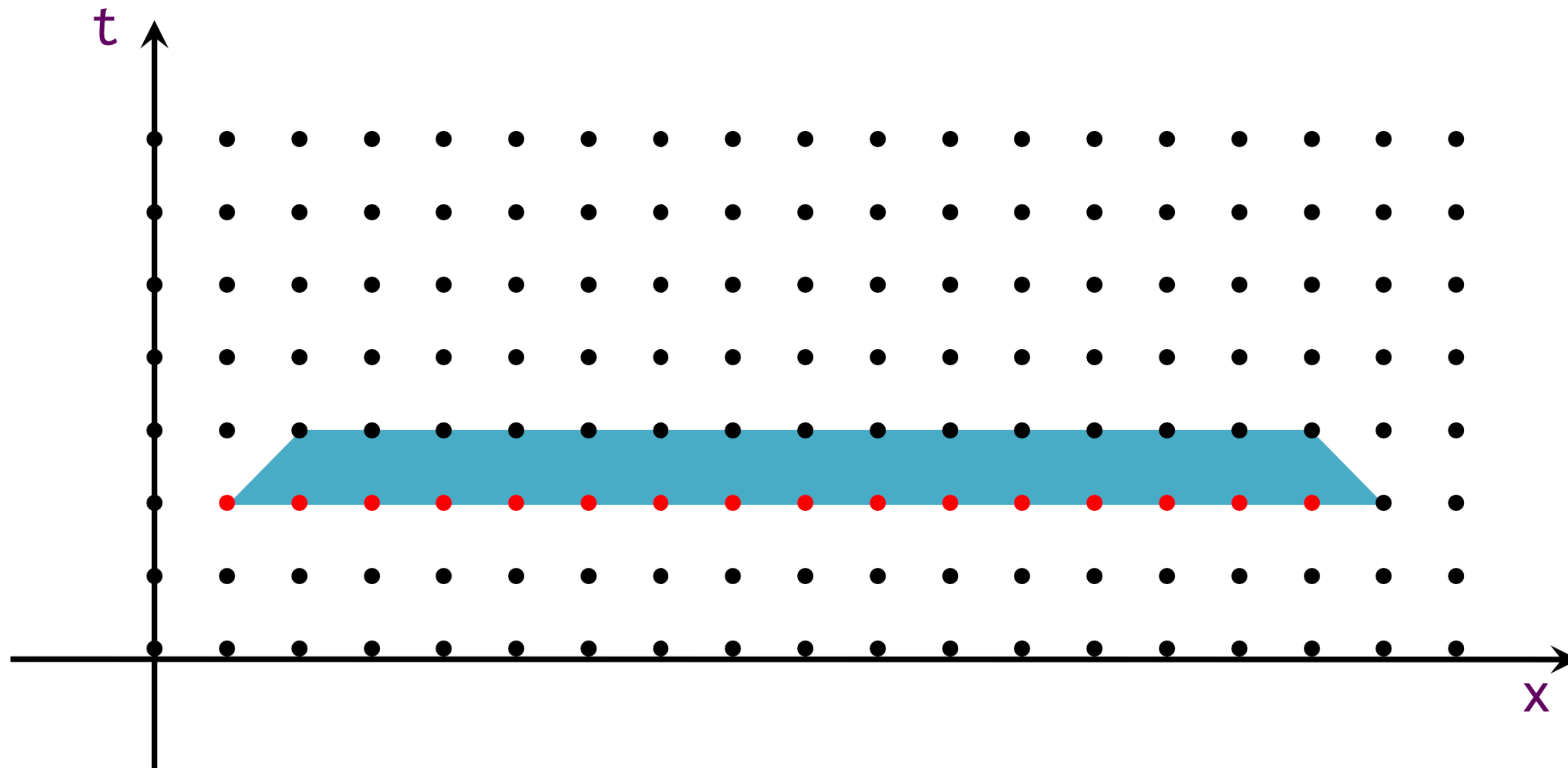
Tall Trapezoid: Time Cut

If $\text{width} < 2 \cdot \text{height}$, cut the trapezoid with a horizontal line through the center. Traverse the bottom trapezoid first, and then the top one.



Base Case

If $\text{height} = 1$, compute all space-time points in the trapezoid. Any order of computation is valid, since no point depends on another.

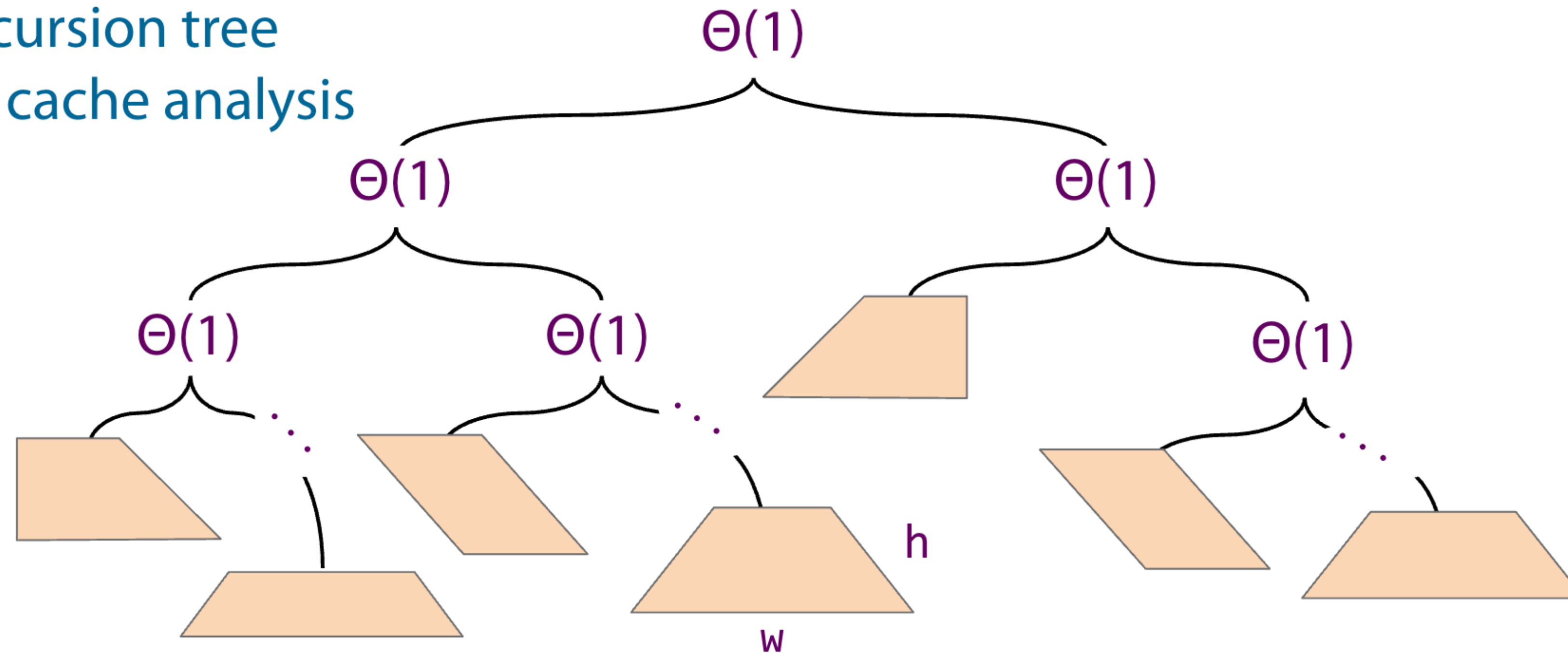


C Implementation

```
void trapezoid(int64_t t0, int64_t t1, int64_t x0, int64_t dx0,
              int64_t x1, int64_t dx1)
{
    int64_t lt = t1 - t0;
    if (lt == 1) { //base case
        for (int64_t x = x0; x < x1; x++)
            u[t1%2][x] = kernel( &u[t0%2][x] );
    } else if (lt > 1) {
        if (2 * (x1 - x0) + (dx1 - dx0) * lt >= 4 * lt) { //space cut
            int64_t xm = (2 * (x0 + x1) + (2 + dx0 + dx1) * lt) / 4;
            trapezoid(t0, t1, x0, dx0, xm, -1);
            trapezoid(t0, t1, xm, -1, x1, dx1);
        } else { //time cut
            int64_t halflt = lt / 2;
            trapezoid(t0, t0 + halflt, x0, dx0, x1, dx1);
            trapezoid(t0 + halflt, t1, x0 + dx0 * halflt, dx0,
                    x1 + dx1 * halflt, dx1);
        }
    }
}
```

Work and Cache Analysis

Recursion tree
for cache analysis



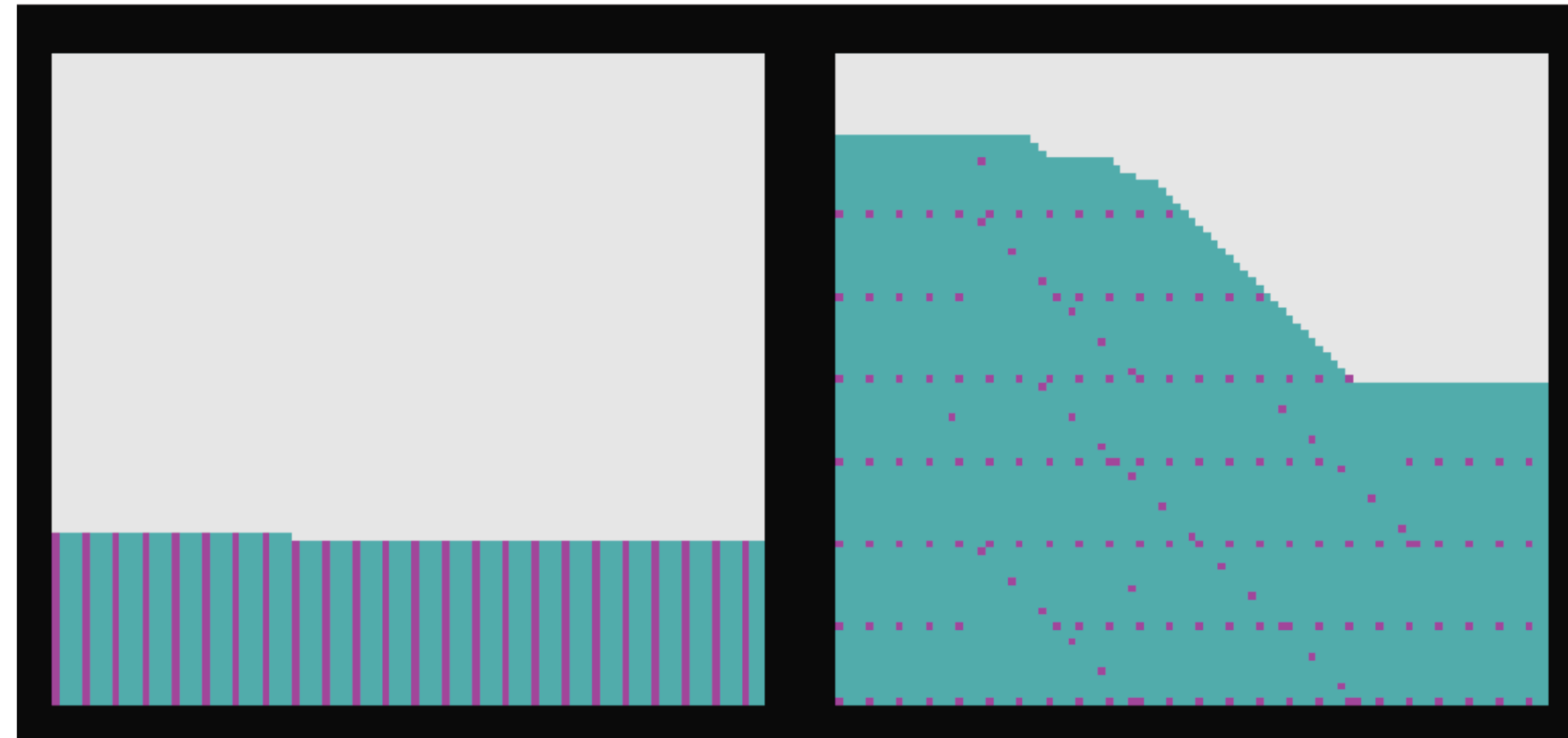
- The bottom of a leaf trapezoid just fits in the cache, so $w = \Theta(\mathcal{M})$.
- A leaf trapezoid contains $\Theta(hw) = \Theta(w^2)$ points and $\Theta(w^2)$ work.
- Since $w \leq \mathcal{M}$, a leaf incurs $\Theta(w/\mathcal{B})$ cache misses.
- There are $\Theta(NT/hw) = \Theta(NT/w^2)$ leaves and internal nodes.
- The internal nodes contribute little to both work and cache misses.
- Work = $\Theta(NT/w^2) \cdot \Theta(w^2) = \Theta(NT)$.
- Cache misses = $\Theta(NT/w^2) \cdot \Theta(w/\mathcal{B}) = \Theta(NT/\mathcal{B}w) = \Theta(NT/\mathcal{B}\mathcal{M})$ (vs. $\Theta(NT/\mathcal{B})$.)

For d dimensions,
 $\Theta(NT/\mathcal{B}\mathcal{M}^{1/d})$

Part 2 ended here

Simulation: 3-Point Stencil

- Rectangular region
 - $N = 95$
 - $T = 87$



Looping

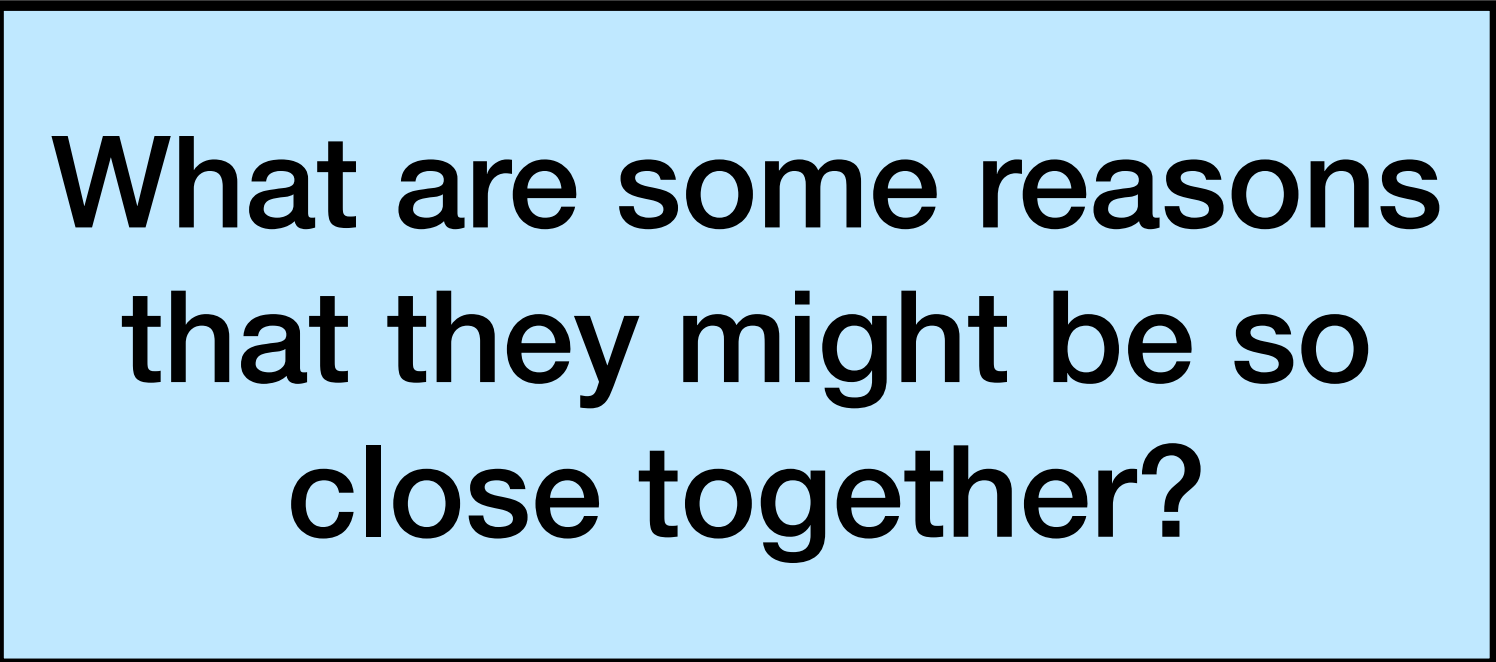
Trapezoid

- Fully associative LRU cache
 - $\mathcal{B} = 4$ points
 - $\mathcal{M} = 32$ points
- Cache-hit latency = 1 cycle
- Cache-miss latency = 10 cycles

Empirical Stencil Performance

For an example 2D heat stencil, we see the following performance:

- Looping ~1560 iterations / s
- Trapezoid ~ 1830 iterations / s

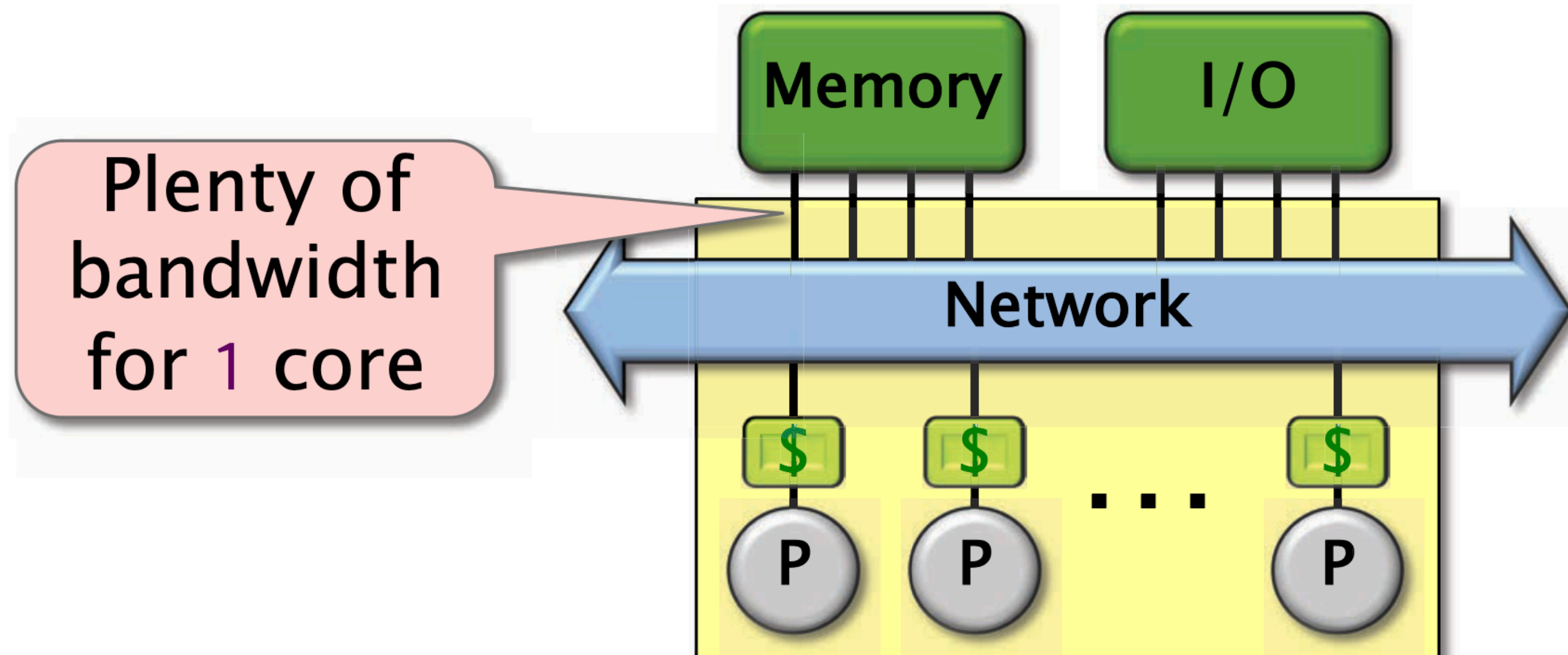


What are some reasons
that they might be so
close together?

Impact on Performance

Q. How can the cache-oblivious trapezoidal decomposition have so many fewer cache misses, but the advantage gained over the looping version be so marginal?

A. **Prefetching** and a good memory architecture. One core cannot saturate the memory bandwidth.



Caching and Parallelism

Rule of Thumb for Caching and Parallelism

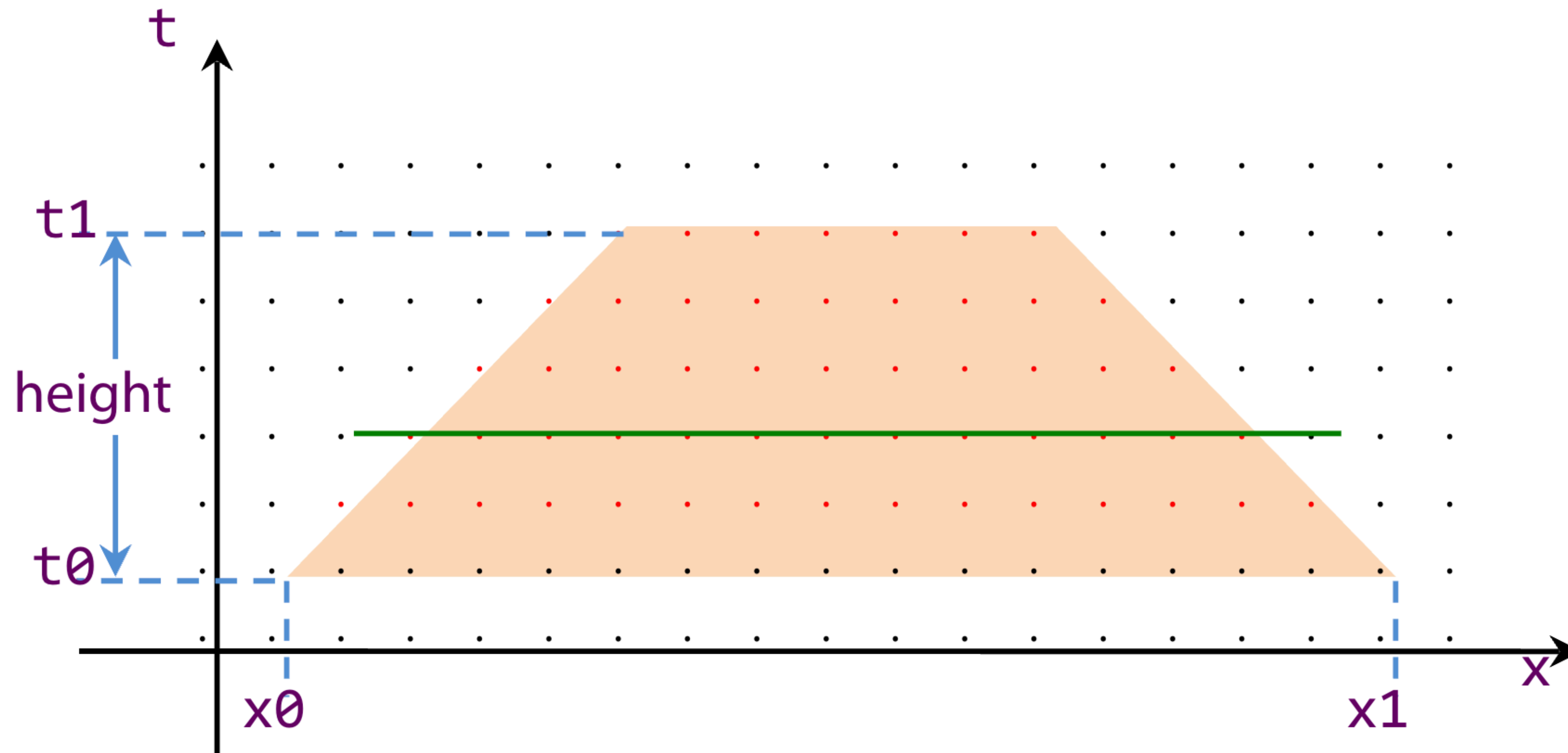
At a high level, minimizing cache misses in the **serial projection** minimizes them in parallel executions.

Serial projection is just the program minus the parallel constructs (spawn, sync, pfor)

cache misses in parallel $\geq$ cache misses in serial

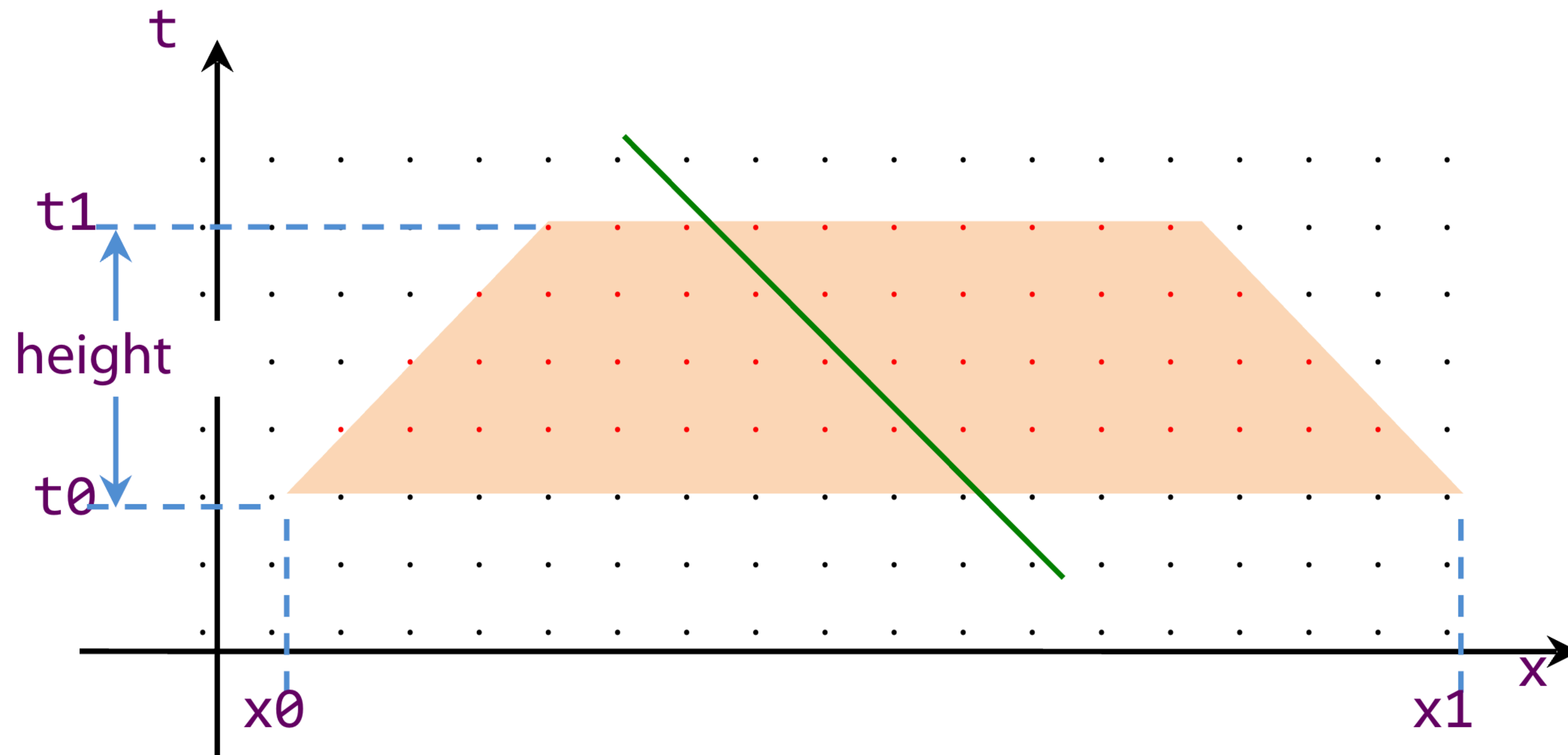
Time Cuts Don't Parallelize

There's no way to parallelize a time cut. The bottom trapezoid must be traversed first, and then the top one.

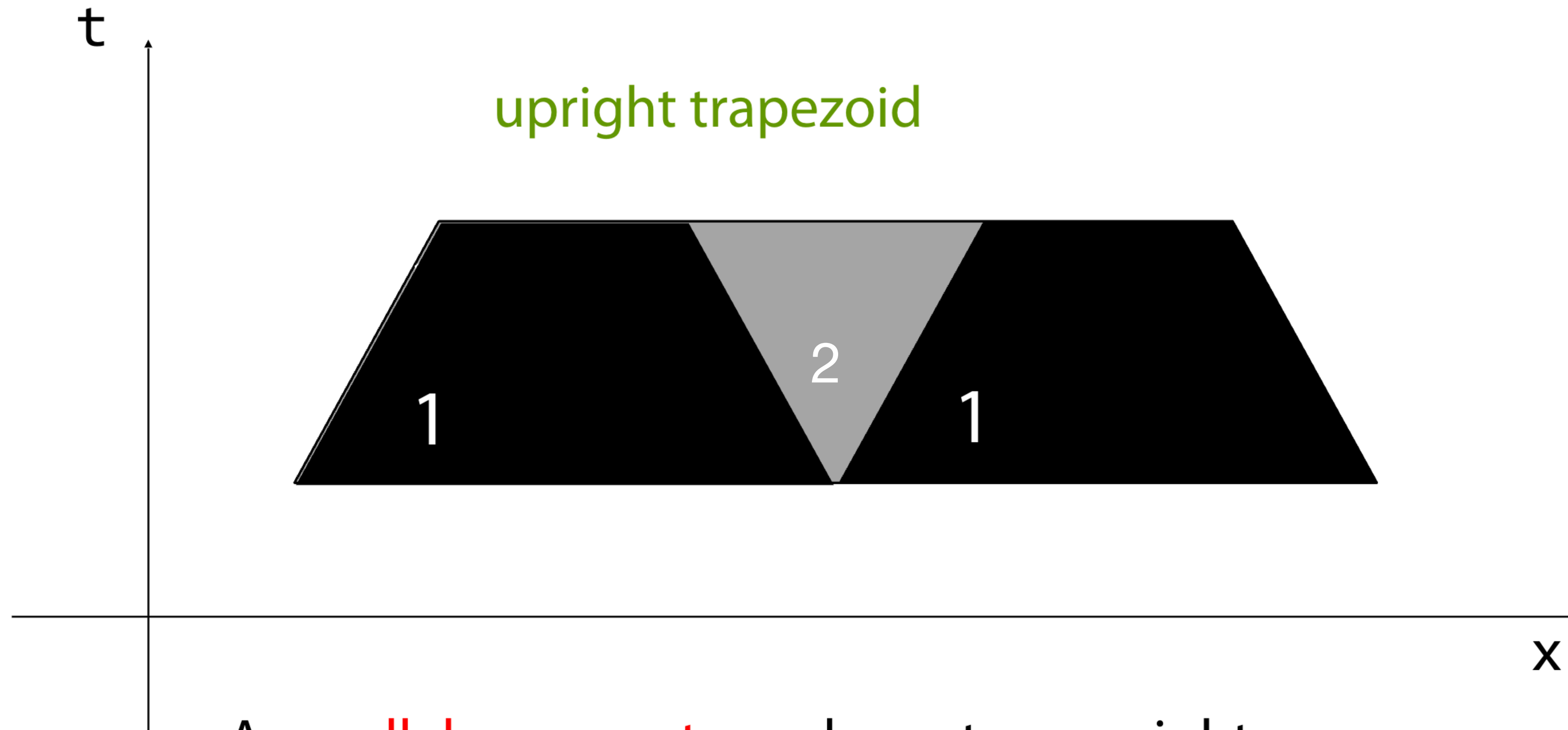


Space Cuts Don't Parallelize, or Do They?

A space cut poses a similar problem. You must traverse the trapezoid on the left before you can traverse the one on the right.

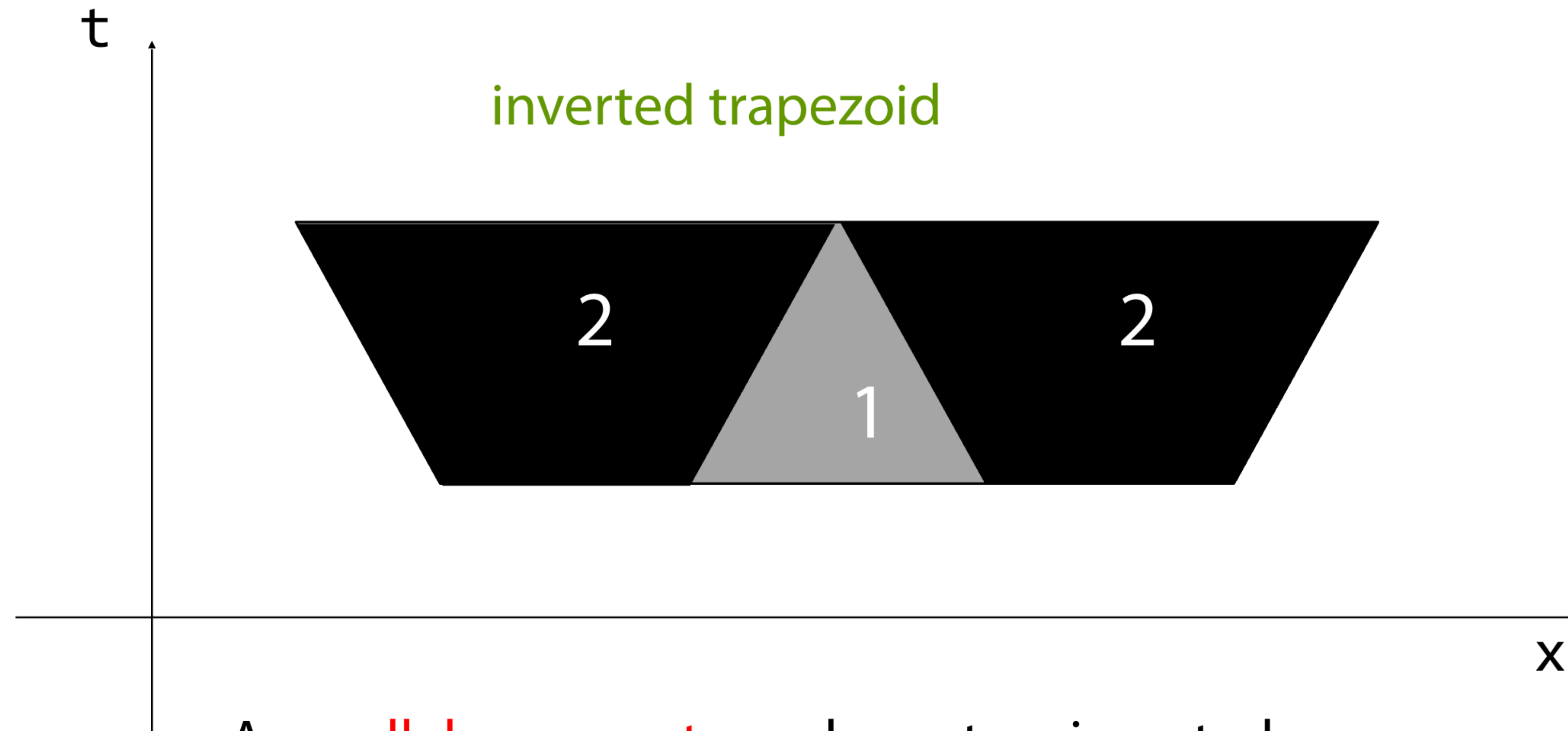


Parallel Space Cuts



A **parallel space cut** produces two upright trapezoids (black) that can be executed in parallel and a third “inverted” trapezoid (gray) that must execute in series after the two upright trapezoids.

Parallel Space Cuts



A **parallel space cut** produces two inverted trapezoids (black) that can be executed in parallel and a third upright trapezoid (gray) that must execute in series before the inverted trapezoids.

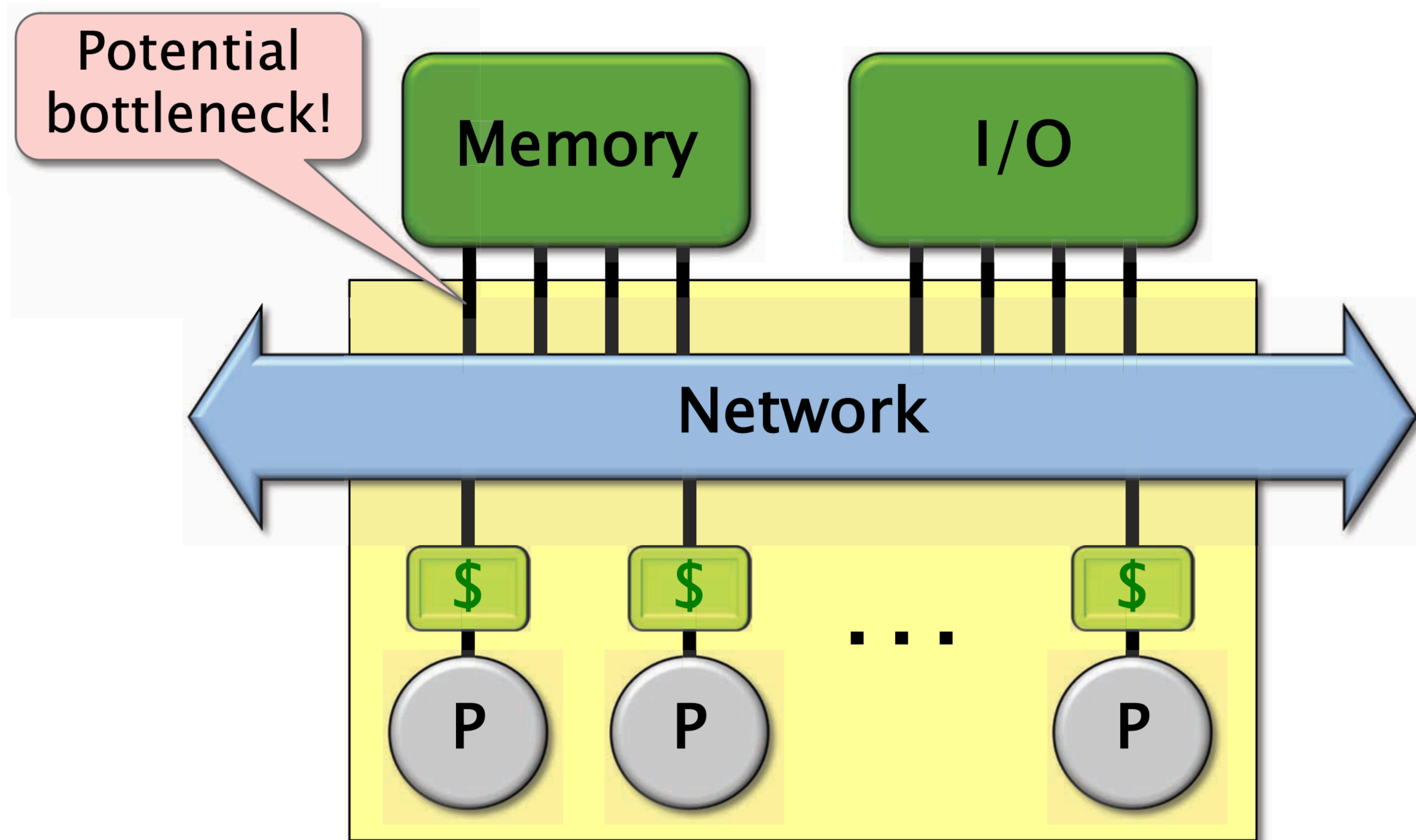
Performance Comparison

Heat equation on a 3000×3000 grid for 1000 time steps (4 processor cores with 8MB LLC)

Code	Time	
Serial looping	128.95s	} 1.93x
Parallel looping	66.97s	
Serial trapezoidal	66.76s	} 3.96x
Parallel trapezoidal	16.86s	

The parallel looping code achieves less than half the potential speedup, even though it has far more parallelism.

Memory Bandwidth



Impediments to Speedup

- Insufficient parallelism

Can get parallelism by computing work and span

- Scheduling overhead

Usually ok if your work chunks are big enough / have low span

- Lack of memory bandwidth

Can run **P identical copies** of the serial code in parallel - if you have enough memory

- Contention (locking and true/false sharing)

Tools exist to detect lock contention in an execution, but not the potential for lock contention. Potential for true/false sharing is even harder to detect.

Other Cache-Oblivious Algorithms

Matrix Transposition/Addition $\Theta(1 + mn / \mathcal{B})$

Straightforward recursive algorithm.

Strassen's Algorithm $\Theta(n + n^2 / \mathcal{B} + n^{\lg 7} / \mathcal{B} \mathcal{M}^{(\lg 7)/2 - 1})$

Straightforward recursive algorithm.

Fast Fourier Transform $\Theta(1 + (n / \mathcal{B})(1 + \log_{\mathcal{M}} n))$

Variant of Cooley–Tukey [CT65] using cache-oblivious matrix transpose.

LUP–Decomposition $\Theta(1 + n^2 / \mathcal{B} + n^3 / \mathcal{B} \mathcal{M}^{1/2})$

Recursive algorithm due to Sivan Toledo [T97].

Cache-Oblivious Data Structures

Ordered-File Maintenance

$$O(1 + (\lg^2 n) / \mathcal{B})$$

INSERT/DELETE or delete anywhere in file while maintaining $O(1)$ -sized gaps. Amortized bound [BDFC00], later improved in [BCDFC02].

B-Trees

INSERT/DELETE:	$O(1 + \log_{\mathcal{B}+1} n + (\lg^2 n) / \mathcal{B})$
SEARCH:	$O(1 + \log_{\mathcal{B}+1} n)$
TRAVERSE:	$O(1 + k / \mathcal{B})$

Solution [BDFC00] with later simplifications [BDIW02], [BFJ02].

Priority Queues

$$O(1 + (1 / \mathcal{B}) \log_{\mathcal{M}/\mathcal{B}}(n / \mathcal{B}))$$

Funnel-based solution [BF02]. General scheme based on buffer trees [ABDHMM02] supports INSERT/DELETE.